



mySAP™ PRODUCT LIFECYCLE MANAGEMENT

STATEMENT OF DIRECTION 2004

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FOREWORD



To succeed in today's competitive environment, you have to adapt to the unexpected. Bad news travels fast, and the stock markets are quick to punish companies that slip up. For example, one company saw its stock drop 10% when delays in product development became public. Another saw a drop of 11% when a new product

ramp-up problem was reported.* Major market forces are at work driving some companies towards greater adaptability while others fall by the wayside. Shorter product life cycles, more agile competitors, the accelerated availability of new technologies, demanding economic circumstances, and a greater compliance burden all add up to a greater need for adaptability and responsiveness. Adaptive companies succeed by sensing the needs of their customers, by bringing leading-edge products to market faster, and by utilizing their extended supplier networks more effectively.

In today's challenging global market, enterprises must innovate to survive. Business innovation must occur across all dimensions — product, process, and organization — to improve competitiveness and business performance. For products to be developed faster and at lower costs, companies must ensure that all product-specific information is consistent and available throughout the company as well as to critical partners and suppliers. This can only happen if the software solution supports all the required processes — those that relate to customer requirements, product specification, product engineering, production ramp-up, sales, and aftermarket services, as well as those that result from quality requirements and environmental factors.

Product life-cycle management (PLM) is not just a piece of technology. It is a business approach to managing all product definition information — creating that information, managing, disseminating, and using it throughout the product life cycle. A PLM solution has to span across the boundaries of the engineering department, it must provide information across all

stages within the life cycle, and it must be seamlessly integrated across the extended supply chain.

For many industries, bringing new products to market is perhaps the most critical business process, because only by having a regular flow of new products to the market can a company maintain or grow its revenue and margins. A recent study of FORTUNE 500 companies showed the correlation between proven success in introducing new products and market capitalization. Companies with strong track records had valuations up to five times the value of their assets; companies who struggled to bring new products to market were valued at close to their asset value.

Successful companies understand this — in a recent interview, the chairman and CEO of a leading FORTUNE 500 company said, “The companies that know how to develop things are ultimately going to create the most shareholder value. It's as simple as that.”** Another recent survey of IT business users conducted by AMR Research showed that new product development and introduction initiatives had the biggest impact on overall business.*** OSRAM, one of the leading lamp manufacturers in the world, is successfully using SAP® solutions to accelerate product development. Thanks to mySAP™ Product Lifecycle Management (mySAP PLM), they are able to leverage advanced product development capabilities that allow collaborative engineering across different countries and divisions within the company. “As a result, we expect to achieve a 30% time reduction for product information retrieval and a 10% to 20% reduction of engineering change cycle times,” says Franz-Josef Bierbrauer, vice president and general manager, ballast and luminaries at OSRAM.

mySAP PLM is working to support the adaptive enterprise with leading capabilities for product and project management, providing up-to-date product and project information; enterprise asset management, increasing asset utilization; environmental health and safety, ensuring legal compliance; and quality management, achieving higher product quality. Close collaboration and communication of product and asset information ensures integration among engineering, production, and sales

and maintenance operations, resulting in greater flexibility to meet customer needs. For leading companies today, mySAP PLM is working with other SAP solutions to provide unmatched support for new product development and introduction, and efficient ramp-up to manufacturing and aftermarket services – all built on a consistent master data foundation.

For many companies, this means building a consistent product structure that integrates product and process engineering, and that can be maintained throughout the life cycle of the product. For a car manufacturer, the challenge was the efficient management of complex products in a broad variety of variants and high volumes. The answer, developed in conjunction with SAP, is a project aimed at restructuring the technical information in the company. The result has been a significant cost reduction by reducing product variants that do not provide value to customers and which in the past led to errors in sales and manufacturing.

In the face of growing market demands and challenges, mySAP PLM enables companies to adapt to change. Testifying to this are the many new and enhanced features of mySAP PLM, which you will find discussed in this *mySAP Product Lifecycle Management Statement of Direction 2004*. These are the innovations that are defining the adaptive enterprise of the future – more flexibility, more integration, and more ability to be ready for anything in the real world of tomorrow.

Sincerely,

Claus Heinrich,
Member of the Executive Board, SAP AG

Sources:

* Prof. Vinod Singhal, DuPree College of Management, Georgia Institute of Technology

**MIT Technology Review, October 2003

***AMR Research; Survey of 100 Business Users, April 2002

ABOUT THE **mySAP™ PLM STATEMENT OF DIRECTION 2004**

This document presents SAP customers, partners, and external influencers with a clear and concise representation of our overall product life-cycle management (PLM) strategy. For example, it describes SAP's commitment to PLM, the key drivers that affect our current strategy and solutions, and detailed product descriptions of key capabilities and key functional elements.

Like the industries it serves, the *mySAP™ Product Lifecycle Management Statement of Direction 2004* is developing and evolving. Therefore, SAP welcomes your feedback on any areas that may be enhanced and improved in the coming months and years.

HOW TO USE THIS DOCUMENT

The *mySAP Product Lifecycle Management Statement of Direction 2004* is designed to meet the needs of many different types of readers. Some people may wish to read it from start to finish. Others may want to review only those sections that are most pertinent to their needs. For example:

- If you are an automotive company executive looking for an explanation of SAP's PLM capabilities for the automotive industry, you may review the introductory section, "mySAP PLM in 2004," as well as other sections of interest, such as "mySAP PLM Program and Project Management."
- If you are the head of customer service and support, you may review "mySAP PLM: A Vision Statement" and "mySAP PLM in 2004," as well as "mySAP PLM for Industries."
- If you are part of a product-development support team, you may use some or all of the document to prepare a business analysis of SAP's commitment and solution offerings.

IDENTIFYING TOPICS OF INTEREST

The best way to identify the topics of greatest interest is to review the table of contents. There, you can see that the document is divided into clearly delineated sections, including:

- **Vision and Market Perspective**
Examines the current PLM market; its needs and trends, values and benefits; and SAP's vision of how the PLM market will develop.
- **mySAP PLM Today**
Examines mySAP PLM as a strategic business solution for companies today and provides an overview of the new and enhanced features of the solution.
- **mySAP PLM Key Capabilities**
Describes the key capabilities of mySAP Product Lifecycle Management. Each section also describes specific new and enhanced capabilities that have been recently implemented.
- **mySAP PLM for Industries**
Describes the PLM concerns of key industries and how mySAP PLM addresses market needs as a strategic business solution.

CONTACT YOUR SAP ACCOUNT MANAGER

This document is not a substitute for an ongoing dialogue with your SAP account manager, who has a unique understanding of how to apply SAP's PLM strategy to your business. But it can complement that dialogue by supplying you with a solid background on SAP's overall approach to providing PLM capabilities to companies worldwide.

We at SAP hope that you find the *mySAP Product Lifecycle Management Statement of Direction 2004* informative and helpful and that the vision it portrays can become a reality for your company.

HOW SAP SETS A MARKET-DRIVEN PLM STRATEGY

SAP SOLUTION PRODUCTION

Customer requirements and satisfaction drive the evolution of SAP® solutions. With customer input, SAP directs a new product development process that manages all phases of planning, development, quality assurance, and production for all SAP products – from initial planning to ramp-up and mainstream rollout. All information about a software solution or development project is stored centrally and coordinated with all other SAP solutions to deliver solutions across our customers’ entire value chain.

A MARKET-DRIVEN PLM STRATEGY

SAP has adopted a unique, outside-in, customer-driven approach to its mySAP Product Lifecycle Management (mySAP PLM) strategy. The planning phase, which kicks off the solution production process, begins with product ideas and requirements that roll in from many channels, including thought leaders, analysts, competitors, partners, customer user groups, industry groups, strategic and customer development projects, Web-based development request systems, and ramp-up customers, as well as SAP’s own field organization and industry business sectors.

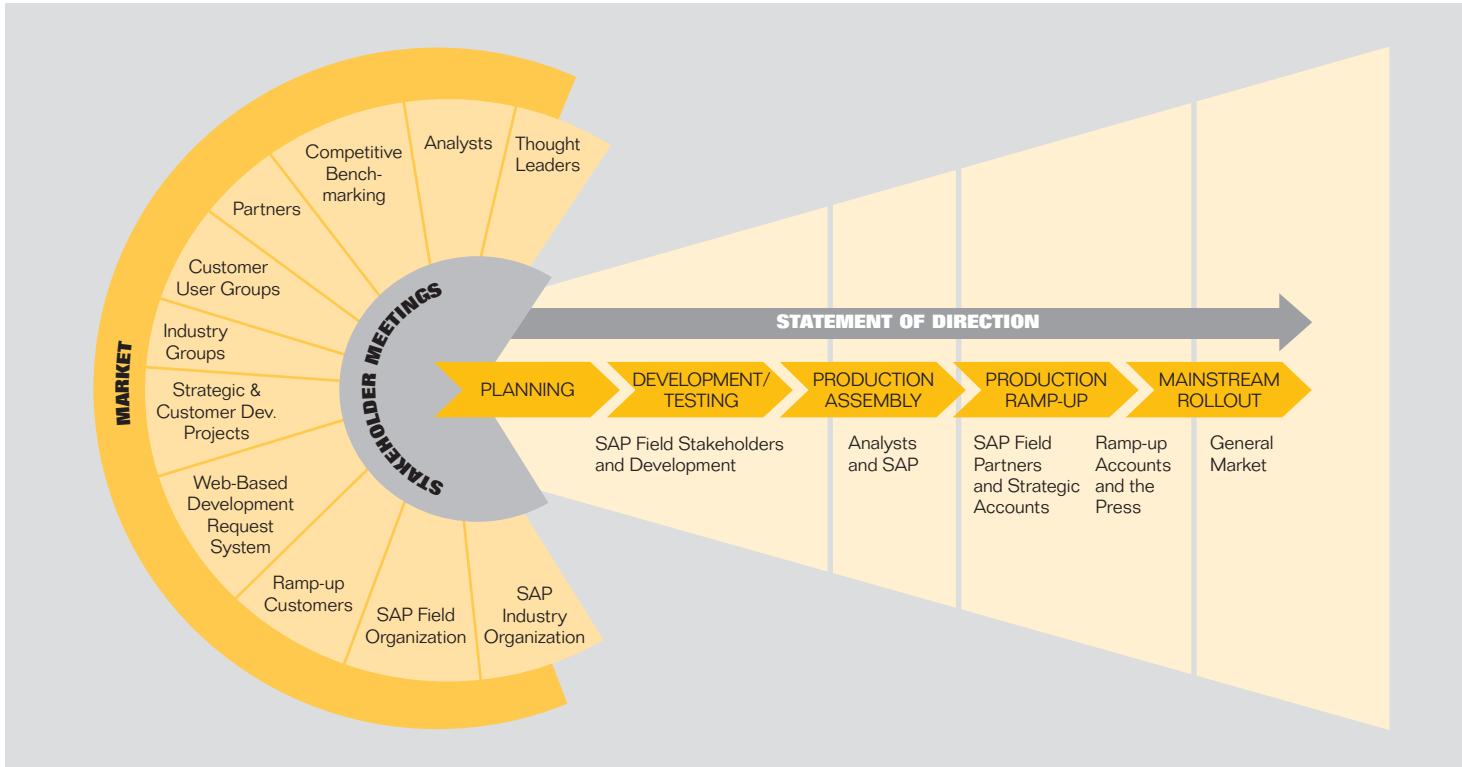


Figure 1: Market-Driven Multichannel Roll-In Process Drives Planning Phase of SAP® Solution Production Process

MULTICHANNEL ROLL-IN PROCESS

SAP continually collects data from roll-in channels. Customers, partners, analysts, thought leaders, and SAP development teams provide input that articulates market demand, which drives product innovation. This input includes market analysis, competitive benchmarks, strategic development projects, and direct feedback from customers and consultants. Throughout the formal planning process, SAP uses a market-driven model to prioritize the ideas and requests it receives.

The following sections describe each of the channels that help make SAP's market-driven, customer-centric PLM strategy a success.

Thought Leaders

Thought leaders provide valuable insights into global business that affect current and future technologies. With its superior reputation and 30 years of business expertise, SAP has access to the world's leading thinkers and practitioners in a variety of forums.

Business and Financial Analysts

Business and financial analysts provide some of the most direct and candid feedback about the market and technology. Through research and public statements, the analyst community provides critical reviews of SAP's software development, as well as market analyses.

Competitors and Competitive Benchmarking

Competitive benchmarking ensures that SAP understands what customers expect from leading vendors. Because customers typically evaluate more than one competitive product when selecting PLM solutions, benchmarks ensure that SAP can gauge how its products compare with those of its competitors. Benchmarks also enable SAP to develop areas of value-added differentiation.

Partners

SAP cooperates closely and openly with its partners, sharing knowledge, ideas, and software solutions. Partner input is an invaluable element of SAP's PLM strategy.

Customer Users' Groups

SAP gathers feedback from customers through a variety of forums, including its customer users' groups. Chief among these international user groups are the Americas' SAP Users' Group (ASUG) and Germany's SAP Users' Group (DSAG).

Industry Groups

By attending and hosting industry group events, SAP can track changing market trends and solicit input from similar companies.

Strategic and Customer Development Projects

SAP undertakes strategic development projects and customer development projects that are in line with its strategic initiatives and its partnerships with key strategic customers, such as SAP Global Alliance partners. These projects often lead to specialized development, which can result in introducing new and superior capabilities, a hallmark of SAP's excellence.

Development Requests

As an additional service, selected SAP customers can request specific developments, which contribute to the continuous improvement of mySAP Business Suite solutions. Such customers include SAP users' groups, customers with a certified customer competence center or global alliance account status, and SAP's development and consulting partners.

Ramp-Up Customers

Through the SAP Ramp-Up program, customers can provide direct feedback on new products. The program also allows SAP to receive detailed information on how customers select, implement, and use mySAP Business Suite solutions.

SAP Field Organization

The SAP sales, consulting, training, and support organizations are important channels for product improvements. These groups have frequent contact with prospects and customers, who provide rich input into short- and long-term product planning.

SAP Industry Organization

Like the SAP field organization, the SAP industry organization provides direct customer input. This group also provides insight into the specific needs of individual industries and translates generic product offerings into capabilities for industry-specific process improvement.

SAP also garners input from the SAP Executive Board and corporate research.

SAP Executive Board

The SAP Executive Board believes that IT spending continues to focus on solutions that can deliver a high and visible return on investment, lower the total cost of IT solution ownership, and leverage existing IT investments. mySAP PLM meets all of these requirements.

Corporate Research

SAP's dedicated research and development organization works in research centers around the globe. Its goal is to establish SAP as the leader in innovative and strategic technologies, such as mobile and electronic business, the Internet, and mobile computing. As a respected member of the worldwide research community, SAP's corporate research division participates in worldwide academic conferences, workshops, and other gatherings of elite R&D professionals to gather and develop the best corporate intelligence available today.

tightly throughout their enterprises. SAP also recognizes that good ideas originate from many places and that its own internally generated ideas may not fit specific customers' needs. By understanding the market and recognizing individual customer requirements, SAP develops the right solutions for the right customers at the right times.

SAP is committed to this customer-driven strategy. Listening to and understanding customers' business needs ensures that SAP will continue to deliver superior solutions that empower its customers to be more successful.

COMBINING RESOURCES FOR PLM SUCCESS

With more than 30 years of success in enterprise business solutions, SAP understands the depth of knowledge and resources required to produce a superior PLM solution. SAP's PLM solutions empower organizations to integrate and synchronize PLM functions

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mySAP™ PRODUCT LIFECYCLE MANAGEMENT: A VISION STATEMENT

WHAT IS PRODUCT LIFE-CYCLE MANAGEMENT?

Over the last quarter of a century, the term “product life-cycle management” (or PLM) has emerged to describe a business approach for the creation, management, and use of product-associated intellectual capital and information. This information is used throughout the product life cycle – from the first idea of a new product, onward through production and sales, all the way to service and support.

The seeds of PLM were first planted in the 1980s, when companies, concerned about how to effectively manage their computer-aided design (CAD) documents within their design departments, began to deploy CAD tools. In the late 1980s, product data management (PDM) applications turned the emphasis to engineering drawings and how to manage them. Limited solutions were sold primarily to the managers of engineering departments. From there, the increasing need to make product information available outside of the engineering domain and to enable a better process integration from engineering to manufacturing, sales, and aftermarket services expanded the scope of PLM toward the extended enterprise. The result was a modern, enterprise-wide PLM solution, which has become a strategic imperative for improved business performance.

Today, CIMdata defines PLM as, “a strategic business approach that applies a consistent set of business solutions in support of the collaborative creation, management, dissemination, and use of product definition information across the extended enterprise from concept to end of life – integrating people, processes, business systems, and information.” (Source: *Product Lifecycle Management, Empowering the Future of Business*, CIMdata 2002, CIMdata.)

Focusing on Product-Led Strategies

When undertaking a PLM initiative, the type of financial benefit upon which a company should focus – cost-efficiencies or profit and revenue growth by developing innovative products – depends upon that company’s basic competitive strategy. Many PLM initiatives fail to meet expectations simply because management

teams have not aligned their PLM goals with their company’s basic competitive strategy.

To understand this issue more fully, consider that customers make three different types of demands on an enterprise – demands for quality products, demands for quality service, and demands for low prices. First, all customers want products that meet or exceed their functional requirements, are reliable, and have a level of quality that is at least on a par with comparable products offered by competing companies. Second, because customers use products to meet a need or solve a problem, they also want the service necessary to ensure their problem is solved, including installation, maintenance, and repair of the product; financing or insurance; and perhaps even guidance regarding the most effective way to use the product. Third, customers want to minimize the price they pay for products and services, which equates to their total economic expense of obtaining the product and putting it to use to meet their needs.

Inside most companies different groups of people are focused on meeting each of these three demands: customer intimacy with sales and marketing teams, product leadership with product or service development teams, and operational effectiveness with manufacturing and logistics teams. Product leadership is often directly related to PLM. However, to be successful at meeting one of these demands often means compromising one or both of the other demands.

Michael Treacy and Fred Wiersema, coauthors of *The Discipline of Market Leaders* (Perseus Publishing, 1997), eloquently describe this dilemma. The authors suggest that to be competitively successful, a company must be disciplined enough to concentrate primarily on just one of three different strategies – product leadership, price (operational excellence), or customer intimacy or service:

- **Leadership:** Product innovators strive to be first to market with new products and services, but success requires these competitors to maintain a creative, agile operation that can quickly reinvent itself to adapt to the next new idea.

- **Price:** An operationally efficient competitor works to drive costs as low as possible, with success depending on scale rather than agility and simplicity and with efficiency achieved partly by discarding all nonessential activities and processes.
- **Service:** A customer-intimate company tries to offer a complete range of services and products to its customers, that is, a complete solution for each customer's problem.

Focusing on Innovation and the Right Products

To meet these challenges, businesses must become more innovative. However, being an innovative business means more than simply creating innovative products. It also means improving the processes a company uses to make its products, and it means improving how the company supports its products with innovative approaches to the complete product life cycle. Today, savvy companies recognize that innovation is crucial to maintain their competitiveness in the marketplace. However, companies must achieve innovation while reducing overall product-related costs across development, production, and service.

In this regard, a primary business driver is the increasing product complexity and customization being witnessed in global markets. Not only are mechanical configurations getting more complex, but increasingly, products are also including complex electronics and software. In addition, customers want to have “their” product configured to their individual specifications. The increase in product complexity, coupled with the desire for personalized configurations, requires an enhanced ability to quickly define new product variations and options and to be able to manage the configurations that are offered. Additionally, companies must manage the entire product or product family, integrating elements like product recipes and packaging to meet regional requirements and regulations.

In the past decade, many business investments, such as enterprise resource planning (ERP), have addressed improving operational efficiency. Those investments continue today with an expanded focus on using PLM product definition information more effectively in both the production and the operation of a product or

plant in service. In addition, to achieve operational excellence, the importance of effectively leveraging supply chain partners for both design and production efficiency has become obvious.

Creating and maintaining product leadership requires generating revenue from a steady stream of new products. Today, expanding enterprises must bring innovative products to existing and new markets more effectively and more quickly to maximize customer interest and sales. To do this they must leverage and reuse the product-related intellectual capital created by their business partners, working together across the extended enterprise value chain. The pressures to reduce time, improve product quality, and lower costs have not gone away. They are being reaffirmed and folded into programs that focus on delivering the “right” product.

Who Uses PLM?

Discrete industries, such as the automotive, high-tech, aerospace and defense, consumer products, and engineering and construction industries, benefit from PLM. However, with the emergence of recipe and formula management capabilities, PLM can now streamline product development processes for process industries, as well. These include such industries as pharmaceuticals, chemicals, and consumer product companies in the food business. In addition, asset-intensive industries, such as oil and gas, mining, and utilities, benefit by using PLM to increase asset utilization and to streamline their service maintenance processes.

PLM Defines Competitiveness

To be competitive today, companies realize that it takes the whole enterprise to design, develop, market, and service a successful product. The view of these organizations is that “the whole enterprise must work as one,” including the business partner ecosystem — with everyone sharing the same information to make timely, cost-effective, quality products — because they know it is the only way to succeed in the marketplace today and to establish the foundation for future growth.

To do this, companies are innovating to improve business performance as never before — not just in product design and customer service, but also in manufacturing processes and the management of strategic programs and organizations. They are using their resources and intellectual assets in ways that help to differentiate them from their competitors. Key to this approach is to establish a closed feedback loop between product design and customers to enable a fast response to changing market conditions and customer feedback.

PLM turns this vision of an enterprise-wide solution into reality. It applies a proven and consistent business approach to solving the problems of making quality products quickly at the target cost, while it drives innovation across the enterprise, resulting in successfully competing products and successfully competing product companies.

A Holistic Approach to PLM

Although piecemeal PLM solutions might be defined by technology, a holistic PLM solution is not. A holistic PLM solution is a business process that manages all product definition information for an entire enterprise. It creates that information, processes and manages it, and makes the same consistent, accurate, up-to-date information available throughout the enterprise and throughout the life cycle of the product.

Implicit in this definition is the solution's full support of all the processes that drive the multiple stages of a product's life cycle. These include all of the new product development and introduction processes, like idea management, portfolio planning, specifications, and product structure management down to the ramp-up to manufacturing, market introduction, and service management. Quality management processes, which span all phases of the product life cycle, as well as the processes that use material specifications to manage environment, health, and safety requirements, also have to be part of a PLM solution.

By supporting all product life-cycle processes — not just engineering — a PLM solution can provide consistent, relevant information on demand, at any time and from any place, across all departments, and across the extended supply chain.

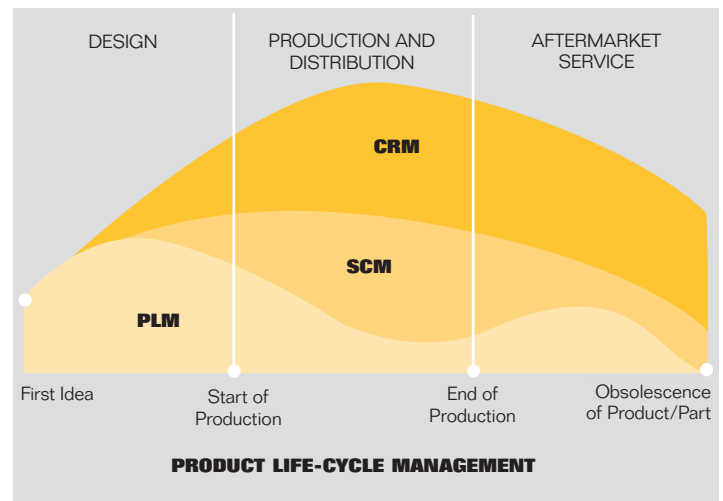


Figure 2: The Holistic View of Product Life-Cycle Management

But in addition to creating, managing, and distributing the product-related information that supports the product life cycle, today's multifaceted PLM solution must also manage large volumes of project-related information. To run a research or development project effectively, it is essential to manage the required project resources, tasks, milestones, and deliverables that come with it using a powerful program and project management solution.

But the management of the product life cycle does not end once a product has been developed. There has to be a close handshake between manufacturing and supply chain management to plan for manufacturability and to enable a smooth ramp-up to production. Service management relies on accurate knowledge of as-maintained product configurations; technical and physical assets need to be operated and maintained; and results from maintenance have to be fed back into product development. Product and asset configurations stored in a PLM system are the foundation for aftermarket services, including service parts

management and customer service. That's why a PLM solution has to be closely integrated with other core business applications, including:

- **Customer relationship management (CRM)**, especially in the early product definition stages for opportunity management, to aid in understanding customer demographics and market statistics, as well as to integrate information from marketing and customer service
- **Enterprise resource planning** to use engineering bills of material (BOMs) and changes to drive the operations processes and to access financial data about product performance as a basis for product portfolio decisions
- **Supplier relationship management (SRM)** to manage the supplier network and to procure direct materials
- **Supply chain management (SCM) and manufacturing** to manage production and provide a smooth and efficient ramp-up

The evolution of the Internet and the adoption of Web-based technologies enables employees, partners, and customers to effectively share information across enterprise boundaries. The introduction of new technologies, such as Web services, along with an enterprise service architecture, provide new ways to flexibly adapt a business process to industry- and customer-specific requirements – building the bridge between different business applications, including PLM, CRM, and ERP, for example, in the product definition and idea management process.

The Holistic Values and Benefits of Product Life-Cycle Management

A holistic approach to PLM streamlines a company's processes, multiplies efficiency, and empowers a fast response to changes, whether in products, markets, or competitors. It can reduce development time (and thereby lower costs) in many ways, for example, by enabling processes to run in parallel instead of sequentially.

Thus, a PLM solution today must be designed to impact an enterprise's bottom line. Specific examples of the values and

benefits achieved by enterprises that have deployed such a solution include the following:

- Reduced time-to-market by using a complete solution to manage and retrieve all product-related information – from concept through design, production, and product obsolescence. Example: a 40% improvement in product change cycle times.
- Reduced cost-to-market by evaluating program and project progress and supporting investment and divestment decisions for products. Example: a 15-30% reduction in unnecessary prototypes.
- Reduced risk exposure and cost of engineering changes by providing an integrated set of capabilities to monitor, implement, and control engineering changes – from customer-requested changes to projects, products, and engineering changes against released production orders. Example: an engineering review process reduced by 83% – from 12 days to 2 days.
- Brought products to market in a more personalized, “market of one” offering, providing products tailored to individual needs; improved knowledge of available parts and components reduces costly design and engineering work; and allowed vendors to enter niche markets. Example: a 25% productivity increase in design engineering.
- Reduced maintenance costs and increased equipment availability by establishing solutions to plan, measure, and track the effective planning, operation, safety, and maintenance of equipment. Example: a 10% higher asset utilization in plant operations.
- Improved decision making at all levels of the organization by using powerful and flexible analytics capabilities to cover such areas as portfolio management, occupational health, product safety, and product quality, as well as enterprise asset management
- Guaranteed that internal, external, professional, and occasional users have centralized, role-specific, context-driven access to all the information, applications, tools, and services needed to accomplish their daily work – anytime, anywhere, and on any Internet device.

- Enabled low total cost of ownership (TCO) through an open and robust infrastructure that provides easy integration of operational systems, such as CAD applications, ERP, CRM, SRM, and SCM solutions.

A PLM Solution for Today and Tomorrow

Many of the PLM solutions applied by companies today were planned 5 to 10 years ago and focus on the life cycle of a product inside the company that owns that product. Their support for the idea of using the complete business network – the complete enterprise, including the extended value chain – to make quality products quickly and at low cost is being only partially implemented. However, solutions to collaborate across the extended supply chain network – with suppliers on request for proposals and quotations, with engineering partners on request for specifications and designs, and with an integral process to feed the design requirements into the early demand planning – are quickly becoming important to companies that want to be more agile, more efficient, and better able to respond to customer demand.

Evidence of the trend toward total PLM solutions is seen in the processes and key capabilities that have been created to enable such solutions, which are described in the following sections.

Support for the New Product Development and Introduction Process

Innovation is paramount to competitive success today. Companies know this. The new product development and introduction (NPDI) process helps companies to be innovative and flexible, enabling them to create new products and get them to market as quickly and efficiently as possible.

SAP is providing a comprehensive, closed-loop solution that helps companies streamline development and new product introduction processes. This solution will help them innovate, design, develop, and introduce new, quality products to market quickly by uniting people with information – all in one, complete,

closed-loop business process. The solution supports the entire NPDI process – from first concept all the way through ramp-up and sell-through.

PLM: A DECISIVE COMPETITIVE EDGE

As companies have mastered improved operational practices since the mid-1970s, the scope of competitive differentiation has narrowed to product excellence. Today, competitive success rests heavily on a company's ability to identify customer requirements, design for purpose, and successfully manage product launches and support in the market.

In the interconnected and keenly competitive business world of today, SAP sees PLM as a strategic business approach designed to empower a business to succeed, to enable product and process innovation, and to enhance both top- and bottom-line business performance. It provides a framework upon which platforms of technology, processes, best practices, and key functional elements combine to enable a complete solution to business problems.

PLM has come a long way in 25 years. For a business to be successful in an increasingly sophisticated global marketplace, PLM is no longer an option. It is a competitive business necessity.

mySAP PRODUCT LIFECYCLE MANAGEMENT IN 2004

mySAP PLM TODAY

To be market leaders, companies must stake out new market space. In today's economy, this isn't easy. It means getting new products to market more quickly and with more precision and efficiency than ever before. Moreover, as competition grows fiercer, product cycle times grow shorter and margins grow thinner. At the same time, customers are demanding that products be produced faster – and with more personalization. Time-to-market is critical. In addition, a company must build and maintain a customer base, and it must keep service costs to a minimum while simultaneously creating new revenue streams. The challenges facing companies today are indeed immense.

Fortunately, product life-cycle management (PLM) solutions are helping companies meet these challenges. To achieve sustained success – that is, to attain the capability to make the right products at the right times and at the right costs – companies must treat the product life cycle as part of their total business strategy. This means that companies in all product-related industries now recognize that using product definition information throughout the product life cycle is critical to their success, and they are considering or have already implemented PLM solutions.

The practice of using product definition data only within engineering and manufacturing operations belongs to yesterday. Today, companies share product information across departments and with other business areas, such as marketing and sales, planning and production, procurement, customer service, and maintenance. They also share this information across corporate boundaries in the extended enterprise, enabling them to integrate their business processes with those of their partners, suppliers, contract manufacturers, service providers, and customers.

At the same time, PLM solution and technology suppliers are creating industry-focused solutions, increasingly helping companies within specific industries to integrate their core product

data with all types of other business information. At the same time, they are providing greatly improved collaboration and visualization facilities.

Companies are also adding other capabilities, such as customer relationship management (CRM) or supply chain management (SCM), to their existing enterprise resource planning (ERP) solution. The question is: which solution offers the best value creation and return on investment (ROI)? The answer is: it depends on the objectives. A company may go with a CRM or SCM solution first. But companies with the goal of being market or product leaders are also concentrating on PLM.

For success with PLM, the solution a company chooses must fit tightly with its existing production systems and environment. To run a complete production environment, a PLM solution must be able to interoperate with other solutions – particularly CRM systems to identify what customers want and SCM systems to control the complete value chain, including vendor and partner relationships. These solutions must integrate with PLM, which is the central solution because it makes the products and manages them for their entire life cycle. How well these solutions integrate dictates the quality of the overall production environment.

Engaging the challenges of leadership is well worth the effort. Driven by the SAP vision of enabling companies to integrate the entire enterprise and to support the business processes that affect a product throughout its life cycle, mySAP™ Product Life-cycle Management helps to bring those challenges down to size.

Meeting Change Head-on

mySAP PLM equips companies for success by seamlessly integrating information and processes, as well as effortlessly sharing project and process data. Unlike piecemeal solutions, it unites and connects critical functions and departments, such as marketing, engineering, manufacturing, quality, and service, so that they operate together – not as disjointed entities. mySAP PLM

makes it easy to securely share product and customer information, and it makes it easy for business partners to access information that companies want to share with them whenever they need it.

mySAP PLM is a comprehensive solution. With it, companies can manage product innovation and new product development, production ramp-up, product change management, and maintenance. And companies can transfer, track, and control product and asset information throughout the extended supply chain to ensure speed, quality, efficiency, and profitability.

A Holistic View of the Product Life Cycle

mySAP PLM is part of the mySAP Business Suite family of business applications, which contains the solutions built to meet the new realities of collaborative commerce. Each solution is designed to work seamlessly with all other SAP® solutions and to integrate with those of other providers. As a member of mySAP Business Suite, mySAP PLM brings this open approach to the management of a company's products, projects, programs, and assets, and it provides the tools, practices, and advantages companies need to succeed in today's fully connected world.

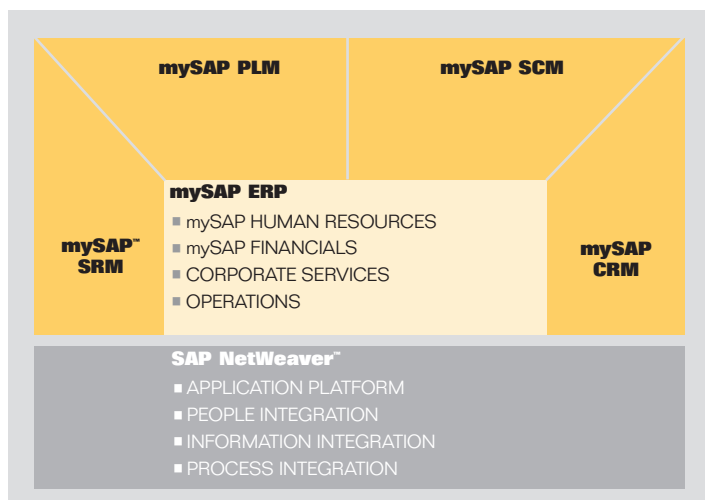


Figure 3: mySAP™ PLM as part of mySAP Business Suite

mySAP PLM begins this open approach by taking product innovation beyond the boundaries of organizational constraints. It integrates products, processes, and people across the enterprise and the supply chain. This means companies can harness the creativity and expertise of internal resources and external partners to quickly turn concepts into successful products.

mySAP PLM facilitates creativity by painting a holistic view of all product-related business processes and information throughout the complete product and asset life cycle. Because it integrates with other solutions, mySAP PLM enables processes to interlock and run in parallel, reducing the time it takes to get to market. The solution enables all life-cycle phases from beginning to end.

By giving open access to all relevant, accurate information, as well as providing the applications, systems, services, and tools to use that information, mySAP PLM improves visibility into both projects and costs. It integrates both product and process information, which changes and improves a product while keeping costs down. Fully integrating product and process information is the only way to manage all phases of the life cycle, which are:

- Product innovation and new product development
- Production ramp-up
- Product change and refinement
- Quality assurance
- Service and maintenance

Answering a Common Need

An enterprise-centric PLM solution is not just for large companies. Its value extends to any enterprise that needs powerful and efficient product development, quality maintenance, plant maintenance, and change management processes.

SAP has analyzed the business processes of numerous industries. This knowledge helps SAP understand how these processes work and how to make them work better. Therefore, aside from its universal support of internal and external PLM processes, mySAP PLM also supports many specific industries.

For example, mySAP PLM provides full, multilevel recipe management capabilities that support the unique needs of consumer packaged goods companies and process manufacturers, such as chemicals and pharmaceuticals. Other industry-specific solutions meet the requirements of the automotive, high-tech, communications, plant engineering and construction, industrial machinery, and aerospace and defense industries. (For more information, see the section entitled “mySAP PLM for Industries” on page 115.)

Common forces influence new product development in diverse industries. In response to these forces, companies in many industries have implemented mySAP PLM because:

- Product innovation and time-to-market are key drivers of product leadership and lasting shareholder value.
- Customers are demanding more personalized choices and higher product quality.
- Products, services, and maintenance offerings are being bundled, increasing the need to have consistent product information available across all life-cycle phases.
- Globalization and lower profit margins are forcing companies to focus on core competencies, requiring the establishment of buyer, vendor, and supplier value networks.

Enabling Discrete and Process Industries

All discrete manufacturing sectors and the process and utilities industries are candidates for PLM solutions. In addition to common market forces, these industries face common challenges: global competition, complex supply chain management issues, often intense cost pressures, and time-to-market demands that can make or break a company's – or even an industry's – profitability.

Today, **discrete manufacturers** have implemented mySAP PLM to help them create innovative products that distinguish them from their competitors. By deploying mySAP PLM, they have enabled their enterprise and everyone in it to become more productive and efficient, and they can now effectively collaborate with and leverage the resources of their worldwide business partners.

The **process and utilities industries** parallel the discrete manufacturing industry in many respects. For example, both industries involve design, manufacture, assembly, support, and maintenance processes. However, other factors influence the PLM needs of these industries, such as compliance with ever-changing government regulations, operational documentation, and extensive functionality testing that must be completed before production can begin. These industries can use mySAP PLM to address those and other essential industry needs. For example, because they are capital-intensive industries, mySAP PLM's powerful enterprise asset management capabilities (such as powerful document tracking, release management, asset structure, and configuration management capabilities) can help them ensure plant reliability and safe operations.

The Right Product, the Right Time, the Right Price

mySAP PLM is the only solution that enables full production process support, including collaborative engineering; custom product development; project, asset, and quality management; and the management of environment, health, and safety requirements among multiple partners.

mySAP PLM mobile capabilities make sure products arrive on time. Seamless integration with enterprise portals, as well as public and private marketplaces, makes the solution easy to use and extends the reach of the business processes it drives.

mySAP PLM is an integrated solution. It works seamlessly with mySAP Customer Relationship Management (mySAP CRM), mySAP Supply Chain Management (mySAP SCM), mySAP Supplier Relationship Management (mySAP SRM), and mySAP ERP – extending and improving communications between all departments in marketing, sales, engineering, manufacturing, and service.

This high level of integration means mySAP PLM enables the end-to-end management of the product life cycle. And that reduces and streamlines overall integration costs, while speeding deployment to customers.

With mySAP PLM, companies can accomplish all of the following:

■ **Manage product information**

mySAP PLM delivers the wide range of product information needed during the life cycle, such as computer-aided design (CAD) models and specifications in the design phase; recipes, bills of materials (BOMs), and routings in the planning and production phases; and as-sold or as-maintained structures for the service and maintenance phase.

■ **Collaborate with suppliers, partners, and customers**

mySAP PLM provides a virtual collaboration environment in which everyone can share, review, and change information using enhanced document management and collaboration capabilities.

■ **Manage product development projects**

mySAP PLM provides advanced capabilities to streamline projects and programs and keep them focused on achieving results.

■ **Maintain and improve product and service quality**

mySAP PLM integrates quality management capabilities, collaborative scenarios, and mobile solutions to ensure quality is built into processes, systems, products, and services.

■ **Improve plant and equipment performance**

mySAP PLM enables product-driven management that helps companies monitor and maintain assets; perform equipment repairs, modifications, upgrades, and refurbishment; and track costs for individual assets, production lines, and entire facilities.

■ **Manage environment, health, and safety processes**

mySAP PLM provides capabilities for monitoring complex regulations, managing data and documents, tracking dangerous materials and products, carrying out waste disposal, enhancing employee and public safety, and supporting preventive healthcare.

■ **Improve access to business content**

mySAP PLM leverages a company's existing investment and delivers tangible cost- and time-saving benefits by providing access to essential business content, at any time and any place via many different devices.

A Complete Solution

mySAP PLM offers a complete solution that addresses all phases of the product life cycle. To discover the full expanse of its capabilities, in addition to this statement of direction, customers and interested readers are encouraged to visit the mySAP PLM Solution Map at www.sap.com/solutions for details on all the PLM-related processes SAP provides and supports.

mySAP PLM is open and compatible and supplies interfaces to non-SAP solutions and a variety of partners – vendors of leading solutions that optimally complement SAP's own portfolio, such as a direct link to the leading CAD interface packages.

Life-Cycle Data Management	Document Management		Product Structure Management		Recipe Management		Integration		Change and Configuration Management	
Enterprise Asset Management	Technical Assets Management			Preventive and Predictive Maintenance		Maintenance Execution			Work Clearance Management	
Program and Project Management	Project Planning			Project Execution		Interfaces			Program Management	
Life-Cycle Collaboration and Analytics	Idea Management & Concept Development	Design Collaboration		Collaborative Project Management	Quality Collaboration		Analytics		Enterprise Portal Content	
Quality Management	Quality Engineering				Quality Assurance/Control			Quality Improvement		
Environment, Health & Safety/ Emissions Management	Basic Data and Tools	Product Safety	Hazardous Substance Management	Dangerous Goods Management	Waste Management	Industrial Hygiene and Safety	Occupational Health	Emissions Management		

Figure 4: mySAP™ PLM Solution Map

Total Information Access

With the advent of the Internet, SAP began its pursuit of a new vision to help companies incorporate PLM as part of their overall strategic strategy for success. Today, that vision is a reality. It uses the power of the Internet and personalized, role-based Internet portals to deliver direct access – from wherever users happen to be – to almost instantaneous information. When this power is applied to mySAP PLM, the result is fully optimized processes involving multiple unit and development groups within one international enterprise.

A State-of-the-Art Enterprise Portal

mySAP PLM is supported by a full-blown enterprise portal that intuitively accommodates the specific needs of each user. Because different users work differently – each placing different and specific demands on their workspace – the individual portals are preconfigured and customized for their role in the organization. The portal provides easy access to the tools and information each person needs to do their job. The portal is ideal for all occasional users, such as marketing managers, project managers, or maintenance engineers.

The role-based enterprise portal gives everyone in an enterprise – including business partners – fast, simple, and user-friendly Web access to all of the internal and external applications, information, tools, services, and resources they need to perform daily tasks as efficiently, as organized, and as productively as possible.

Support for the New Product Development and Introduction Process

For discrete and process industries, an effective process that can streamline development and transform an idea into a new product and launch it in record time is important. In addition, a single process that can foster a development environment of innovation to keep pace with today's demanding requirements and fast-changing consumer habits is essential. A single, tightly integrated, closed-loop process that can support the entire new product development and introduction (NPDI) process – from the first idea to sell-through – is critical to lasting success.

SAP supports that NPDI process. The following sections describe the benefits of SAP's support for the NPDI process.

A SINGLE PROCESS FOR NEW PRODUCT DEVELOPMENT

SAP addresses all facets of the NPDI life cycle, including:

- Defining the product strategy
- Managing the program
- Gathering all requirements
- Developing the product
- Sourcing for suppliers
- Managing ramp-up

A HOST OF CAPABILITIES

To accomplish this, mySAP PLM offers all the diverse capabilities needed to ensure new product success, including:

- Product portfolio and life-cycle analyses
- Concept management
- Project management
- Document and change management
- Product structure and recipe management
- External partner collaboration
- Integrated quality management
- Integration into mySAP SRM for strategic sourcing
- Integration into mySAP SCM for launch of production
- Integration into mySAP CRM for trade promotion and campaign management
- Integration into mySAP ERP for financials and human resources

THE COMPETITIVE ADVANTAGES OF NPDI

SAP's solution ensures a complete, efficient, and faster process for new product development and introduction, resulting in benefits like these:

- Reduced development time, which reduces time-to-market
- Reduced unproductive effort through more efficient collaboration processes, internally and externally
- Improved product quality, leading to customer satisfaction

- Fewer costly and time-consuming errors through improved information quality, which reduces data redundancy and inconsistency
- Increased profit through streamlined processes, lower costs, and most significantly, improved sales

The Steps of NPDI

The SAP closed-loop NPDI business process consists of distinct steps. The following sections briefly describe these steps.

PORTFOLIO MANAGEMENT AND IDEA ASSESSMENT

Using the SAP xApp™ Product Definition (SAP xPD) packaged composite application, users can analyze the existing product portfolio to identify products or product categories that need enhancement to increase market share or profitability. (SAP xApp composite applications compose existing functions into cross-functional business processes.) Product management teams use reports to plan and simulate the life cycle of existing or future products. Ideas submitted by sales, consumers, retailers, suppliers, and employees held in mySAP CRM are gathered, evaluated, and classified. Anyone can check the status of ideas in the idea submitter, and discussion threads facilitate team evaluations. Finally, promising ideas are consolidated.

PROJECT MANAGEMENT

With the Collaboration Projects (cProjects) project management application of mySAP PLM, users coordinate all development, marketing, sales, and production activities. Projects are derived from templates and adjusted to actual needs. The project scope is fixed, including milestones (quality gates), all project phases, all deliverables, and all project tasks and costs. The project team is defined via role-based qualification profiles and availability.

PRODUCT DEVELOPMENT

SAP solutions manage all product and project-related documents that stand behind any new product development and introduction – from specifications to CAD drawings, from meeting minutes to test protocols. Discrete manufacturers use life-cycle data management and product structure management capabilities,

whereas process and consumer product manufacturers use recipe management capabilities to develop the physical product. CAD integration, together with engineering change and configuration management, is key in the discrete industry; recipe management, which enables the creation of recipes, specifications, formulas, materials, and production operations, is the centerpiece in the process industry. Using the Collaboration Folders (cFolders) collaboration application, OEMs can collaborate effectively on part design with key suppliers.

STRATEGIC SOURCING

Tight integration into mySAP SRM allows engineers and procurement specialists to work with preferred suppliers and manage bid invitations with the mySAP PLM cFolders collaboration application. Bringing engineering and sourcing closely together integrates commercial and technical negotiation, which significantly shortens the strategic sourcing process.

PRODUCTION RAMP-UP

If market introduction of a product isn't synchronized with production ramp-up, companies can lose sales. Similarly, if advertising isn't synchronized with stocks in the supply chain, consumers can be disappointed. Integration with mySAP SCM aligns these processes.

MARKET LAUNCH

All the work comes to fruition when the market accepts a new product. The integration with mySAP CRM is used to plan and define marketing campaigns and trade promotions. The solution equips sales reps with collateral materials and sends pricing information to their mobile devices. An online and offline order management system provides fast, efficient order processing. Top-down and bottom-up demand planning deliver the best possible forecast.

POST ANALYSIS

With mySAP PLM, product-related costs and sales are fully monitored, often in real time, and compared with planned data. Is the product successful? Should the price change? Are promotions needed? Should production increase? Answers to these questions are enabled through the solution's integration with the data warehouse component of the SAP Business Intelligence solution.

THE KEY CAPABILITIES OF mySAP PLM

To enable this, mySAP Product Lifecycle Management is tightly integrated with other mySAP Business Suite solutions – mySAP ERP, mySAP SRM, mySAP SCM, and mySAP CRM – and complemented by packaged composite applications built on the Enterprise Services Architecture paradigm. The key capabilities of mySAP Product Lifecycle Management are described in the following sections.

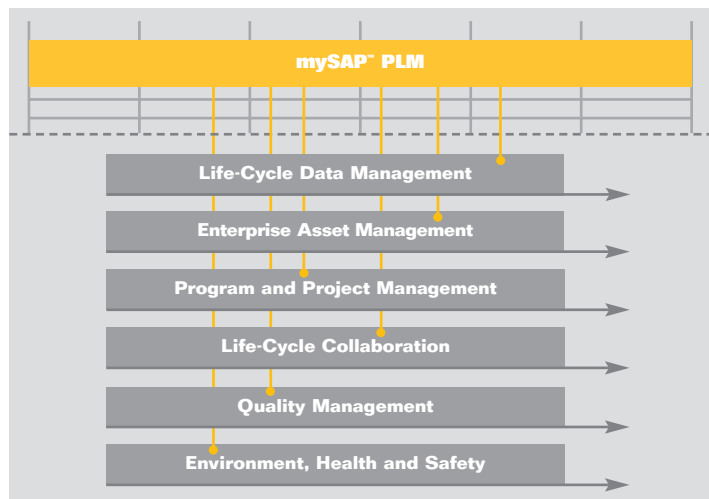


Figure 5: mySAP™ PLM Key Capabilities

Life-Cycle Data Management (LDM)

Information is the lifeblood of business and processes, including the NPDI process. Information is one thing; but how companies access and use that information is a major influence on market dominance and success.

The life-cycle data management (LDM) capabilities of mySAP PLM help companies manage all product-related data, including product structures, documents, recipes, and all related configuration and change management processes. More important, mySAP PLM makes life-cycle information seamlessly accessible and useful to users across the enterprise. It supports product development with complete integration of authoring tools, such as CAD, office applications, and analysis tools. It also seamlessly supplies accurate, up-to-date information on the actual product configuration – from the start to the finish of the product life cycle. (For details, see the section titled “Life-Cycle Data Management with mySAP PLM” on page 38.)

Program and Project Management (PPM)

mySAP PLM provides the advanced program and project management capabilities needed to plan, manage, and control programs and projects of all types, including the complete product development process. Its project management capabilities let users efficiently manage the project structures, dates, costs, and resources needed in any size project and in any industry. With its program management capabilities, users can see the big picture. A comprehensive overview helps manage a company's entire product portfolio. (For details, see the section titled “Program and Project Management with mySAP PLM” on page 52.)

Enterprise Asset Management (EAM)

For enterprises (especially those in asset-intensive industries like oil and gas and utilities) to meet today's demands for innovation, quality, timeliness, and value, they have to get the most out of their facilities and equipment. They have to manage these assets to assure peak performance, but at the same time keep capital and maintenance costs down. That is the purpose of the enterprise asset management (EAM) capabilities of mySAP PLM.

These capabilities manage the complete life cycle of any asset – including fleets, buildings, and equipment – from selecting it, to buying it, to installing and operating it, right up to its useful end. Integration with mySAP Supplier Relationship Management's e-procurement tools keeps companies on top of service parts,

buying supplies, and processing returns and repairs. Support for a wide range of mobile and wireless devices enables companies to stay in touch with field personnel and to manage maintenance operations anytime and anywhere. (For details, see the section titled “Enterprise Asset Management with mySAP PLM” on page 62.)

Life-Cycle Collaboration (LCC)

With the growth in partnering and outsourcing, the ability to support effective collaboration has emerged as a definitive competitive advantage. Superior collaborative capabilities enable companies to gather, organize, and use product and project information in a logical manner and to share and exchange this information with powerful collaboration tools.

mySAP PLM offers complete, integrated collaborative solutions to companies that deal with and must collaborate on complex products, assets, and project information. For example, it provides cProject Suite – which is composed of cFolders and cProjects – a complete platform for design and project management collaboration. Its collaborative capabilities enable the high level of logical, but rapid decision making found in the intensive NPDI process. (For details, see the section titled “Life-Cycle Collaboration with mySAP PLM” on page 74.)

Quality Management

Customers are demanding more and better products. To them, quality goes without saying. But it’s not easy to continuously improve quality. It takes a comprehensive approach – one that integrates processes and information across the enterprise and the extended supply chain, ensuring product quality by seamlessly executing all product-related processes.

mySAP PLM is an integrated, powerful quality management solution that applies a comprehensive and broad-based approach to total quality management. A full menu of quality management features and functions enables companies to control and maintain product and asset quality throughout the entire product life cycle. Quality management capabilities also equip companies to react quickly when unexpected events or problems occur. From

product development, ramp-up to production, change management to service and maintenance processing, the quality management capabilities of mySAP PLM result in higher-quality products faster and at the lowest attainable cost. (For details, see the section titled “Quality Management with mySAP PLM” on page 90.)

Environment, Health, and Safety (EH&S)

For many companies – and even whole industries – the ability to effectively plan and manage operational processes that involve hazardous raw materials and by-products of finished goods is essential. Local, national, and international laws and regulations stringently control such potential dangers as hazardous substances and dangerous goods. Many of these laws and regulations are constantly changing. The need to provide the proper safeguards to protect customers, communities, and environments is absolute.

The environment, health, and safety (EH&S) capabilities of mySAP PLM are designed for EH&S professionals and support them in the many tasks they must carry out in the following areas: industrial hygiene and safety, occupational health, product safety and hazardous substance management, dangerous goods management, and waste management. (For details, see the section titled “Environment, Health, and Safety with mySAP PLM” on page 100.)

mySAP PLM is complemented by a new breed of packaged composite applications that can span different enterprise solutions and provide a high degree of business process integration – the SAP xApps™ portfolio. SAP xApps composite applications include solutions for project portfolio and resource management, product definition via concept management, and emissions management.

THE UNIQUE ADVANTAGES OF mySAP PLM

In addition to providing companies with the key capabilities that enable an end-to-end PLM solution, mySAP PLM is unique in the following ways:

- mySAP PLM allows companies to turn sequential processes into parallel processes so they can drive new products to market faster, better, and at lower cost. It enables companies to integrate the processes of vendors, partners, and customers into their own processes. Because the same information is available to everybody, processes are accelerated and run in parallel, speeding the production cycle and rushing time-to-market.
- mySAP PLM is the only solution that manages the complete life cycle of technical assets, from first planning to disposal. Better life-cycle management leads to higher system availability and lower maintenance costs.
- mySAP PLM integrates all participants into the development process. It provides a collaborative environment for managing, tracking, and controlling all product and project information throughout the entire product life cycle.
- mySAP PLM improves the communication of product, quality, and asset information by creating close links between product development, production, quality, and maintenance operations.
- With mySAP PLM, companies can foster collaboration within the enterprise and beyond in product innovation, design and engineering, total quality management, asset management, and services across the extended supply chain.
- And companies can easily improve their decision-making process with mySAP PLM by using enterprise portals, PLM analytics, and data warehousing.

THE KEY BENEFITS OF mySAP PLM

With mySAP PLM, a company can achieve the objective of successfully getting quality products to market quickly at low cost and reap the following benefits along the way:

- **Better business decisions**
mySAP PLM improves decision making at all levels of an organization by providing powerful and flexible analytics for such areas as portfolio management, occupational health, product safety, product quality, and enterprise asset management. With universal access to relevant, up-to-date information about anything they need to know – at any point in the product life cycle – users can make quicker, fact-based decisions and intervene wherever and whenever necessary.
- **Better business results**
mySAP PLM helps companies increase their revenues by enabling them to develop innovative products and explore new market opportunities. The results are higher market share and increased customer satisfaction.
- **Reduced risk**
mySAP PLM integrates external stakeholders, such as engineering partners and suppliers, into a virtual network and allows a company to focus on its core competencies. Tasks can be outsourced, while keeping the cost of relationship management under control.
- **Higher product quality**
mySAP PLM manages product quality throughout every phase of the product life cycle. An integrated quality management solution covering product development, production, and maintenance processes results in higher-quality products, less waste, and higher customer retention.
- **Lower costs**
mySAP PLM reduces cost-to-market by managing the cost of changes and evaluating the progress of projects across product lines.
- **Faster development**
mySAP PLM reduces the time-to-market and time-to-volume by closely integrating supply chain management and procurement.

- **Improved manufacturing operations**

mySAP PLM reduces maintenance costs and increases equipment availability by providing the means to plan, measure, and track equipment planning, operation, safety, and maintenance.

- **Higher productivity**

mySAP PLM delivers all necessary content through an easy-to-use, role-based enterprise portal, increasing workers' satisfaction and productivity.

- **Lower cost of ownership**

The robust, open infrastructure of mySAP PLM allows easy integration of operational systems, such as CAD, ERP, CRM, SRM, and SCM solutions. The modular approach of mySAP PLM allows for gradual implementations that match companies' evolving needs.

Because of these benefits, mySAP PLM has become an integral solution contributing to the success of a string of companies around the world – more than 3,200 successful companies – and in diverse industries. Many of the world's largest construction, service, and R&D projects and programs in current operation use mySAP PLM to achieve their goals – evidence enough that companies can enjoy all the benefits of an integrated, end-to-end PLM solution today.

NEW AND ENHANCED FEATURES FOR mySAP PLM IN 2004

SAP is committed to expanding its vision of a PLM solution that integrates the entire enterprise. In response to the market trends that help to guide this vision, SAP has incorporated many new and enhanced features into mySAP PLM for 2004. Included among these are expanded PLM collaboration capabilities, increased ease of use, and other usability and functional enhancements, tighter integration of core business processes, an increased focus on the NPDI process, the inclusion of a complete recipe management capability, and additional use of the SAP NetWeaver technology foundation.

For more detail about the following new and enhanced features, see the section on new and enhanced features for each specific key capability of mySAP PLM.

New and Enhanced Life-Cycle Data Management Features

RECIPE MANAGEMENT

mySAP PLM now includes a comprehensive and versatile development environment for recipes and formulation to optimize development time in the process industry. The enhancement integrates all recipe and formula development data, together with research and development, production, and EH&S data, into a single environment to provide enhanced development capabilities, including faster product development and improved product consistency.

The new recipe management capabilities support the NPDI process with recipes, specifications, formulas, materials, and production operations generated by the recipe development environment.

Recipe management features give users faster access to objects and functions, faster recipe creation, faster calculations, better control through integration with project management and workflow, faster time-to-volume, and better reuse of experiences gained with existing recipes and formulations.

In addition to the new single-environment recipe management workbench, other new and enhanced recipe management features include the following:

- **Enhanced recipe management functions**, such as managing formulas, process descriptions, in process-control (IPC), and equipment requirements
- **Recipe building blocks** to predefine parts of process description steps to help staff quickly build recipes
- **New calculation functions** to calculate nutritional composition, contribution, product costs, product losses and gains during processing, and formula comparisons

- **Integration with PPM features** for a comprehensive set of project management capabilities
- **Master recipe functions** to enable user-guided conversion of recipes from the company or area level to the plant level
- **New search abilities** to help users locate specifications, recipes, and formulas
- **Integration of recipe management in SAP R/3® Enterprise**, making it easy to install and upgrade recipe management. Recipe management is no longer shipped as an add-on.

Future enhancements to recipe management features will include trial management functions.

THE NEW SAP EASY DOCUMENT MANAGEMENT INTERFACE

A new interface to SAP Document Management System (SAP DMS), called SAP Easy Document Management, makes it easy to use the solution's powerful DMS capabilities with features like drag and drop. Users can manage documents with ease and flexibility. The solution displays folders hierarchically for easy handling. Users enjoy tight integration with the Microsoft Windows environment, combined with the advantages of SAP document management.

New and Enhanced Life-Cycle Collaboration Features

cProject Suite provides the collaborative design and project management capabilities of mySAP PLM. It offers the cFolders application for design collaboration and the cProjects application for project management. SAP has enhanced both cFolders and cProjects features and functions for 2004.

cFOLDERS

Enhanced authorizations include local object authorizations that now override inherited authorizations only of the same type (user, user group, or role); an authorizations tab in the user settings; guest authorizations in the competitive scenario; authorizations for fields in generic objects and document categories; and object authorizations per status.

New and enhanced general functions include **new history and overview functions** for improved transparency in collaboration; **new archiving support** for previous collaborations; and **enhanced document handling** for multiple file transfers.

New **generic object types** and **document categories** help users reflect business needs by creating company- or industry-specific object types, such as a document structure or technical asset. Users can also create their own document categories that contain additional data, such as certificates.

Enhanced status management features include **restricted status profiles** that protect specified fields, **object authorizations per status** that determine who is authorized on an object, **protected versions** to protect object versions, and a **status log** that ensures transparency in status management.

Future plans for cFolders include integration with mySAP SRM, with mySAP CRM service order functions, SAP Enterprise Portal knowledge management capabilities, user management functions, and Microsoft Windows Explorer.

cPROJECTS

Subtasks enable more flexibility in task management. Users can now share information with partners through Microsoft Project integration and uploads. **User status** lets users model company processes. **Usability enhancements** afford direct navigation to documents, collaborations, and object links, and users can now change the user interface. **Accounting integration** enables numerous enhanced project controlling functions. The **control plan** feature lets users create plans in an SAP R/3 back-end system. A **resource search** capability lets users determine the best resource for each role and task. The staffing feature lets users advertise in the enterprise portal, and managers can stay on top of **staffing** and resources with **workforce analytics**. **Appointment planning** features allow for complete appointment management, and **resource assignments** let users flexibly manage assignments.

Future plans for cProjects include enhanced DMS integration, exception-based workflow, new project status reporting capabilities, multiproject management capabilities, and project versions.

New and Enhanced Program and Project Management Features

RESOURCE-RELATED DOWN PAYMENTS AND BILLING DOCUMENTS

Resource-related billing down-payment requests can now be based on the service rendered and the billing rules. The solution includes down payments when invoices are created. In addition, services between company codes can now be billed within one controlling area. Using a resource-related billing document instead of a ledger avoids delays.

New and Enhanced Enterprise Asset Management Features

COMBINED REPORTING USING ORDERS AND OPERATIONS

This new development allows the combined and concise mapping of order header information with data from the corresponding operations. The improved selection options let users select orders using operation data or use objects from the object list to search for orders and their corresponding operations. Users simply select and evaluate all the operations and orders for a particular object in the object list.

MAINTENANCE, REPAIR, AND OVERHAUL (MRO) COMPONENT

Using the new **maintenance event builder**, planners can view outstanding maintenance requirements, due dates for work, work orders, and other information relevant to the maintenance of complex technical objects.

MULTIPLE COUNTER PLAN

The enhancement of the **multiple counter plan** allows users to define and schedule different cycle sets and different items in a sequence with one plan.

CONFIGURATION CONTROL

Configuration control is the new main transaction for engineers, inspectors, or other people who participate in the configuration control process. Using this transaction users can display both current and allowed configurations.

REFERENCE MEASURING POINTS

Users can create functional locations on the basis of reference functional locations.

New and Enhanced EH&S Features

ENHANCEMENTS IN INDUSTRIAL HYGIENE AND SAFETY (IH&S)

IH&S enhancements include **integration of exposure profile ratings** with transfer exposure ratings for the work area to respective infotype 1403 organizational management objects.

Risk assessment features now also assess tasks. A new **measurement management** component allows users to monitor different agent types to help minimize risks and comply with regulations. The **assigned costs** feature allocates costs to IH&S measures. Users can determine an overall exposure rating, as well as a received rating using **work area management** features, and they can now create period-based accident reports, such as Occupational Safety and Health Administration (OSHA) 300 and 300A reports, with **incident and accident management** features. In the **personal exposure log**, users can now specify the person to whom a measurement applies.

ENHANCEMENTS IN OCCUPATIONAL HEALTH

Occupational health enhancements include a **vaccination service** to quickly enter medical information and a **brief consultation service** that allows users to document and evaluate routine, daily medical tasks.

European Union Directive 95/46/EC

mySAP PLM fully supports this directive, which requires access to sensitive personal data to be monitored and documented.

Enhancements in Basic and Logistics Functions

SAP has made enhancements to basic data and tools, product safety, dangerous goods management, and waste management features. They include a **new specification workbench** to edit specification data from a single interface; new **global label management functions** to simplify and automate labels; **integration with Microsoft Word**; easier **legal reporting** for EH&S reports, including Superfund Amendments and Reauthorization Act (SARA); support for **dangerous goods classes 1 and 7** – explosive and radioactive dangerous goods. An **explanation component** in the EH&S Expert system help users understand rules, and **integration with SAP for Utilities** allows that solution to use the waste management functions in mySAP PLM.

mySAP PLM IN 2004: CONCLUSION

SAP's commitment to enhancing mySAP PLM's capabilities is helping companies in all industries fuel growth and increase their competitive edge in the marketplace.

To learn more about how mySAP PLM can be adapted to specific PLM requirements, go to the SAP Web site at www.sap.com/plm. The site contains in-depth discussions of the solution, including solution briefs, solutions in detail, white papers, brochures, and other collateral. For detailed discussions on the key capabilities of mySAP PLM, be sure to refer to the solution maps and collaborative business maps on the site.

mySAP PRODUCT LIFECYCLE MANAGEMENT: FEATURE OVERVIEW

Feature	Benefit
Collaborative engineering and design	■ Integrate all participants into the development process ■ Take advantage of industry standard protocols, such as HTML, XML, and WAP, to communicate information within a virtual development team ■ Make the transition from a linear value chain to a three-dimensional, collaborative community focused on a common goal
Custom product development	■ Plan, manage, and control the entire product development process and allow for concurrent production process design ■ Automatically transfer product changes and customer order changes to manufacturing – even for production orders that are already running ■ Reference previous projects and predefined templates to identify the optimal mix of human and material resources
Collaborative enterprise asset management	■ Control costs through reduced maintenance and procurement expenses and closer tracking of operations, safety, and maintenance ■ Improve plant and equipment performance with preventive maintenance and fast repair capabilities that help reduce downtime
Project management	■ Increase revenues by efficiently managing projects and closely integrating supply chain management activities to shorten time-to-market ■ Reduce costs related to redundancy and errors with the seamless sharing of consistent and accurate information ■ Adapt projects to changing markets and customer needs
Quality management	■ Stay competitive by delivering high-quality products and services ■ Reduce costs by automating processes, reducing errors, and increasing the accuracy of quality performance data ■ Continuously improve quality by satisfying legal requirements and industry standards, such as ISO 9000:2000 and good manufacturing practices (GMP)
Environment, health, and safety management	■ Ensure regulatory compliance through improved processes, tracking, and documentation ■ Control costs and save time by automating EH&S processes and operating with consistent, accurate information ■ Keep up with changing regulations on a global basis with a flexible solution and open architecture
Mobile business	■ Enhance customer satisfaction by improving the responsiveness of the field staff ■ Provide accurate, up-to-the-second information on customers, equipment, and assignments ■ Automate all relevant processes so there are fewer mistakes, fewer lost opportunities, and less time spent on routine tasks
Knowledge management	■ Increase effectiveness through fast access to consistent, up-to-the-minute product and project information ■ Streamline processes that automate information exchange with partners and suppliers, as well as within the company ■ Integrate flexible global processes that run smoothly and efficiently to meet changing conditions and requirements
Analytics	■ Improve decision making through analytical capabilities that cover such areas as portfolio management, occupational health, product safety, product quality, and enterprise asset management ■ Conduct complete product-value analyses, cost-revenue analyses, and cost-of-sales projections ■ Merge structured and unstructured information in a role-based, context-driven fashion – anytime, anywhere
Open technology framework	■ Accelerate product development by integrating engineering and development with sales, marketing, manufacturing, service, and maintenance ■ Work with a comprehensive, accurate view of assets and activities across the enterprise by linking all systems ■ Leverage investment with support for open Web standards for communication, document processing, and security

LIFE-CYCLE DATA MANAGEMENT WITH mySAP PLM

LIFE-CYCLE DATA MANAGEMENT WITH mySAP PLM TODAY

Most companies understand how today's economy works. It's all about collaborating, creating value, building solid customer and partner relationships, and using information to increase productivity and profitability. But if a company can't consistently get the right data to the right place at the right time – throughout the company and outside its walls – the benefits of collaboration, value creation, strong relationships, and increasing profits can disappear in a hurry.

Today, the control and dissemination of structured information is paramount to continuous business success. And in large and small companies around the world, a vast array of software programs for managing product design and production is gaining ground.

But do these programs integrate the traditional data stovepipes that hinder efficient product development processes, such as inconsistent product data stored in different systems, the sequential flow of data, or lack of internal and external collaboration? If they do, why then are companies unsatisfied with their success rate? Why are they still on the lookout for new ways and means to speed up innovation cycles?

To solve the challenges they face – the constant time pressure compounded by the unrelenting demand for both better product choice and better quality – companies are on a constant quest for improvement. But are they looking in all the right places?

For example, a surprisingly significant portion of the cost of a new product is determined in the initial design phase. Flexible, innovative processes in the design phase increase the value added across the entire life cycle of a product.

The truth is, you would be amazed at the enormous rationalization potential a full business analysis can find for improvement. Every company has untapped potential for streamlining and optimizing development, production, and maintenance.

All companies want to be market or market-segment leaders. However, to achieve this goal, they need to find not just ways, but the best ways to innovate, engineer, and design products. They need to collaborate efficiently and meaningfully, both internally and with the extended business community. And they need to deliver unsurpassed support and maintenance services. By doing these things, they can bring the right product to market at the right time and at the right cost.

The key to achieving market dominance is information and how it is used. Managing information to make it useful is the purpose of the life-cycle data management (LDM) capabilities of mySAP PLM:

- LDM features manage all product-related data – start to finish. They supply up-to-date information throughout the complete product and asset life cycle straight through the extended supply chain, which ensures legal compliance.
- LDM capabilities fully support the development process, including complete integration with direct access to the leading computer-aided design (CAD) software packages.
- To anyone who needs it, any place and any time, LDM capabilities provide fast and reliable access to product-related data. They form the foundation that supports all reporting and decision-making processes.

Mastering the product life cycle means mastering product information, and companies can't do that without a solution that connects the whole picture, not just a part of it. That's why the tight integration of the information processing and delivery capabilities of LDM with other mySAP PLM capabilities provides the best global solution for reducing time-to-market, cost-to-market, and cost-of-changes today.

OVERVIEW OF LIFE-CYCLE DATA MANAGEMENT WITH mySAP PLM

Within the closed-loop mySAP PLM solution, which supports all aspects of the product life cycle, is a comprehensive set of versatile LDM tools for evaluating the entire spectrum of product and project data and distributing it wherever it is needed – inside and outside of any company. LDM capabilities enable transparent data storage and provide a dynamic, consistent global view of a product throughout its life cycle.

Sound Decision Making

The changing and managing of product and project data dictates the most important day-to-day tasks. The reality is simple: Every day, designers, work schedulers, and project managers need a sound basis for making not just some decisions, but all the right ones. Immediate, reliable access to the up-to-date, relevant information they need is critical to the success of products and to the continuing success of a company.

Access Throughout the Life Cycle

Providing this level of access isn't easy – although the results of LDM capabilities make it look that way – because the volume, complexity, and range of information about a product is dynamic, varying from phase to phase as it travels through its life cycle:

- Documents and specifications in the early development phase
- Engineering documents, materials, bills of material (BOMs), recipes and routings in the design, planning, and production phases
- Installed base information and asset and equipment data for organizing work in the service and maintenance phases

Support for NPDI

mySAP PLM supports this access in the SAP new product development and introduction (NPDI) process. The NPDI process brings new products to market efficiently and successfully by orchestrating people and information into one streamlined process.

Consistent Data for Everyone

Life-cycle data management not only cuts administrative costs and helps to eliminate errors, it also systematically makes product and process knowledge available in all systems at all times and to everyone who needs it. The result is an environment of highly visible, accessible, transparent data storage for improved decision-making abilities.

Enabling a Global View

These capabilities also provide a consistent, global view of a product, which is key to ensuring quality in all phases of that product's life cycle. By combining the capacity of a global product view with the collaborative working environment that life-cycle data management capabilities make possible, mySAP PLM enables all the key functional elements needed to succeed throughout the product life cycle. The following sections describe those key elements.

KEY ELEMENTS OF LIFE-CYCLE DATA MANAGEMENT WITH mySAP PLM

The key functional elements of life-cycle data management include the following:

- Document management
- Product structure management
- Recipe management
- Integration
- Change and configuration management

Document Management

The document management system (DMS) of mySAP PLM is applicable to all areas of product development and logistics. SAP delivers one of the most complete document management systems in the industry.

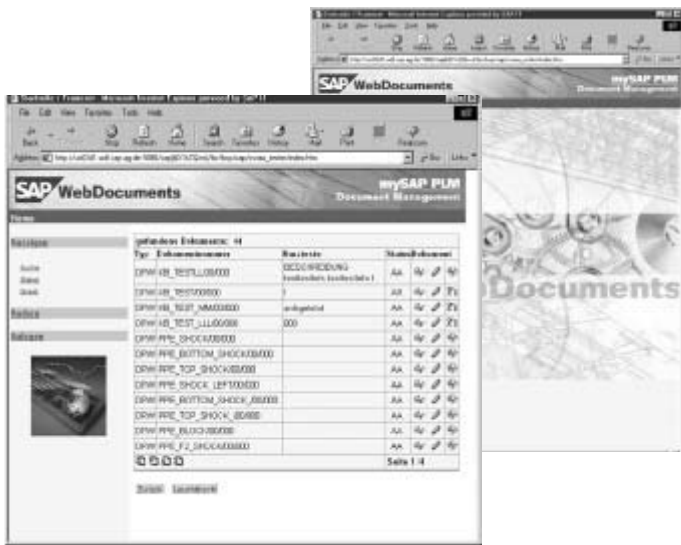


Figure 6: The mySAP™ PLM Document Management System

Document management functions can handle the enormous number and kind of documents that flow through a company each day – from meeting minutes to product descriptions, specifications, and CAD documents. The document management capabilities of mySAP PLM have the power to seamlessly integrate this massive influx of information into all product-related business work processes, allowing companies to cross boundaries without noticing them.

The document management capabilities of mySAP PLM include the following:

- **A status network for control**

The DMS of mySAP PLM employs a flexible, status-based control network to indicate the up-to-the-minute status of documents. Version control and engineering change management (ECM) capabilities support secure change processes under formal control. In addition, status-based workflows significantly speed up processes.

- **Document conversion**

Any kind of conversion – for example, a CAD drawing converted into a read-only 2D or 3D viewing format – can be triggered and controlled.

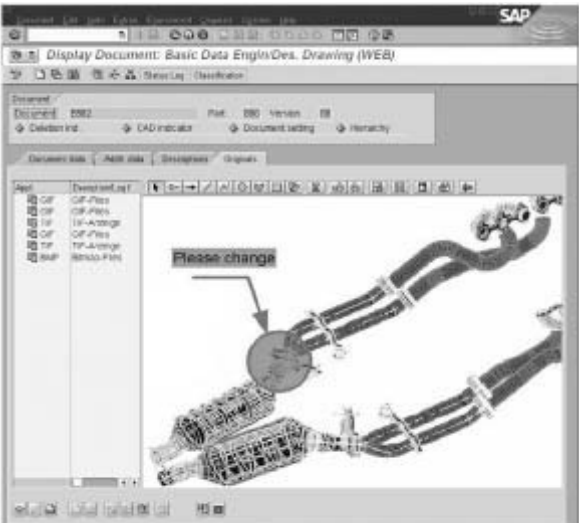


Figure 7: mySAP™ PLM Integrated Product Structure Viewer

- **Integrated viewing**

An integrated viewer can display most 2D pixel, vector, and text formats, as well as 3D model formats. Digital mock-up viewing enables anyone in the company to display a new or modified product in the development phase as a graphical 3D model. Redlining functions support digital processing and the change process for original files.

- **Multiple interface access**

Document management capabilities provide a broad range of access interfaces, including traditional GUI, Web, SAP Enterprise Portal (SAP EP), and the new SAP Easy Document Management interface (see the “New and Enhanced Life-Cycle Data Management Features” section on page 33), which allows users to work with the DMS in an easy-to-use, intuitive new way.

- **Openness**

The DMS of mySAP PLM easily integrates authoring environments, such as office or external mechanical or electrical CAD and computer-aided engineering (CAE) systems. Open interfaces are also provided for optical archives and document input and output systems.

- **Event-based workflows for paperless efficiency**

When changes to a document occur or a new version becomes available, people must know about it. By combining the DMS with SAP's comprehensive workflow capabilities, users can easily send electronic, event-triggered notifications or workflow items.

- **Comprehensive distribution of documents**

Through the use of automatic or manual routing and transmission, users can distribute original documents to customers and suppliers via mail or the Internet. The solution also provides complete auditing of the distribution information (who received what and when).

- **Electronic vaults for original documents**

For secure storage and retrieval of documents, original application files are managed by secure electronic vaults, such as content and cache servers, which can be easily administered using a standardized administration workbench.

- **Easy document search**

Document search capabilities include index-based search via metadata, classification data, object links, and document content (full text retrieval).

- **Tight integration with business objects**

The DMS can be used as a stand-alone application. However, it can also provide tight integration between business documents and business objects, such as materials, BOMs, equipment, projects, assets, customers, and production orders.

With these capabilities, the document management system delivers all of the following benefits:

- Reduced time-to-market by speeding up development processes
- Reduced costs across the board for archive administration, paper-based reproduction, document distribution, and external viewing tools
- Reduced time wasted for document search and access, as well as approval and change processes
- Eliminated data redundancies because it provides a single repository of up-to-date, relevant information
- Secure work content
- Increased work efficiency and reduced search and file reformatting time because the solution handles almost any type of file and saves documents directly from applications

Product Structure Management

A pivotal mySAP PLM capability for managing the product life cycle — one that directly affects its results — is product structure management. In essence, product structure management is the engine that transfers relevant product information into the sales, planning, manufacturing, and service processes.

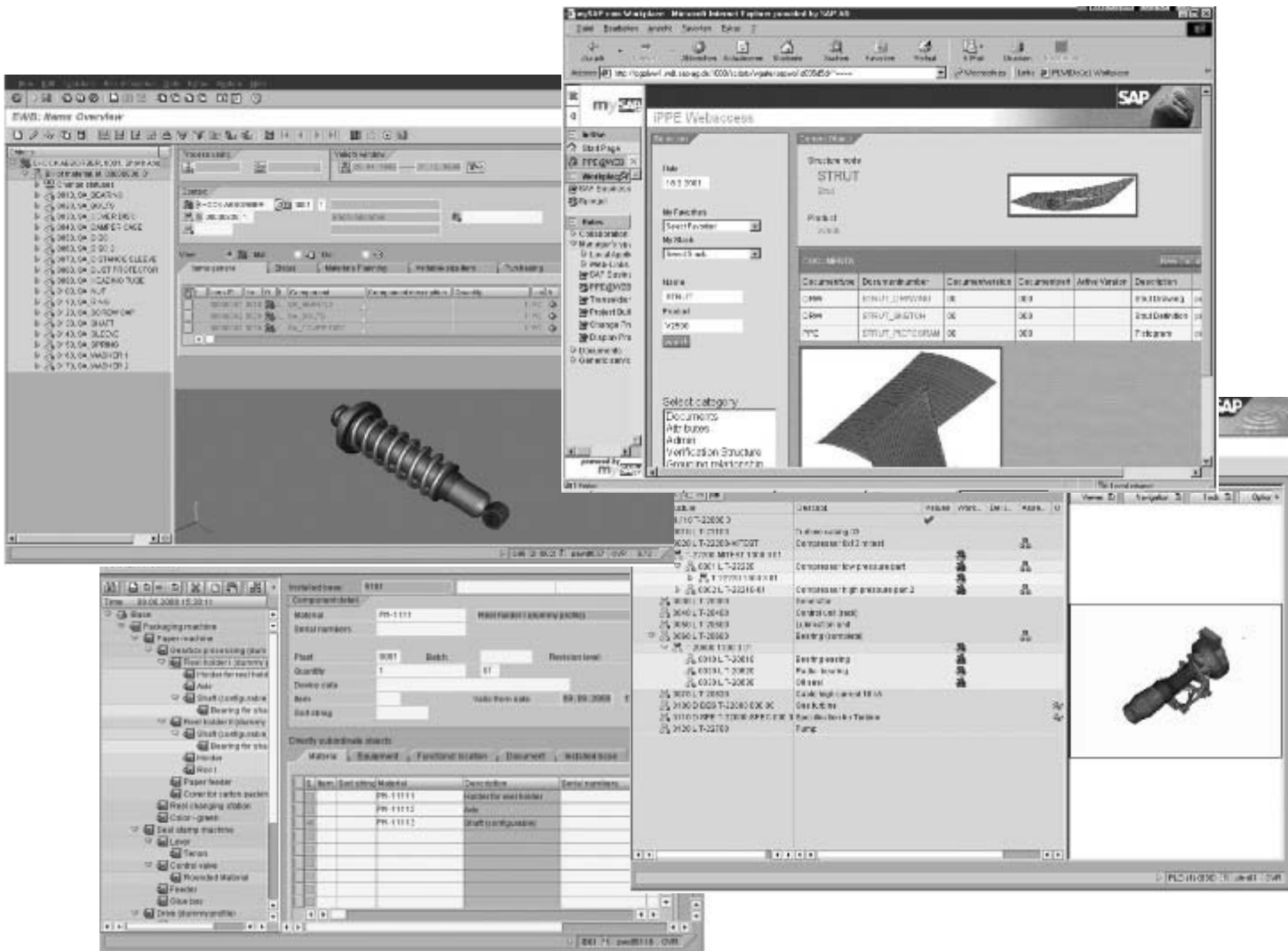


Figure 8: Product Structure Management with the Life-Cycle Data Management Capabilities of mySAP™ PLM

The product structure management capabilities supported by mySAP PLM are unique. They manage the product structure through the entire product life cycle. In keeping with the holistic approach of mySAP PLM, product structure management capabilities support product structures for all life-cycle phases: as-required, as-engineered, as-sold, as-planned, as-built, and as-maintained.

Also in keeping with this open view is mySAP PLM's support of product structures for both discrete industries, via BOMs, and process industries, via a powerful recipe management capability (see the section entitled "New and Enhanced for Life-Cycle Data Management Features" on page 48).

The following are the key product structure management capabilities of mySAP PLM:

- **Product designer**

Right from the start of product development, mySAP PLM offers a range of structures – such as feature and requirement, functional, and product structures – to optimize engineering processes.

- **Document structure management**

Complex sets of product documents can be maintained with or without engineering change control and revisioning.

- **BOM management**

With mySAP PLM, users can define and modify separate BOMs for different areas within a company, such as engineering or production, giving each area its own view of the product with the specific data it requires.

- **Engineering workbench**

A powerful workbench easily maintains product and process data. Users can even process BOMs and routings simultaneously, saving time and simplifying work scheduling. The engineering workbench also contains an integrated viewer, which supports digital mock-up (DMU) viewing of product structures.

- **Classification system**

To speed searches, the classification system allows users to work with characteristics like color or size to describe all types of objects – from materials to documents to equipment – and then to group all similar objects in classes so they can find them quickly.

- **Multiple-variant products**

Using the variant configurator, users can describe and easily manage all the possible variants of a product. mySAP PLM handles multiple-variant products with ease. In the ordering process, customers can configure products to exactly match their requirements.

Recipe Management

Recipe management defines both the product and the product formulation process. mySAP PLM manages the recipes for manufacturing a virtually limitless universe of product possibilities.

SAP enhanced the recipe management capabilities of mySAP PLM to offer a development environment for recipes in the process industry. Among the enhancements is a recipe management workbench that provides an environment for all recipe and formula development information. For complete information about the recipe management enhancements, see the section titled “New and Enhanced Life-Cycle Data Management Features” on page 33.

Recipe management is also a key element of the NPDI process. SAP supports the entire NPDI process – from concept to sell-through – with every capability companies need to assure new product success. Because recipe management defines the products and creates the recipes, it is a central capability of NPDI.

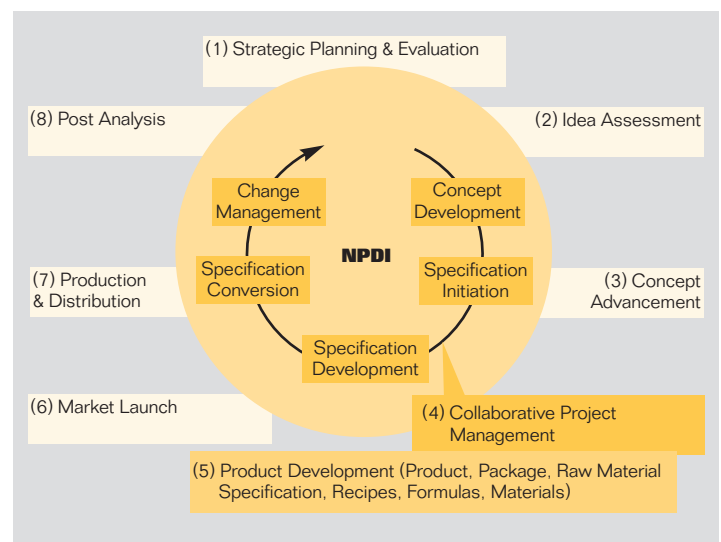


Figure 9: mySAP™ PLM Recipe Management Capabilities and the NPDI Process

In addition, the multilevel functions of recipe management support new product development on not just a local, but also an enterprise-wide basis. This enables companies to create enterprise-wide general recipes, area-specific site recipes, and machine-related master recipes.

mySAP PLM's recipe management functions are also tightly linked with the environment, health, and safety (EH&S) functions. This assures an up-to-date, consistent basis for making product safety and dangerous goods classifications. It also informs all responsible departments early on about product changes, enabling companies to get to market quicker and at lower cost.

Likewise, recipe management involves the manufacturing processes early in development. It then coordinates the manufacturing input, together with the knowledge from other departments, to precisely adapt the product to the production lines. Additionally, mySAP PLM can incorporate the knowledge used with previous products, which can shorten development time and significantly cut costs.

By improving collaboration and knowledge management and by facilitating the daily management of research and development departments, these recipe management capabilities provide more time for creative and experimental work — work that can lead to future products and product refinements.

With the recipe management capabilities of mySAP PLM, the following improvements occur:

- Better knowledge exchange and sharing inside a business and with suppliers and customers
- Better control over development time and costs through effective process and project management
- Better communication and information control to reduce time-to-market and time-to-volume

The key recipe management capabilities of mySAP PLM include the following (For enhanced recipe management features, see the section titled “New and Enhanced Life-cycle Data Management Features” on page 48):

■ **New product formulation and recipe development**

mySAP PLM provides a single recipe development environment from which users can produce product definitions, invoke the product formulation process, and create recipes for manufacturing products. Using structural product and raw material specifications, developers can drive product development forward, independent of existing material masters. Users can predefine parts of processes and quickly build new recipes by copying predefined building blocks.

■ **Multilevel recipe management and scale-up**

Recipe management makes available company-wide recipes, which can then be adjusted down to country- and line-specific recipes. The solution supports the transition processes, allowing production to be linked with laboratory standards early on. This speeds up the conversion from laboratory to production standards, which in turn speeds up time-to-market.

■ **Comprehensive recipe management**

Users can manage formulas, process planning, and equipment requirements, as well as inspections, which can be incorporated even during the development phase. Master recipes are easily converted from the company to the plant level.

■ **Complete calculation functions**

Users can make a variety of calculations, from determining a product's nutritional composition to calculating input and output materials and product costs. Calculation functions can be used as templates to create new, custom, company-specific functions.

■ **Flexible reporting capabilities**

mySAP PLM provides extensive and flexible reporting capabilities, and users can generate customer specifications in multiple languages.

■ **External collaboration**

Companies can collaborate with their vendors and suppliers to jointly define raw material and product properties.

■ **Change management**

mySAP PLM combines recipe management with change management and versioning capabilities so it can manage and document recipe development changes and approvals.

■ **Access control**

Detailed access control ensures that only authorized users can access sensitive data.

■ **Cost and time controls**

Project-based cost and time controls can be applied to the entire product life cycle.

Integration

In a complex, diverse environment, a company can't afford technical or organizational stovepipes. Information silos can cause delays, and a late release can cripple the potential commercial success of a new product.

That's why mySAP PLM provides an open and robust infrastructure that enables seamless integration between products, processes, and people. mySAP PLM integrates the entire product life-cycle management process – from product development to product sell-through – bringing all engineering and business processes into one open, flexible, powerful, fully integrated solution. The result is efficient management of product design documents and structures for the extended enterprise, without information islands.

Another result is that current and consistent product data can be accessed by everybody early in the process, enabling concurrent and collaborative engineering almost from the get-go.

The following sections describe the life-cycle data management integration capabilities in mySAP PLM.

DIRECT LINK TO LEADING CAD PACKAGES

By using direct (CAD) integration with mySAP PLM, the work output of a design department is integrated immediately into the complete business flow and is available to the extended business community. All CAD system results – such as documents, assembly structures, materials, BOMs, and engineering changes – can be stored directly in mySAP PLM, which ensures data consistency and direct access for everybody to the same CAD-related data.

Each of the following integrations is delivered as an SAP product – backed by SAP quality and full global support – which simplifies things by eliminating the need to deal with several system integrators. The ready-made SAP integrations are:

- mySAP PLM Integration for AutoCAD
- mySAP PLM Integration for Autodesk Inventor
- mySAP PLM Integration for Pro/ENGINEER
- mySAP PLM Integration for Unigraphics
- mySAP PLM Integration for CATIA V4 and V5
- mySAP PLM Integration for I-DEAS
- mySAP PLM Integration for SolidWorks
- mySAP PLM Integration for Solid Edge

BIDIRECTIONAL DATA CONSISTENCY

CAD integration enables designers to use production system data in their CAD applications, or they can create data in the production system directly from CAD data. Because data is transferred between a CAD system and mySAP PLM in both directions, data inconsistencies are avoided.

- Online connection between mySAP™ PLM and external applications with bidirectional data transfer
- Standard functionality for:
 - Material master
 - Bill of material (BOM)
 - Document
 - Engineering change management
 - Classification and finding objects
 - Mail and workflow
 - Plant maintenance
- Can be extended to suit individual requirements
- Independent from the specific CAD system

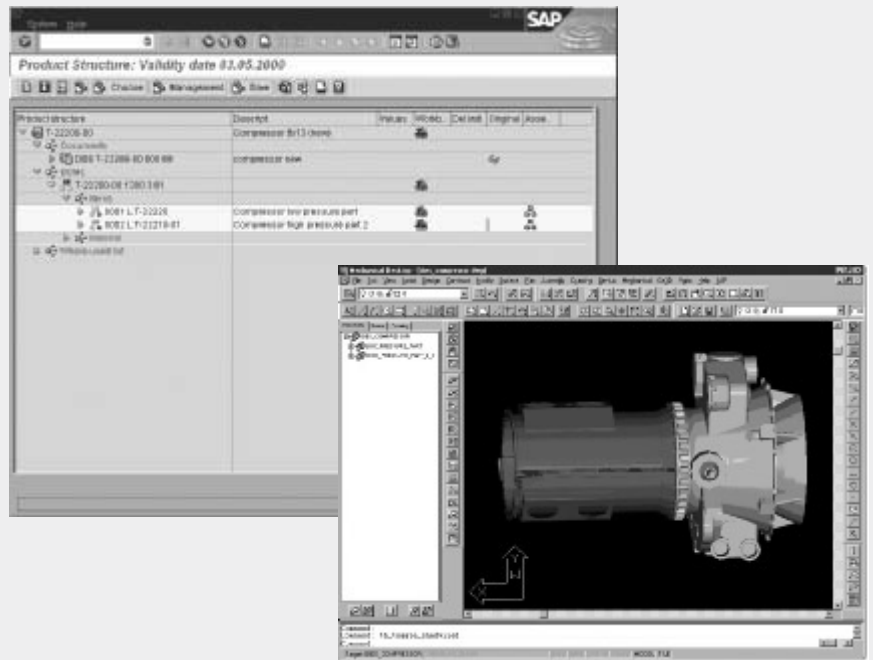


Figure 10: Features of the CAD Interface in mySAP™ PLM

INTEGRATION WITH MULTIPLE CAD SYSTEMS

SAP uses the same data model and philosophy for all of its CAD integrations. Therefore, companies can even integrate multiple CAD systems from multiple vendors with mySAP PLM, securing their existing investments. This gives companies the ultimate freedom of using the CAD system that best fits the design challenge at hand.

OPENNESS

To enhance engineers' performance and productivity, mySAP PLM also provides an open interface to link to other external mechanical or electrical CAD, CAE, and digital manufacturing packages or to link indirectly to external team data management (TDM) and product data management (PDM) systems. A variety of integrated third-party solutions are available through the SAP

Partner Program. More information can be found at

<http://www.sap.com/partners>

INTEGRATION WITH GIS AND SCADA SYSTEMS

mySAP PLM integrates with more than just CAD systems. To help users focus on important information and avoid information overload, mySAP PLM can be linked to geographical information systems (GIS) and supervisory control and data acquisition (SCADA) systems. The GIS interface allows a GIS and the document management system of mySAP PLM to be linked to an SAP R/3 object, such as a functional location object in maintenance management, to locate the asset. A SCADA system recognizes events that have been predefined using maintenance management capabilities. The SCADA interface passes filtered data to a document that is linked to an asset.

INTEGRATION FOR PROJECT-ORIENTED MAINTENANCE

Through its enterprise asset management capabilities, mySAP PLM offers an interface to Primavera P3 software, which supports customers using Primavera for their project-oriented maintenance procedures to plan production outage events. Project-oriented maintenance supports the key processes of work identification and selection; planning, budgeting optimization, and simulation; and execution and support for unplanned work.

Integration with mySAP Business Suite Solutions

mySAP PLM shares the engineers' product design and development information with other business functions. It creates an information exchange environment that transcends organizational boundaries. The solution integrates product innovation and development directly into the extended business processes driven by mySAP Business Suite.

mySAP PLM integration with the following applications and processes enables specific capabilities.

Integration with mySAP Supply Chain Management

(mySAP SCM) enables all routings, product structures, and recipe data to be released both to production and supply chain planning. To introduce and plan product and sales order changes in production orders, the solution's integrated engineering and order change management capabilities ensure a closed-loop process.

In engineered goods sourcing scenarios, **integration with mySAP Supplier Relationship Management** (mySAP SRM) enables engineers and purchasers to link a bid invitation using the Collaboration Folders (cFolders) application of mySAP PLM. By bringing engineering and sourcing closely together, integration with mySAP SRM helps companies to better optimize product costs at the early design stage. The integration also increases process efficiency and shortens time-to-market by enabling better communication between engineering and purchasing teams.

Integration with mySAP Customer Relationship

Management (mySAP CRM) allows complete knowledge bases designed with variant configuration to be exported to the sales configuration engine of mySAP CRM's mobile business application. This is important, because knowledge bases contain all the information about the classes, characteristics, characteristic values, and dependencies belonging to a configurable product.

CHANGE AND CONFIGURATION MANAGEMENT

Anyone familiar with new product development knows change is as constant – and sometimes as variable – as the wind. Requests for inconsistent customer requirements and changeovers to new designs or materials are commonplace. Nevertheless, if product change processes are inadequate or not in place, change orders can wreak havoc in an unprotected production environment.

To better manage the change process, mySAP PLM provides the flexibility to design company-specific product change processes that become an integral part of best practice operations. This results in a development and production environment that is conducive and adaptable to change – an environment that can act and react at any time with logical and methodical precision.

Integrated Change Management

Only mySAP PLM offers the integrated change management processes required to ensure quality and enable the efficient management of engineering change requests and orders – not just in the design phase, but in all phases of the product life cycle.

HOW CHANGE IS PROCESSED IN mySAP PLM

The key processes that support mySAP PLM change and configuration management capabilities are the following:

■ Change notification and claim management

Change notifications can trigger the change process via a simple problem description. Change notifications can be linked to engineering change requests (ECR) to process a change, if necessary. Claims are a type of notification used to trigger and control corrective actions. Besides cost calculation, a link to engineering change requests is supported.

■ Engineering change management processes

mySAP PLM uses ECM functions to change various aspects of production master data, such as BOMs, task lists, materials, and documents. Changes take effect only under precisely defined conditions. The changed object is saved twice: in its state before and after the change. An engineering change request or engineering change order (ECO) controls and documents the changes.

■ Order change management (OCM) processes

The order change management processes of mySAP PLM processes provide change management for production orders. mySAP PLM can simulate the effects of change and identify existing procurement elements, such as production orders, planned orders, or purchase orders, that would be affected by either an engineering change or a change to a sales order for a configurable product.

■ Configuration management

Objects that describe a product in a particular phase of the life cycle are collected in a configuration folder in mySAP PLM. The configuration folder enables companies to manage the configuration of products and projects across the life-cycle phases.

A WIDE PRODUCT VIEW

By enabling these processes, mySAP PLM supports product structure management and change and configuration management capabilities throughout the product life cycle. This allows companies to track the complete evolution of a product. Configuration folders and baselining capabilities are additional tools to ensure consistent configuration management throughout the entire history of a product.

COMPLETE CHANGE CAPABILITIES

Product changes can be minor or major. mySAP PLM change and configuration capabilities can handle any change, no matter how simple or complex. These changes can be released step-by-step for simulations, which helps individual areas of the company to determine the cost of the change before a product enters the modified production process.

THE STEPS OF CHANGE

Typically, the initiation of a change in mySAP PLM follows these steps:

- A change is requested.
People inside and outside of a company – external partners and internal colleagues – can each report an issue by easily creating a change notification on the Web.
- The change is considered.
mySAP PLM workflow immediately routes the change notification to the responsible agent in configuration management, who checks the feasibility of the change.
- The change is approved.
If the decision is made to make the change and modify the product, an ECR is created, and the affected objects are assigned.
- The change is made.
Using workflow, the internal change process is executed quickly, smoothly, and efficiently. And it is a flexible process – another hallmark of mySAP PLM. After the final release of the change, production orders can be quickly adapted to accommodate any last-minute changes.

NEW AND ENHANCED LIFE-CYCLE DATA MANAGEMENT FEATURES

SAP added many new LDM features in 2004. Some are discussed below. Each feature shares a common purpose: to continually improve mySAP PLM capabilities and to enable companies to launch new, high-quality products into the marketplace faster, more efficiently, and more profitably than they could a year ago. Individually and collectively, these features add to the completeness of mySAP PLM as the only true product life-cycle management solution.

Enhanced Recipe Management Features

To optimize development time and accelerate time-to-market, mySAP PLM now offers a comprehensive and versatile development environment for recipe management in the process industry.

This enhancement integrates all recipe and formula development information into a single, complete, function-rich development environment to enhance a company's development capabilities at every step – from the first idea to the final product. It integrates structured product and raw material specifications with research and development, production, and environmental product safety information – giving companies all the information they need to quickly, efficiently, and more accurately innovate and develop new recipes and formulas.

The new development environment results in faster product development, fewer errors and changes, and improved product quality and consistency. It also helps to ensure legal compliance in all applicable areas.

The recipe management workbench, together with the other new features, also enhances the closed-loop business solution that supports the entire NPDI process.

To support the consumer packaged goods industry, the SAP NPDI process accesses the recipes, specifications, formulas, materials, and production operations generated by the mySAP PLM recipe development environment, as well as all of its other management capabilities.

The new recipe management enhancements provide:

- **Faster access:** Users have quicker access to all relevant objects and functions, increasing the productivity of daily development work.
- **Faster recipes:** New recipes and processes can be set up quickly, reducing development time.
- **Faster calculations:** Users get out-of-the-box calculations, which reduce the time and cost of implementing products.
- **Better control:** The integration of recipe management with project management and workflow provides intrinsic control of all processes. The new development environment provides an accurate information base for decisions and better control of time, resources, and costs throughout development.

- **Faster volume:** Recipes can be easily converted from development into plant-specific recipes, reducing time, costs, and time-to-volume.
- **Better information:** Time and development costs are further reduced by the ability to use previously developed recipes, formulas, and specifications.

Among the specific new and enhanced features of recipe management are the following:

■ **Workbench**

The recipe management workbench integrates all relevant information for recipes and formulas, specifications, and materials into one environment, introducing new dimensions of flexibility and precision to the new product development process.

■ **Enhanced recipe management functions**

Recipes now include enhanced functions for managing formulas, process descriptions, in-process control (IPC) inspections, and equipment requirements – in both structured and unstructured formats. Inspection characteristics can be used in the process description to incorporate the IPC inspections as early as the development phase.

■ **Recipe building blocks**

Building blocks for process, stage, operation, and actions let users predefine parts of process description steps. Users can then easily build new recipes quickly by copying the building blocks to recipes using drag and drop.

■ **New calculation functions**

New functions for calculating nutritional composition, contribution, and product costs, as well as product losses and gains during processing, are now included. The comparison of formulas is also supported.

■ **Project System**

The integration of recipes into Project System – the project management component of mySAP PLM – has been enhanced to enable a comprehensive set of project management capabilities.

■ Master recipe functions

New master recipe generation functions enable user-guided conversion of recipes from the company or area level to the plant level.

■ New search abilities

Enhanced information system features enable new search capabilities to locate specifications, recipes, and formulas with combined search parameters and process information.

■ Integration of recipe management in SAP R/3 Enterprise

To make it easier to install and upgrade recipe management, it is now included in the new SAP R/3 Enterprise release with all of its new functions. It is no longer shipped as an add-on.

Finally, future mySAP PLM recipe management feature enhancements include trial management functions. Recipe management capabilities and the management of samples in the quality management capabilities of mySAP PLM will be integrated into the workbench to manage trials. Trial management will allow developers to combine management of their creative work and the definition, testing, and comparison of related samples in a single environment.

The New SAP Easy Document Management Interface

The document management system provides a new interface, called SAP Easy Document Management. It makes it easy for users to access and use the powerful SAP document management capabilities. The document management system becomes a virtual drive in file explorer, providing exactly the same features and handling people are used to in this environment, such as drag and drop.

Anybody who has to file and manage documents will appreciate the simplicity and speed of using SAP Easy Document Management. Just as its name implies, SAP Easy Document Management empowers users to handle all of their document needs with a new degree of ease and flexibility. For example, SAP Easy Document Management displays the hierarchical structure of the documents and folders in Windows Explorer so users can copy, move, rename, delete, and search for files and folders with speed

and ease. Users appreciate the tight integration of SAP Easy Document Management into the Microsoft Windows environment without sacrificing the advantages of SAP document management.

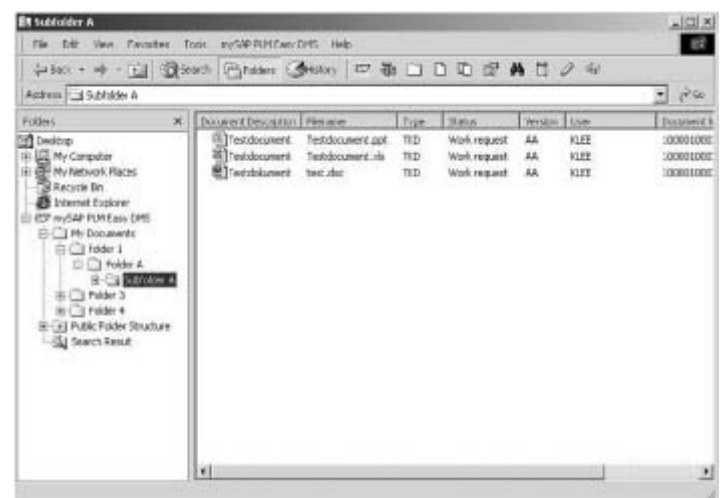


Figure 11: Hierarchical Structures with SAP Easy Document Management

With the capabilities provided by SAP Easy Document Management, both occasional and frequent users can do the following:

- Use drag and drop to check in documents from Windows Explorer
- Enjoy intuitive handling similar to Windows Explorer
- Using the save-as function, check documents in and out directly from Microsoft Office applications and send them to the SAP document management system
- Use document search tools to search for documents and add them to a work list
- Work easily with public folder structures
- Manage and maintain classification data
- Create links to other business objects
- Learn how to use the application in a short time

LIFE-CYCLE DATA MANAGEMENT WITH mySAP PLM:**FEATURE OVERVIEW**

Document management	■ Document vaulting ■ Document search ■ Integrated viewing and redlining for 2D and 3D viewing ■ Document distribution ■ Status management and status networks ■ Document conversion ■ Easy document management
Product structure management	■ Material management ■ BOM management ■ Product designer ■ Routing management ■ Classification system ■ Variant configuration ■ Document structure management
Recipe management	■ Product formulation and recipe development ■ Global multilevel recipe management ■ Trial management
Integration	■ CAD integration ■ CAx, PDM, GIS, and SCADA integration ■ Primavera integration ■ Data replication ■ Integration with mySAP CRM ■ Integration with mySAP SCM ■ Integration with mySAP SRM
Change and configuration management	■ Change notification and claim management ■ Engineering change management ■ Order change management ■ Configuration management: Identify the objects that describe a product in a particular life-cycle phase and collect them in a configuration folder.

PROGRAM AND PROJECT MANAGEMENT WITH mySAP PLM

PROGRAM AND PROJECT MANAGEMENT WITH mySAP PLM TODAY

A strategy for success is to be fast – and then get faster. That’s because change has never been as fast or more demanding than it is today. Keeping up with change is good, but not good enough. Staying ahead of change is the hallmark of successful companies.

Neither one is easy. Keeping pace with the almost daily, rapid-fire developments and technological advances, together with the increasingly sophisticated customers and competitors these developments are enabling, is an ongoing and major challenge for global businesses. But staying ahead of these changes requires something extra. Only companies dedicated to developing effective strategies to manage and respond to change faster than the competition will achieve sustainable success.

Helping companies to accomplish that strategy is the whole purpose of mySAP Product Lifecycle Management and its key capabilities. Among those key capabilities, program and project management (PPM) gives companies the advanced tools they need to see change so they can manage change instead of being managed by change.

In keeping with the holistic view of the product life cycle supplied by mySAP PLM, the solution’s PPM capabilities enable a holistic view of projects. mySAP PLM provides the advanced PPM functions needed to plan, manage, and control the complete product development process – from initial idea to creation and beyond.

These capabilities provide a view of the product development process from every angle. Using mySAP PLM’s project management capabilities, companies can efficiently and flexibly manage the project structures, dates, costs, and resources needed in any project of any size and in any industry. Simultaneously, using mySAP PLM’s program management capabilities, companies can get a handle on the bigger picture with a comprehensive overview of their entire product portfolio.

This view is enhanced with PPM controlling functions, which are integrated with SAP Business Intelligence and the wealth of business information contained in this data warehousing and analysis solution. These functions allow users to consolidate controlling data on an enterprise-wide basis throughout the complete life cycle of a project, supporting both progress and profitability analysis.

The view becomes broader: mySAP PLM’s program and project management capabilities are integrated with the SAP Strategic Enterprise Management (SAP SEM) component, which provides portfolio management solutions to support business planning processes.

In the worldwide arena of business competition, no successful company can afford not to have such visibility. For this reason, program and project management capabilities have become a crucial influence in the competitiveness between companies – especially companies whose success depends on project-oriented business processes. That’s why the comprehensive project and program management capabilities of mySAP PLM – capabilities that provide a consistent, integrated approach to carrying out projects and programs of all kinds – are in use around the world today, such as:

- With mySAP PLM, some of the world’s largest construction companies are building some of the world’s largest projects.
- With mySAP PLM, global service companies are managing multiple offerings and billing customers in diverse countries.
- With mySAP PLM, R&D departments in both process and discrete manufacturing are creating products that will shape our lives tomorrow.

OVERVIEW OF PROGRAM AND PROJECT MANAGEMENT WITH mySAP PLM

It can be stated simply enough: the daily challenge is to bring quality products to market quickly while keeping costs down. Here's what it takes to meet that challenge:

- To focus on the right areas, companies need comprehensive product portfolio information so they can analyze the effects of change early on in the development process.
- To deliver successful products on time and within budget, any number of complex programs and projects have to be managed, including product development and manufacturing projects, in addition to a diverse range of investment and maintenance projects and activities.
- To meet deadlines and schedules and to contain costs, the progress of all programs and projects must be monitored. At the same time, executives must keep an eye on the wider picture to make sure all programs and projects are working in sync.
- To communicate with supply chain networks, companies must cross corporate boundaries to collaborate with partners in planning and executing projects.

mySAP PLM helps companies manage projects successfully as they plan, execute, and control their goals with a targeted, cost-effective approach to achievement. With these capabilities, companies can:

- **Plan and manage programs and projects at a strategic level** using state-of-the-art analytical tools and business warehouse capabilities to generate complete, optimized product plans
- **Manage the product portfolio** and support business planning processes with SAP SEM integrated with mySAP PLM
- **Analyze profitability throughout the product life cycle**, allowing costs and revenues to be planned all along the way – from development through service and maintenance

- **Manage and control projects collaboratively and proactively** by monitoring cross-project information, maintaining time-tracking information with the Cross-Application Time Sheet (CATS) component, comparing planned-to-actual progress to stop problems fast and using Web-based collaboration to easily manage documents among partners
- **Monitor and manage costs** for one or multiple projects with capabilities that allow planning in various degrees of detail and tools for managing budgets, availability, cost forecasts, and actual costs and commitments
- **Manage capacity** early and react flexibly to changes, matching capacity and resources, including people, positions, and work centers, with project needs
- **Manage materials** with supply chain partners using integrated procurement processes to determine project materials; trigger procurement and internal production; track materials, costs, and stock levels; and group requirements from several projects
- **Enable a single point of access** for everybody, including customers and suppliers, using SAP Enterprise Portal's Web-based technology to open the door to the full range of applications, tools, and services people need to work and collaborate
- **Manage content** for occasional users by using portal technology and business packages for projects, which enables fast and easy access to personalized, project-related information, such as project structures, documents, and time or exception reports

New Levels of Effectiveness

With the underscoring challenge of making quality products quickly and at a lower cost, mySAP PLM capabilities work in symphony to increase efficiency and effectiveness wherever they extend, reducing costs and time-to-market.

mySAP PLM optimizes project execution by reducing routine activities and speeding up business processes. In addition, the multifaceted functionality of the solution's comprehensive project management capabilities enables new dimensions of effectiveness in progressive project management.

In the Final Analysis

What mySAP PLM and its program and projects management capabilities do is keep programs and projects moving forward, despite the challenges of change, to deliver new products on time and within budget. In doing this, they also deliver a wide band of business benefits:

- When projects are finished quickly and efficiently and integration with supply chain management has ensured distribution, time-to-market goes down and revenues go up.
- Sharing accurate, up-to-the-minute information among all project members allows managers to evaluate the progress of projects across product lines – and that can mean beating the competition to market.
- Redundancy and errors cost money, but seamlessly sharing consistent, accurate information that is stored in repositories of nonredundant data reduces both.
- The integration of project information with supply chain management and procurement systems minimizes material costs and maximizes purchasing efficiency.
- The constant, precise overview of costs – for even the most complex projects and across multiple projects – enables fast and accurate responses to unexpected expenses.
- Accurate profitability analysis helps companies allocate resources to the most strategically important initiatives.
- The competitive advantages of collaborative communications are at users' disposal. They can use them throughout the company and across the supply chain to adapt projects to changing markets and customer needs, drive product innovation, and explore new market opportunities.
- To make sure the benefits keep coming, templates and tools allow everyone to easily build on lessons learned as projects are executed, which helps companies cope with future change.

KEY ELEMENTS OF PROGRAM AND PROJECT MANAGEMENT WITH mySAP PLM

Program and project management capabilities provide the features and functions needed to plan, control, and execute projects of any number, kind, and size, including:

- Research and development projects
- Make-to-order production projects
- Engineer-to-order projects
- Investment projects
- Maintenance projects
- IT and organizational projects

The key functional elements that address the challenge of making high-quality products quickly and at lower cost also support and enable companies to do all of the following activities.

Plan and Manage Programs and Projects

With mySAP PLM, companies can plan and manage programs and projects at a strategic enterprise level using a full range of analytical tools enhanced with SAP Business Intelligence.

Whereas the solution's **project management** capabilities focus on doing things right, mySAP PLM's **program management** capabilities focus on the predevelopment phase and the right things to do. With program management capabilities, users can:

- **Analyze and evaluate an existing product portfolio**
Program management provides an overall view of all existing products, including overall product life-cycle performance individually and as a whole. This enables decisions to be made on a tactical level. To support decision making, mySAP PLM takes all available facts into account. Using sophisticated analysis methods, users can compare products.

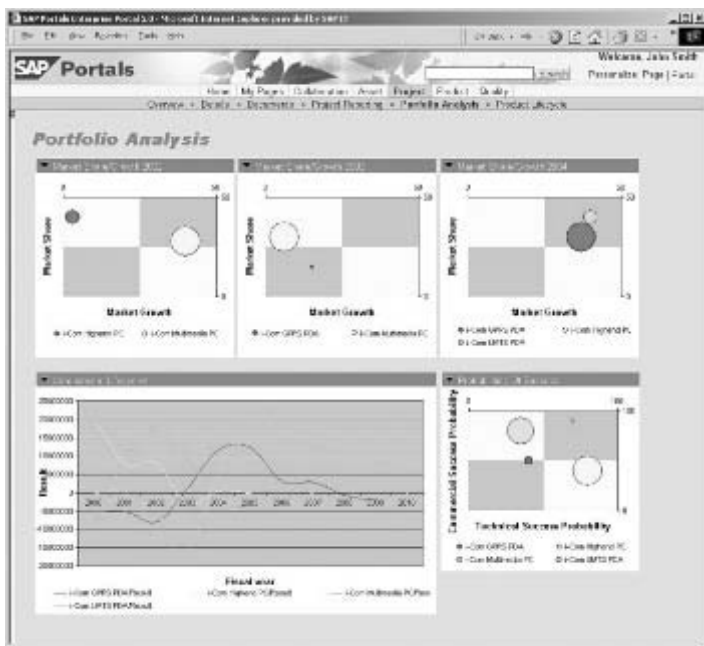


Figure 12: Portfolio Analysis with mySAP™ PLM

■ Enable comprehensive planning scenarios

With the analytical and business warehouse capabilities, users can develop comprehensive planning scenarios to evaluate an existing product portfolio and make crucial decisions concerning costs, resources, and scheduling.

■ Focus resources where needed

With mySAP PLM, answers to such questions as, “Do we discontinue this product?” or “Should we implement a new product?” or “Should we intensify production?” are easily within grasp.

■ Stay informed with portal-enabled portfolio management

With mySAP PLM, users can view all relevant business information through a Web-based portal – a single point of access for anyone. This access is particularly important to product managers, who must have information to monitor the product mix and to adapt it to external changes (like markets and prices) and internal changes (like costs and capacities). Using the portal, managers can access multidimensional portfolio analyses, such as market growth, market share, and sales

volume – plus many more dimensions. From the data warehouse – with its powerful Microsoft Excel and Web-reporting facilities and its ability to integrate SAP and non-SAP information – product managers have complete portfolio management information at their fingertips.

■ Plan and simulate current and future product life cycles

With mySAP PLM, users can create complete, optimal product plans. When plans are completed for all relevant products, they can evaluate the complete product program on differing aggregation levels. They can then use the resulting reports to plan and simulate the life cycles of both existing and future products in the company. During execution, they can follow progress using actual versus planned data.

■ Plan life-cycle costs and revenues

Life cycles can be planned in step series, which can be integrated into existing applications. Users can plan quantities, prices, and revenues and, from the revenue plan, they can generate cost targets. The cost targets are compared to the results of the life-cycle cost planning, and the costs and revenues can be interactively aligned. The life-cycle costs can then be planned, considering partial cost plans for research and development, manufacturing, administration, sales, and service and maintenance. The final result is the planned cash flow.

With project management users can use central structures and tools. The central structures of the project management capabilities in mySAP PLM are work breakdown structures (WBS) and networks with their attendant activities and milestones. These structures are used to model complex projects in the system for use with production and procurement bills of material (BOMs), sales orders for sales and distribution, and other needs.

The central tool used for setting up and monitoring projects in mySAP PLM is the project builder. Using a user-friendly working environment, the project builder integrates all required application-specific functions. The graphical project planning board tool simplifies project planning by combining tabular and graphical data to provide a global overview.

Analyze Profitability

With mySAP PLM, users can analyze profitability and plan costs and revenues throughout the life cycle of one or multiple products. Users can:

- **Collect actual costs over the complete life cycle**

Knowing the actual life-cycle costs reveals how a product is actually doing and whether or not it will achieve long-term profitability. To know, a profitability analysis produces point-of-sale data, which provides product-specific sales and costing information. This data is combined with product-specific investment and project development data to reveal product cash flow and all other necessary information across the life cycle.

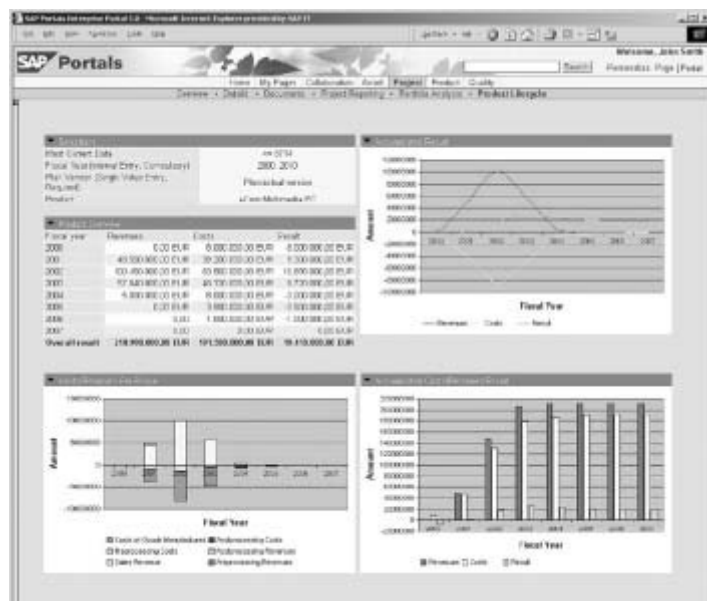


Figure 13: Analyzing Profitability with mySAP™ PLM

- **Compare plan data with actual data**

Users can compare plan data and actual data during every phase of the product life cycle. Planning can be verified by identifying the differences between the life-cycle plan and the actual situation at any point in the life cycle.

- **Compare costs and revenues**

Because revenue planning is a prerequisite for efficient revenue control, users can continually compare costs and revenues. This analysis provides for early assessment and, if necessary, corrective actions of the project results in terms of profit and loss. mySAP PLM can provide reliable information on the likely profit or loss of a project as early as the quotation phase and subsequently throughout a project.

- **Minimize costs and optimize profits**

The solution's profitability analysis tools and functions help users see where to minimize costs and thus optimize profits at any point in the product or project life cycle. Especially in the early stages of development projects, future costs can be influenced, optimizing the life-cycle contribution of a product to a company's profitability.

- **Determine product performance**

During production and sales, users can continuously monitor actual data to evaluate product performance so they can take any necessary corrective actions quickly.

Plan Projects

The range, intensity, and depth of detail required to fulfill certain project aims depends on the nature of the project. In one project, costs may take precedence; in another, the emphasis might be on planning and monitoring dates and resources.

To carry out a project, users require specific resources. In addition to materials and funds, the availability and optimal use of capacities, such as personnel, production, and assembly capacities, are important factors in determining the success of the project. Sufficient resources need to be available at the right time and place.

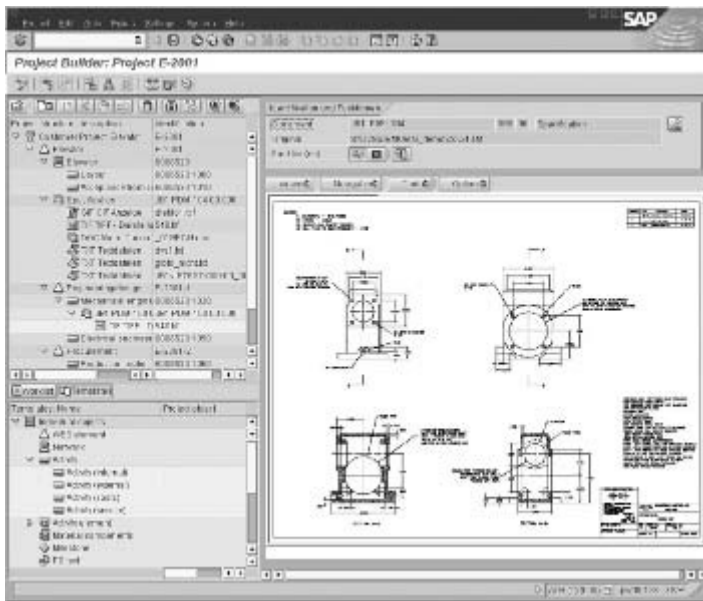


Figure 14: Set Up Project Plan Using the Project Builder

mySAP PLM enables users to draw accurate, up-to-date information into planning processes. From that point, a host of planning features and functions allows users to:

- **Match capacity with project needs**

A capacity planning function can recognize, pinpoint, and identify capacity requirements at any point in the life cycle, but particularly in the early stages of the project. This allows quick, flexible decisions to fast-changing demands for capacity. Users can analyze the capacity load, and capacity leveling functions optimally match the capacity load to existing resources.

- **Use the workforce planning tool to distribute work**

The work distribution functions of mySAP PLM are based on work centers that have predefined capacities in different categories, including the identification of which people work in which locations. The flexible workforce planning tool distributes people and resources into the appropriate positions.

- **Schedule the project**

The Gantt chart in the project planning board helps to flexibly schedule projects.

- **Track the project's progress**

Users can view costs, resources, and dates across an entire project so they can recognize variances from the project plan early on. mySAP PLM provides a complete range of functions for following project progress, including date planning, scheduling and network scheduling, confirmation, and progress analysis functions that include milestone trend analysis (MTA).

- **Interface to Microsoft Project and Palm Pilot**

The Open Project System (Open PS) interface links mySAP PLM and other project management systems and personal organizers. Users can transfer projects from Microsoft Project to mySAP PLM and vice versa, and projects can be synchronized between them. Open PS enables mobile project management. Users can take a project with them – on a plane, on-site, or at a customer's premises. They can download activity work details, such as planned, actual, and forecast work, onto a Palm Pilot or laptop, then confirm the activities on-site directly on the remote device. Back at the office, they can simply transfer the laptop or Palm project data into mySAP PLM and incorporate it into the organizational project structure.

Execute Projects

Successful project management means planning, executing, and monitoring important undertakings economically and efficiently – with company objectives always in mind.

Comprehensive functions and integrated project processing are the hallmarks of a high-performance project management system. These functions and processes are essential to smooth, efficient project execution and material management, and to work effectively with the supply chain.

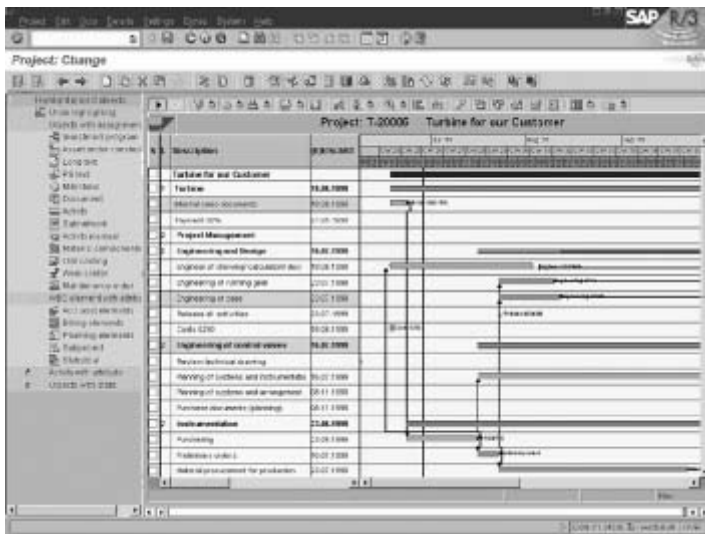


Figure 15: Manage Project Schedules Using the Gantt Chart

mySAP PLM procurement and production capabilities provide advantages that are decisive for successful project management. These advantages include the following:

- Project-oriented material management for all industries

Production facilities are not the only beneficiaries of mySAP PLM's integrated procurement processes. Both material-intensive and nonmaterial-intensive industries, such as engineering, R&D, and service industries, can take advantage of its powerful features.

- Flexible determination of all project materials and components

Users can plan and assign materials and components in the early stages of a project. Later, they can start procurement processes to ensure on-time delivery. Alternatively, users can plan internal production using material requirements planning (MRP), for example, so that materials are available when needed. mySAP PLM can trigger the procurement or production of individual parts that have long lead times.

- Tight integration with auditing

At any time, the demands of controllers and auditors can be met. Users can evaluate the planned versus the actual costs of components down to the line-item level.

- Trigger procurement or internal production

Once materials and components have been assigned, mySAP PLM integrated procurement processes can be triggered. Procurement can be internal using MRP requirements, or external, which generates purchase requisitions in MRP or directly in the project. Components can be procured in advance, or procurement can be started at any time during the project. Materials can be released when necessary, allowing greater flexibility to change component details.

- Calculate aggregate volumes

All materials and components, costs, and stock levels can be tracked. mySAP PLM gathers aggregate requirements from several projects so users can take advantage of the benefits of volume purchasing.

- Monitor material orders

Users can follow orders for materials throughout the procurement process using the monitoring dates function.

- Automate processes with workflow tools

A set of workflow tools helps to automate and accelerate many procurement processes. In addition, project-driven procurement and production management capabilities provide an overview of the materials and services documents required in a project, as well as stock levels, dates, and quantities.

- Manage project issues with claim processing

Claim notifications document variances in the project and trigger the appropriate corrective actions. Claim management functions also include the ability to generate general notifications to, for example, a subsupplier on a joint project.

■ **Control information with the project information system**

Projects generate huge amounts of data. The project information system within mySAP PLM is an efficient and flexible tool for evaluating and displaying that data. With it, users can control a project schedule and examine the business aspects of a project in detail. The system's information reports range from overviews of progress in multiple projects down to single line items within one activity. This advanced system, which is integrated with SAP R/3, also offers predefined views of schedules and milestones, costs, resources, and capacities, as well as an integrated structure view — a hierarchical representation of the entire project.

■ **Access to complete business information**

SAP Business Intelligence offers an application environment that draws on both SAP and external information sources, summarizes them, selects data, and presents the results in an easy-to-understand Microsoft Excel or Web report format. To create comprehensive reports, users can combine purchased data from analysis services with internal SAP data. SAP Business Intelligence offers predefined reports, available with a simple mouse click, and the reports are Internet capable. The integration of mySAP PLM with SAP Business Intelligence assures that the information needed for planning and decisions is available throughout the company in a form that is easy to use, quick to access, and user friendly.

■ **Easy document management**

The program and project management capabilities of mySAP PLM use the highly efficient document management system (DMS) to manage the extensive documentation and to provide the technical documents that projects require. The document management system enables access to documents enterprise-wide and links them with SAP objects in various parts of the enterprise, for example, program and project WBS elements, activities, and material components. The document management system assures project-related information flows to all departments either via the Internet or the company intranet.

Monitor and Manage Financials

A distinguishing mark of successful project management is successful financial management and control. Throughout a product's life cycle, mySAP PLM not only supplies all functions and tools needed to keep a sure, constant, and flexible eye on costs, it also provides the means to keep costs down. The functional cost management capabilities of mySAP PLM allow companies to:

■ **Track any cost**

Costs can be tracked for individual projects or across several projects simultaneously. Project expenditures can be restricted and controlled using project budget approval and releasing functions.

■ **Analyze costs daily**

Costs can be monitored continuously and analyzed at the touch of a key with user-defined reporting tools.

■ **Plan costs in various degrees of detail**

Powerful, precise cost-controlling features allow planning at the detail level required by the phase of the life cycle. For example, hierarchical planning allocates basic values and is used to make a first, rough, quick cost estimate. For more advanced cost management, the mySAP PLM easy cost planning capability provides simple, fast, quantity-based planning using templates specifically designed for individual planning purposes. Detailed planning tools, as well as network and general cost planning, enable a complete cost-planning environment.

■ **Efficiently manage budgets and forecasts**

mySAP PLM includes important tools for managing budgets. For example, the budget update function adapts an original budget to a current project. mySAP PLM provides both automatic and manual budget allocation capabilities, as well as both passive and active availability controls.

■ **Record actual costs, commitments, and cost forecast**

Variances between planned cost and actual cost are inevitable. mySAP PLM stores the information from actual data and commitments so it can be used for further planning. When data is entered into the Controlling or Financials components, mySAP PLM automatically records the information in the project. The cost forecast function adapts cost planning to the changing circumstances.

■ Efficiently manage project payments

Cash management functions help to maximize profits by optimizing payment flows on a project-related basis. Cash flow can be planned, then monitored and evaluated against actual flow.

■ Create resource-related billings

mySAP PLM integrates billing plans and resource-related billing, which allows resource-related billing requests to be generated. For more information about resource-related billing enhancements, see the section titled “New and Enhanced for Program and Project Management Features” on page 61.

A ONE-STOP PORTAL FOR OCCASIONAL USERS

For occasional users, including supply chain and other business partners, mySAP PLM provides a Web-based portal that serves as a single point of access to all the applications, information, and tools they need to do their jobs.

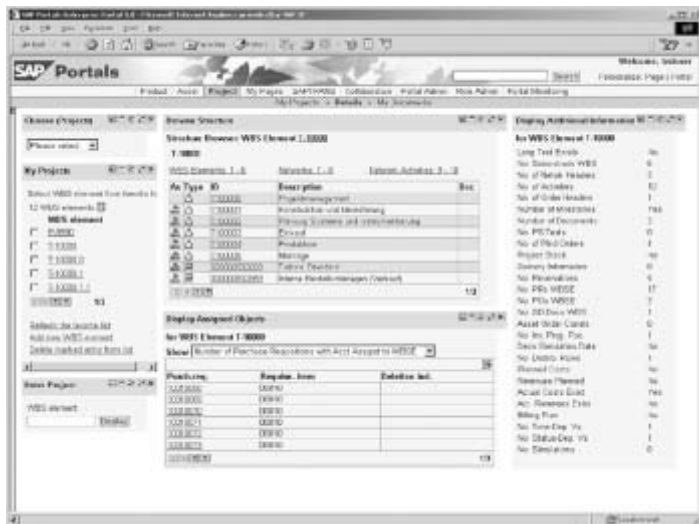


Figure 16: The Business Package for Projects

Role-Based Content

The highly functional, but easy-to-use and modifiable portal interface enables users to quickly reach all content and services. The interface is customized to the specific business role of users

and every other user, enabling seamless access to the role-based content and services they all need. The content is delivered through iViews, small applications that assemble information from various sources according to each specific role.

Content Tailored to Support Project Management Processes

SAP created specific portal interfaces and menus to fit the roles encountered in project management, including roles for project manager, controller, planner, accountant, subproject manager, and project worker. Project management professionals use the portal to access mySAP PLM's Web-based collaborative product and project management applications.

EXTENSIONS TO PROGRAM AND PROJECT MANAGEMENT CAPABILITIES

mySAP PLM handles many program and project management functions, but for certain aspects, the solution is enhanced with additional offerings.

SAP xRPM

The SAP xApp Resource and Program Management (SAP xRPM) packaged composite application is a strategic solution that offers project proposal and prioritization through strong, preconfigured portfolio management capabilities that are especially useful for R&D projects. SAP xRPM consolidates project information not only from SAP's project management applications, but also from third-party products, such as Microsoft Project. Besides project portfolio management, SAP xRPM also addresses project resource management.

Advanced Product Quality Planning

For the management of projects in research and development, an approach called advanced product quality planning (APQP) is becoming popular. APQP is an approach to new product development based on stage-gate process methodologies. This process is supported by the Collaboration Projects (cProjects) application of mySAP PLM.

cProjects and mySAP PLM

cProjects complements the program and project management capabilities of mySAP PLM to support all steps in project planning, execution, and the close of the project.

cProjects is used when companies require a software solution that manages stage-gate process methodologies. In addition, this completely Web-based application stresses collaboration. cProjects is the application of choice – especially for APQP-based processes.

The program and project management capabilities of mySAP PLM are used if a company requires deep logistic integration (such as billing, materials management, or sourcing) and tight financial integration. Large and complex projects – such as building power plants or aircraft carriers, engineering and construction projects, and investment projects, as well as maintenance-centered projects (shutdowns) – all require the power of deep integration.

Companies that need all of these aspects of project management can combine both applications, employing tight integration between them to replicate all data.

NEW AND ENHANCED PROGRAM AND PROJECT MANAGEMENT FEATURES

As mySAP PLM and its program and project management capabilities evolve to meet the diverse needs of project-oriented businesses, SAP continually adds new features and functionalities to the comprehensive capabilities that mySAP PLM already possesses. SAP R/3 Enterprise Product Lifecycle Management Extension 2.0 now supports resource-related billing documents and resource-related billing between company codes.

Resource-Related Down Payments and Billing Documents

In long-term projects, services rendered can often be billed only after they have been approved or after the customer confirms project progress. In most contracts, customer and contractor agree on down-payment dates. Often, these payments do not generate revenue.

Users can now create down payment requests analogous to the functions offered by resource-related billing. Based on the dates specified in the billing plan, users can create down payment and billing requests based on the service rendered and the billing rules. The system takes down payments into consideration when invoices are created.

Resources that belong to different organizational units often render services. It was not previously possible to bill these services between company codes within one controlling area; instead, a reconciliation ledger was used, leading to delays in invoicing. Now, companies can bill the requesting company code for the services provided using a resource-related billing document.

mySAP PLM PROGRAM AND PROJECT MANAGEMENT: FEATURE OVERVIEW

Feature Group	Feature
Program management	■ Portfolio management ■ Evaluating strategic product developments ■ Scenario simulation ■ Life-cycle planning ■ Life-cycle costing actuals
Project planning	■ Project structures ■ Project builder and the project planning board ■ Resource planning and workforce planning ■ Documents ■ Costs ■ Revenues ■ Budget ■ Scheduling functions
Project execution	■ Confirmation ■ Project-oriented procurement and production ■ Claim management ■ Payments ■ Project progress ■ Progress tracking ■ ProMan ■ Project information system
Interfaces	■ Open Project System ■ Collaborative project management ■ Balanced scorecard

ENTERPRISE ASSET MANAGEMENT WITH mySAP PLM

ENTERPRISE ASSET MANAGEMENT WITH mySAP PLM TODAY

To succeed in the world of global business today, most companies are maximizing every advantage they have. For many of these businesses – those in which the life cycle of assets plays a central role – this includes the major advantage of having optimized assets throughout the enterprise. When this happens – when facilities and equipment are operating at peak performance and efficiency – cycle times grow shorter and profits grow taller.

It is no wonder, then, that most manufacturers and distributors today share a distinguishing characteristic – their universal commitment to tightly controlled and efficiently executed enterprise asset management (EAM) processes. To these companies, the value of EAM extends beyond the fact that it helps them to get the most out of their assets. Because it avoids downtime and speeds time-to-market and profits, companies view EAM as a strategic solution.

For the maintenance engineers and asset and plant managers who manage physical and technical assets every day, meeting the challenge of customer demands quickly and cost-effectively means wringing everything possible from the available assets:

- Assets must perform at top levels. At the same time, companies have to keep capital and maintenance costs down.
- Optimizing asset investment, service, and repair requires a constant eye on a complex and changing array of assets – across locations and over time.
- Companies have to ensure that equipment and maintenance procedures comply with all safety and work regulations.
- Because processes now extend beyond corporate boundaries, asset management means working closely with external partners in the extended supply chain.

One-dimensional, off-the-shelf asset management products cannot meet such extensive needs, nor can they provide the consistently reliable, up-to-date asset information required to efficiently manage and maintain a company's complex asset base. These professionals know that to fully optimize their assets,

only a comprehensive, full-circle, asset management solution – one that optimizes assets throughout the entire life cycle of the assets – can meet their strategic requirements and objectives.

OVERVIEW OF ENTERPRISE ASSET MANAGEMENT WITH mySAP PLM

Asset-intensive companies in all industries choose mySAP Product Lifecycle Management when they require a reliable and flexible, as well as strategic and long-term, software solution to their asset management needs. mySAP PLM is a complete, cross-industry enterprise asset management solution for achieving the highest possible performance of the technical assets, as well as the optimal condition of all assets at the lowest attainable cost.

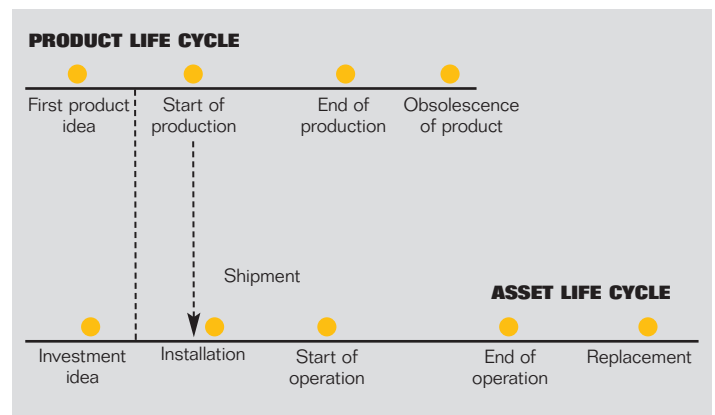


Figure 17: The Management of Product and Asset Life Cycles

Companies from diverse industries use the enterprise asset management capabilities of mySAP PLM, including:

- Automotive
- Aerospace and defense
- Banking and insurance
- Chemical
- Consumer products
- High tech
- Industrial machinery and components
- Engineering, construction, and operations
- Mill products
- Mining

- Oil and gas
- Pharmaceuticals
- Public service
- Utilities
- Media
- Retail
- Transportation

End-to-End Asset Management

mySAP PLM provides a full array of functions needed for beginning-to-end enterprise asset management. This rich set of functions helps asset managers and maintenance engineers handle any type of asset or object – from the initial investment idea through the start of the operation to the end of the operation and the replacement of the asset.

The enterprise asset management capabilities of mySAP PLM help companies manage the selection, purchase, and installation of equipment. Users can monitor and maintain assets; perform equipment repairs, modifications, upgrades, and refurbishment; and track costs for individual assets, production lines, and entire facilities. They can share drawings, manuals, service bulletins, and parts information, and they can collaborate across the company and with external partners to optimize the use of assets.

The enterprise asset management capabilities of mySAP PLM encompasses all components of an integrated computerized maintenance and management system (CMMS). Its integrated e-procurement capability makes it easy to order supplies and services, and inventory management capabilities help to improve the efficiency and performance of equipment throughout a plant.

Because truly cost-effective asset management requires a detailed understanding of the costs caused by an individual asset, manufacturer, facility, or production site, mySAP PLM helps to identify problem areas and analyze their origin. In addition, integrated technical reporting and detailed cost reporting help to optimize maintenance and support decision making on all staff levels.

Tight Process Integration

All enterprise asset management capabilities of mySAP PLM are integrated with other capabilities of mySAP Business Suite. That means the solution can deliver a complete overview of all product- and asset-relevant business processes, and it can provide all of the information needed about the product and asset life cycles, including all partner information. And it can do so at any time.

In addition to integrating with all SAP solutions, mySAP PLM integrates with many non-SAP solutions and software, making total information and access to it a hallmark of mySAP PLM. This includes access to geographical information systems (GIS), as well as systems for plant data collection, measurements, and counter readings.

Convenient Access: SAP Enterprise Portal

A role-specific, content-rich, and highly functional user environment empowers employees with the information and tools they need to succeed at their jobs, every day.

The portal is tailored to the specific roles of the employees in an organization. Once logged on, the maintenance manager, maintenance engineers, maintenance planners, maintenance supervisors, and technicians instantly have the information and tools they need to perform their daily tasks. And the portal can be further customized to suit personal user preferences. The portal is readily accepted by users, who are often more productive with it than they have ever been. In addition, SAP development teams have direct contact with users to continually improve and enhance the portal for them.

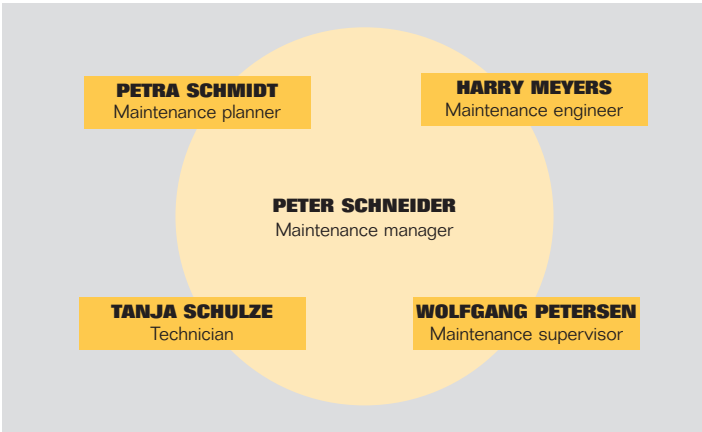


Figure 18: Enterprise Asset Management with mySAP™ PLM: Focusing on People

To enhance this portal technology, SAP has developed content packages, called business packages, which are predefined collections of role-based portal content. Many multilingual business packages are available, including the **Business Package for Assets**, which provides information about equipment, functional locations, documents, malfunction reports, and maintenance orders.

Mobile Support

In addition to the enterprise portal and support for all standard Web-based applications and services, mySAP PLM uses state-of-the-art technology for all types of communications.

For example, mySAP PLM supports mobile users, including mobile asset management, with support for a full range of mobile and hand-held devices and pagers. Furthermore, the integration of mySAP PLM with other SAP and non-SAP solutions and applications lends total support for collaborative engineering design and procurement communications requirements.

A Customizable Enterprise Asset Management Solution

To precisely fit a company’s maintenance management requirements, SAP and its partners can work closely with a company to customize the solution, while still allowing the use of a standard system. This means companies can upgrade or adapt their systems at any time to meet their changing needs.

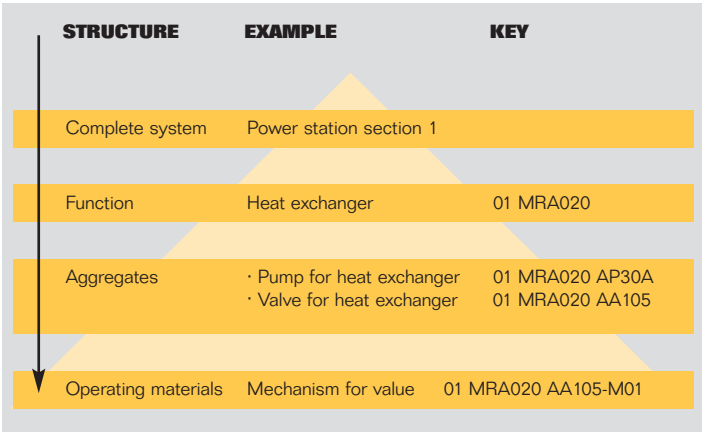


Figure 19: Standardized Structuring of a Power Station in mySAP™ PLM

mySAP PLM also allows a company to structure its organization individually and flexibly. It can use the solution’s organizational units to structure maintenance work according to company-specific criteria. Whether those criteria are functional, based on location or responsibility, or driven by business, the solution’s versatility gives companies the freedom to realize their own unique solution.

Uniformity Across the Company and Internationally

Because mySAP PLM is integrated into the mySAP Business Suite family of solutions, lean, effective business processes are assured. Consistent, accurate, up-to-date information from each enterprise area is available on the spot, whether that information comes from human resources, controlling, production, or purchasing. This uniformity allows a company to connect the entire enterprise – even multinational groups – with the same maintenance management information. Because a single language for all users is not required, all users can work in their own native language, assuring rapid system acceptance.

Improve Asset Profitability

The enterprise asset management capabilities of mySAP PLM lead to the improved profitability of technical assets throughout all phases of the asset's life cycle:

- During the selection procedure and procurement process, staff can use evaluations of the current asset portfolio as the basis for defining technical and business selection criteria for new and replacement procurement measures.
- During plant operation, preventive inspections and maintenance – managed by a complete and reliable plant maintenance solution – leads to higher system availability and reduced breakdown costs.
- Integrated plant maintenance enables more transparent and simplified processes in maintenance. Therefore, companies can reduce material and labor costs while maintaining the same level of system quality, and they can optimize costs for the use of external service providers.
- The relationship between business data and technical information supports management in making decisions about repair, refurbishment, and the scrapping of technical assets.

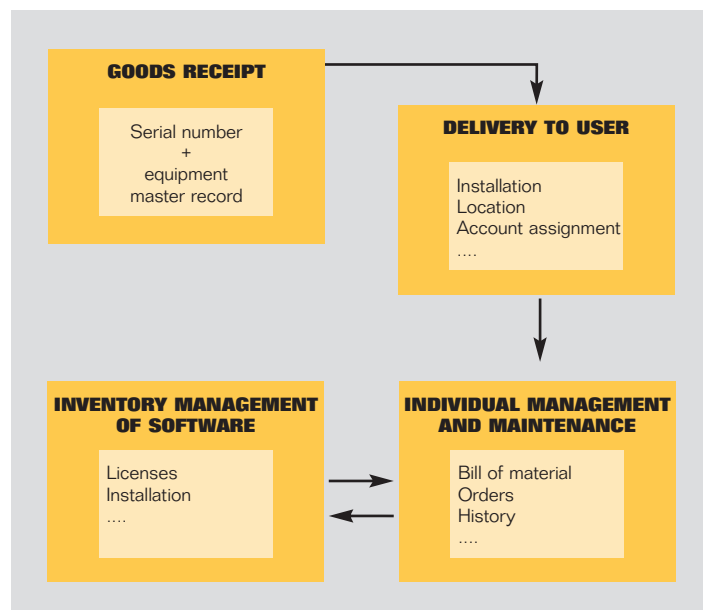


Figure 20: Example Process for Inventory-Based Management of Objects with mySAP™ PLM

The Benefits of Assets that Work

mySAP PLM puts assets to work. With enterprise asset management capabilities, users can sort through the complexities of tracking and maintaining equipment and facilities. They can share accurate, integrated information and work with a holistic view of assets. They can collaborate across the enterprise and the supply chain to ensure their companies get the most out of those assets.

The enterprise asset management capabilities of mySAP PLM help users:

- **Control operating costs and better manage capital expenditures** by improving operational excellence and reducing maintenance and procurement expenses; allowing closer tracking of operations, safety, and maintenance; and helping users make better-informed capital investment decisions
- **Improve plant and equipment performance** with maintenance planning, preventive maintenance, and fast repair capabilities that help reduce downtime; closely match assets with production requirements through improved equipment planning, operation, maintenance, and replacement processes; and reduce service and spare parts inventory
- **Respond to change** with mobile access to information, improved management of upgrades and maintenance, the rapid repair of inoperative equipment, and faster decision making based on accurate, up-to-date information and global enterprise portal access
- **Improve environment, health, and safety performance** with capabilities that allow a company to organize and plan the maintenance of all technical objects so it can to meet strict industrial health and safety measures, as well as comply with regulations and industry standards
- **Support competitive strategies** with assets that are performing cost-effectively at peak levels to help a company reduce time- and cost-to-market, streamline product launches and customization, enhance quality, and increase revenues and profits

Key Elements Optimize EAM Processes

mySAP PLM helps companies optimize all their maintenance processes in a wealth of ways.

To save time, when saving time is money, the solution's easy-to-use **breakdown maintenance** functions allow users to enter malfunction reports quickly and simply. When they need to act immediately, they can enter all the available malfunction data on a single screen, add order data, and print the necessary shop papers for technicians instantly, saving valuable time.

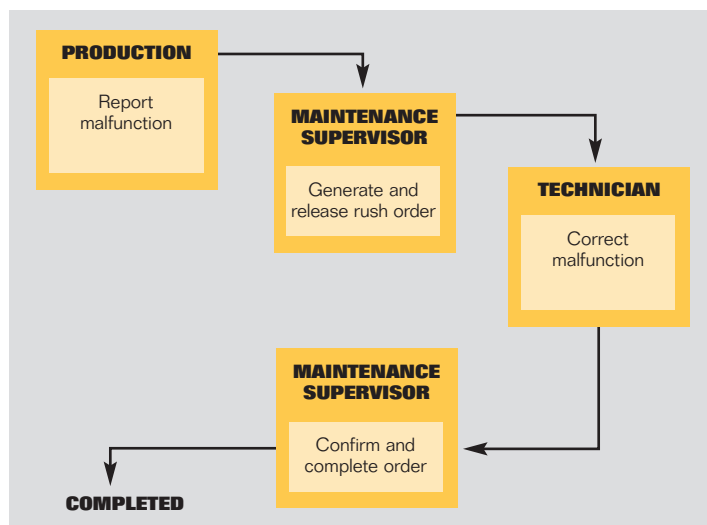


Figure 21: Breakdown Maintenance using EAM with mySAP™ PLM

The solution provides comprehensive **work planning** functions that are clear, easy-to-use, and can provide any level of detail desired.

mySAP PLM helps companies establish **universal processes** that enable the exchange and refurbishment of repairable service parts. This core maintenance process requires direct communication with materials management. The wide enterprise asset management capabilities of the solution, together with the business know-how and considerable experience of SAP means that the software is certified as ergonomic by international norms ISO 9241 and ISO 12407. This ensures that mySAP PLM offers

universal, seamlessly integrated processes that track each phase of the refurbishment cycle – from procurement to warehouse storage to productive use to continued refurbishment to final scrapping.

The solution's external **resource management** functions allow employees to work together successfully. At times, to reduce capacity overloads or for financial reasons, most companies use specially qualified personnel from external companies. To enable internal and external employees to work with one another, the enterprise asset management capabilities of mySAP PLM seamlessly work with procurement functions. This means companies can represent the distribution of tasks at their chosen level of detail:

- Simple reference to the required material
- Seamless inclusion of services into the required work
- User-friendly, standard service catalogs that are available online
- Flexible processing using outline agreements

Because safety is the first priority, the **work clearance management** functionality allows users to quickly plan and implement work safety measures, such as lockout or tagout, fire, or radiation protection for maintenance work on technical objects. Work clearance management is essential in highly sensitive technical systems. It guarantees a safe working environment for the maintenance staff, enables a company to meet environmental regulations, and ensures the reliability of technical systems.

If a company makes regular revisions and invests in new technical systems and modifications, mySAP PLM enables **complete project planning**, from conception through execution, for such projects. The project work can be broken down into individually linked project steps, and users can always display dependencies. Investment management is used to monitor accounting; budget management can be used for all work.

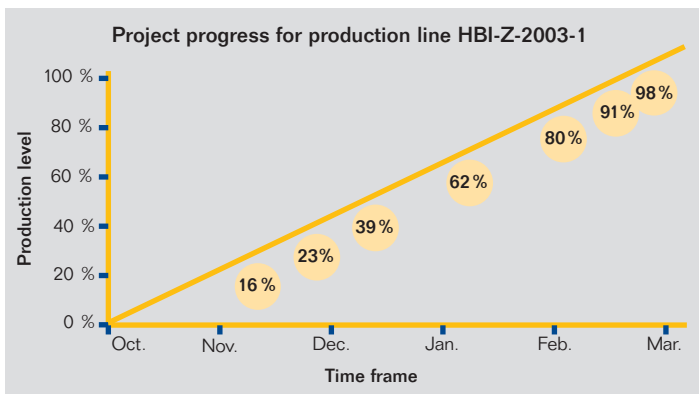


Figure 22: Controlling Project Progress Using the Enterprise Asset Management Capabilities of mySAP™ PLM

mySAP PLM offers versatile **analysis and evaluation** options to help assess strategies and identify potential for improvement. The solution provides key performance indicators (KPIs), an early warning system, and a maintenance history. These differentiated analyses make decisions easier. They support the continual improvement of maintenance processes – the goal of every maintenance management team.

KEY ELEMENTS OF ENTERPRISE ASSET MANAGEMENT WITH mySAP PLM

With the key functional elements of enterprise asset management, companies can manage assets throughout their life cycle to control costs, enhance performance, and compete in all of the following ways:

■ Manage all assets with precision

mySAP PLM lets companies manage fleets, buildings, equipment – virtually any asset – in detail. Users can track equipment and serial numbers; manage warranties, bills of material, and change orders; monitor service level agreements; and analyze equipment history.

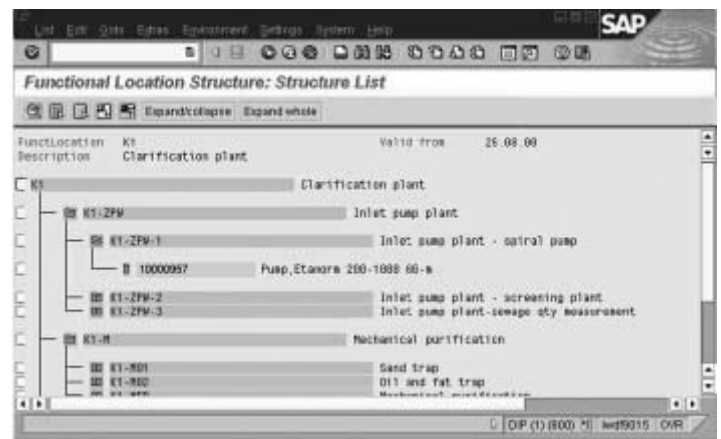


Figure 23: Installation Locations and Objects Using the Enterprise Asset Management Capabilities of mySAP™ PLM

■ Perform preventive maintenance

Companies can avoid malfunctions and breakdowns with preventive maintenance features that help to identify potential problems early on. mySAP PLM supports test equipment management, automatic measurement transfer, and various maintenance approaches, including performance-based, time-based, condition-based, and predictive maintenance.

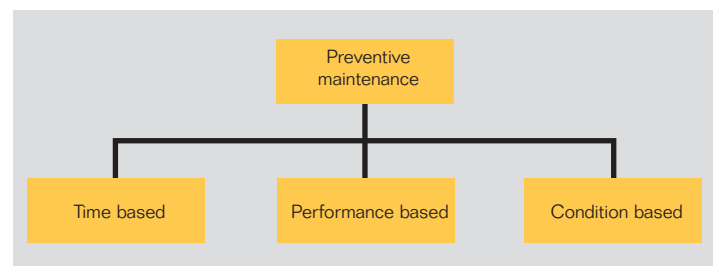


Figure 24: Types of Preventive Maintenance

■ Ensure safety during maintenance procedures

mySAP PLM helps companies comply with safety and environmental regulations so they can provide a safe working environment for their maintenance staffs. Work-clearance management functions handle lockout or tagout processes, maintenance processes, and work approvals, and users can draw on a time-saving work clearance document library.

■ Perform repairs with speed and accuracy

mySAP PLM provides work-order-cycle features that help maintenance staff repair malfunctioning equipment quickly and efficiently. Maintenance workers can see the complete maintenance process flow, including all the equipment, tools, and techniques involved. The solution also supports notification processing, work order resource planning, work order scheduling, work order costing, and execution and completion.

■ Collaborate with manufacturers and MRO suppliers

With e-procurement tools and tight integration with supplier relationship management (SRM) applications, companies can work with their business partners to check the availability of service parts, conduct business-to-business procurement of supplies, use Web-based illustrated parts catalogs, and process returns and repairs. They can also take advantage of online exchanges to find and compare vendors to ensure quality sources of supply.

■ Stay in touch with field personnel

mySAP PLM supports a wide range of mobile and wireless devices, including personal digital assistants (PDAs), mobile computers, smart phones, and pagers. That means companies can confirm orders and collect data and measurements directly from the plant floor, quickly dispatch employees to respond to problems, empower decision makers in the field, and manage maintenance operations – anytime, anywhere.

■ Deliver asset management information where it is needed

The solution's Web-based enterprise portal technology and content are tailored for asset management processes. The portal serves as an information cockpit that gives a company's employees and supply chain partners a single point of access to everything they need to work and collaborate.

The key functional elements of enterprise asset management with mySAP PLM are:

- Technical assets management
- Preventive and predictive maintenance
- Maintenance execution
- Work clearance management

Technical Assets Management

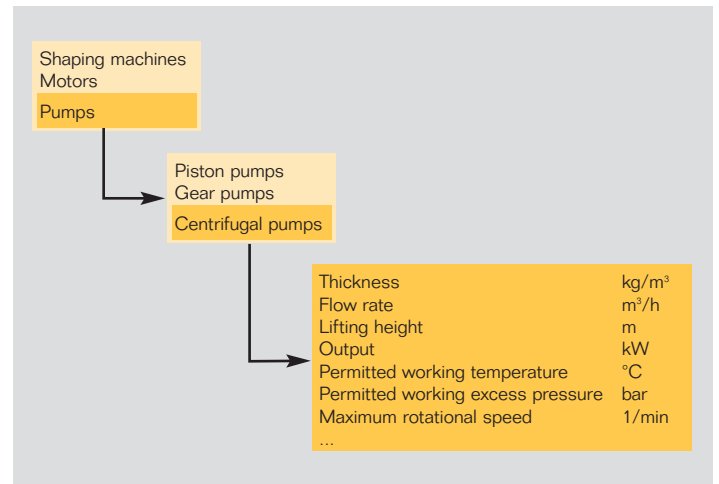


Figure 25: Classification and Representation of Technical Features Using the EAM Capabilities of mySAP PLM

mySAP PLM helps **structure technical objects** in a company with ease. **Serial number management** lets users integrate individualized items with maintenance processes, and the **classification system** can describe all types of objects.

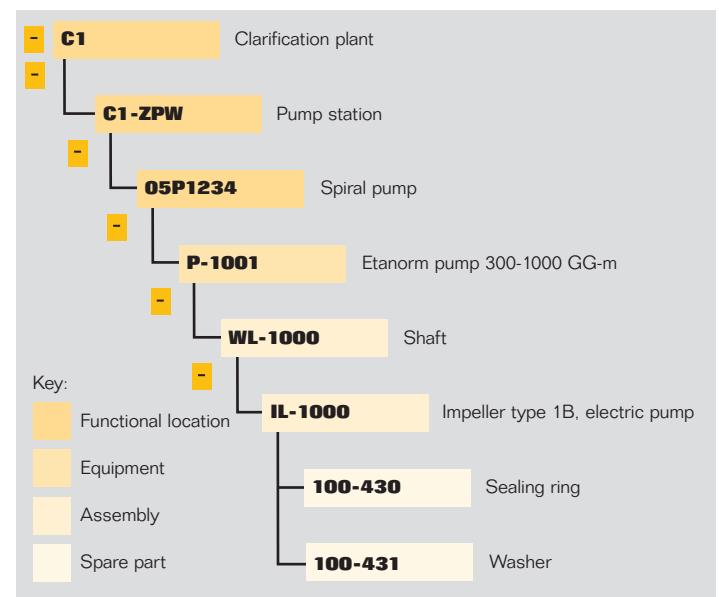


Figure 26: Structuring of Objects Using the EAM Capabilities of mySAP PLM

Using **change management** features, complex technical structures can be changed, and with the **variant configurator**, companies can configure parts to match exact requirements.

Companies can manage the master data for all the materials it procures, produces, stores, or sells with **material management** functionality. They can define separate bills of material (BOMs) for different company areas, including BOMs for service parts. And all this information is perfectly integrated with the **document management system (DMS)** in mySAP PLM.

Process control mechanisms include **approvals** and maintenance execution processes can be tailored to specific needs with, for example, a **notification service**.

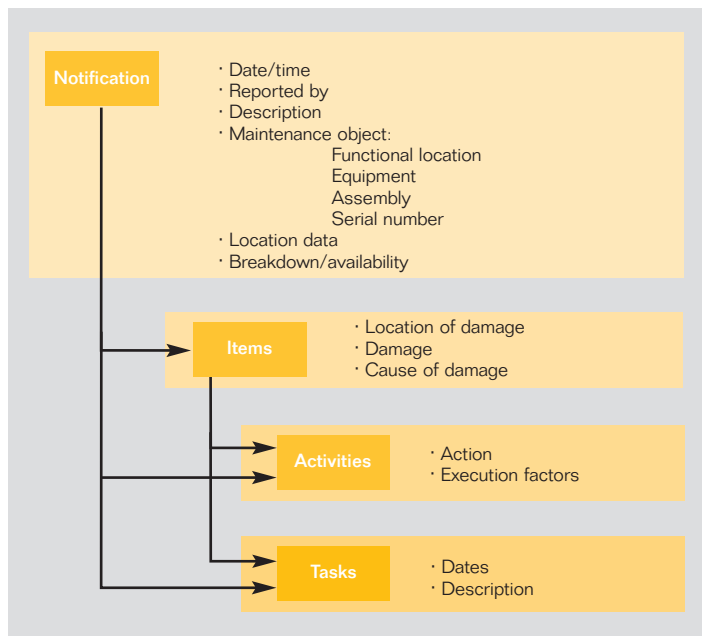


Figure 27: Notification Structure Using the EAM Capabilities of mySAP™ PLM

Companies can control maintenance costs with the help of **warranty management** functions, and they can plan and accurately process orders by assigning **production resource tools** to specific maintenance operations. They can also apply

performance- and conditioned-based maintenance with **counter and measurement readings**.

CAD-generated engineering and design documentation can be displayed in EAM master data. mySAP PLM interfaces with industry-specific systems including geographical information systems, supervisory control and data acquisition (SCADA), and CAD systems.

An open interface to asset accounting lets companies **connect technical and commercial assets**, and they can display both financial and maintenance-specific views of assets.

Fleet management functionality supports resource planning for vehicles and employees, procurement, refueling, and maintenance.

mySAP PLM manages assets for life – not only the life cycles of maintenance-intensive production assets, but also the life cycles of moveable assets, such as IT equipment. **IT asset management** features allow companies to make a complete map of the life cycle of their hardware and software so they can monitor and support it effectively. Companies can not only map which devices and software are installed at which location, they can also process all IT-related services smoothly and promptly. Only through this complete documentation of all the activities in the IT area can companies achieve a high cost transparency, which allows them to plan more precisely and constructively.

Preventive and Predictive Maintenance

mySAP PLM includes **time and performance-based preventive maintenance** features, which automatically determine which preventive maintenance work to perform.

To help maintain assets in optimal condition, the solution provides **conditioned-based maintenance** so process control systems can be interfaced to SCADA systems to monitor critical data. Conditioned-based maintenance functionality also

includes calibration inspection features for systematic equipment examination.

Tasks lists for maintenance planning help companies avoid redundancy by defining a series of standardized steps in a task list, which can be included in the maintenance plan.

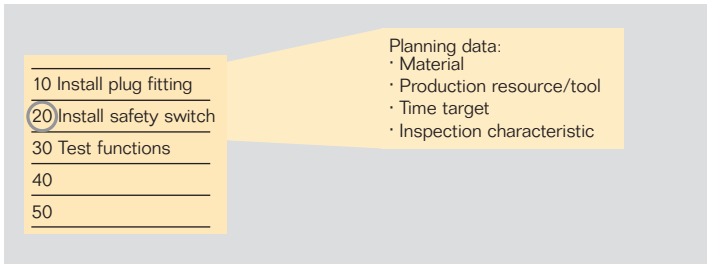


Figure 28: A Task List Using the EAM Capabilities of mySAP™ PLM

Maintenance Execution

Companies can use the notification service to document problems and streamline follow-up measures related to **malfunction-based maintenance**. The notification service also includes **employee self-service tools**, such as the **Internet service request** to find solutions quickly.

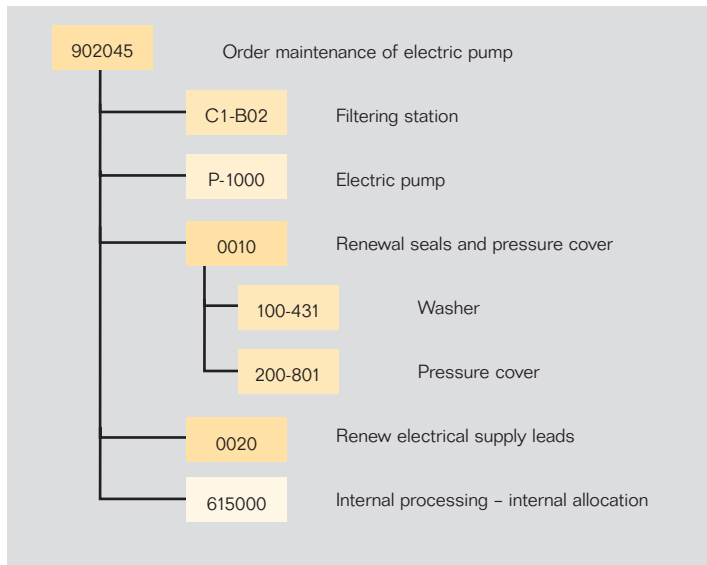


Figure 29: Order Structuring Using the EAM Capabilities of mySAP™ PLM

Notification and order processing ensures an effective maintenance process throughout the order process to the completion of the order.

Maintenance execution support in mySAP PLM includes extensive **mobile asset management** capabilities and support for business applications processed on all kinds of mobile and hand-held devices, such as PDAs, ruggedized devices, and laptops. With mobile asset management, a permanent connection to the back-end system is not necessary; the application is platform independent, and users only have to connect to the back end if they need data or want to write entries, such as material consumption or hours worked, to SAP solutions.

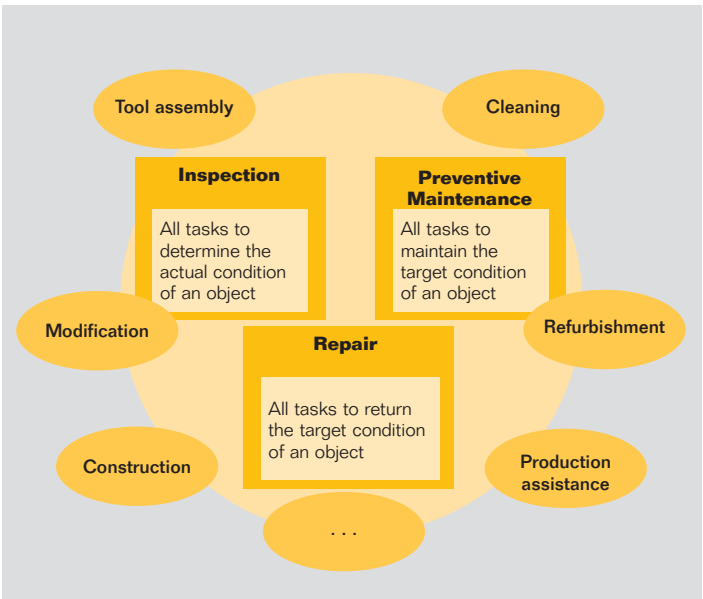


Figure 30: Overview of Plant Maintenance Using the EAM Capabilities of mySAP™ PLM

Enterprise asset management offers versatile solutions for **Web-based maintenance execution** – from notification processing to online recording of counter and measurement readings.

The solution **prints shop papers**, which include all relevant documents. **Catalog management** capabilities help users enter

structured information, such as damage and cause codes, when completing maintenance and notification orders.

With **capacity and requirements planning** functions, capacity data is derived from the responsible maintenance work center, and requirements planning provides time estimates for operations and orders completion.

Service parts management is fully integrated with materials management, including identification of stock and nonstock material, availability checks, and procurement processes. The

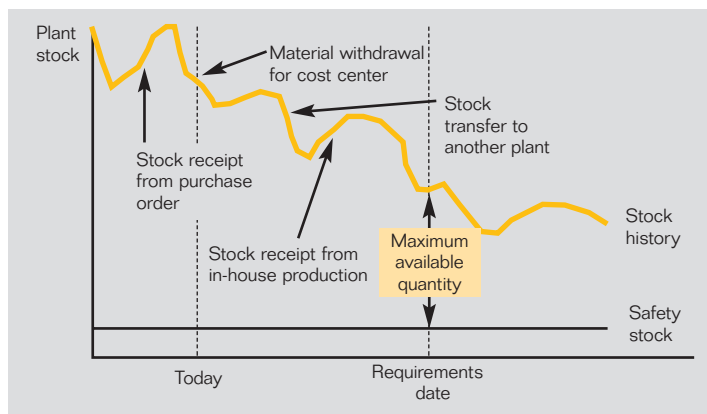


Figure 31: Availability Checks Using the EAM Capabilities of mySAP™ PLM

solution offers versatile options for spare parts handling, including BOM management and material where-used lists. In addition, complete **catalog integration and e-procurement of MRO parts** assures the comparison and selection of the correct parts and reduces the time, cost, and administrative effort for ordering.

Work Clearance Management

mySAP PLM offers fully integrated **work clearance management functions** so companies can observe strict health and safety regulations, as well as directives for hazardous substances. Work clearance management functions offer everything needed to thoroughly plan and organize the maintenance for all technical objects. **Work clearance management objects** control and

monitor safety measures, and **work approvals** control and regulate critical processes.

Industrial health and safety measures can include fire protection, radiation protection, or **lockout or tagout**. In the case of lockout or tagout, technical objects are isolated electrically or separated mechanically from other parts of the technical system while work or special tests are performed.

Integration with maintenance ordering processing assures that aspects of workplace safety are considered when executing maintenance work. Safety measures can include permits for radiation protection, fire protection, or lockout or tagout.

Work clearance management capabilities include a **work clearance document library** that provides complete logic to handle lockout or tagout processes for security-relevant areas or objects.

NEW AND ENHANCED ENTERPRISE ASSET MANAGEMENT FEATURES

There are numerous new and enhanced enterprise asset management features for 2004. For complete details, visit www.sap.com/plm. The following sections describe the key new and enhanced features.

Combined Reporting Using Orders and Operations

The solution now allows the combined and concise mapping of order header information with data from the corresponding operations in one list, providing a holistic view of the order or order package with all of its operations.

The selection options in particular are a major improvement.

Users can select orders using operation data or use objects from the object list to search for orders and their corresponding operations. This way users can simply select and evaluate all the operations and orders for a particular object of the object list – for example, a notification or a piece of equipment. They can use the list as a planning overview for future tasks and to initiate an array of order processing functions.

Enhancements for Maintenance, Repair, and Overhaul (MRO)

Using the **maintenance event builder**, planners can view outstanding maintenance requirements, due dates for work, work orders, and other information relevant to the maintenance of complex technical objects.

The enhanced **multiple counter plan** allows users to define and schedule different cycle sets and different items in a sequence with one plan.

Configuration control is the new main transaction for engineers, inspectors, or other roles participating in the configuration control process. Here, users can display both current and allowed configurations (represented by the Plant Maintenance structure and a master parts list in SAP R/3), browse and navigate through these structures, and maintain structure gaps in the actual configuration.

Reference Measuring Points

Users can create functional location structures on the basis of reference functional locations. With the definition of reference measuring points in reference functional locations, users can copy data into the measuring points of functional locations and create measuring points automatically.

mySAP PLM ENTERPRISE ASSET MANAGEMENT: FEATURE OVERVIEW

Functional location/ equipment/installed base/ serial number/ object network	■ Action log ■ Addresses ■ Alternative labels ■ Classification ■ Counter readings ■ Counter replacement ■ Documents ■ Inventory information ■ Long text ■ Measurement reading transfer ■ Measuring points and counters ■ Multilingual texts ■ Object information ■ Partners ■ Permits ■ Serialization data ■ Structure graphic ■ Structure list ■ Superior locations
Notifications	■ Action log ■ Activities ■ Addresses ■ Administration data ■ Basic dates ■ Causes of damage ■ Damage codes ■ Documentation of telephone calls ■ Document flow ■ Note database ■ Object information ■ Paging/communication ■ Partner determination ■ Sending confirmations of receipt ■ Task determination ■ Tasks ■ Task simulation
Material/bill of material order	■ Action log ■ Addresses ■ Budget/commitments ■ Completion confirmations ■ Components ■ Credit limit check ■ Delivery addresses of external material ■ Document flow ■ Estimated, planned, and actual costs ■ Execution factor ■ External processing ■ Internal processing

Material/bill of material order, cont'd.	■ Location ■ Material availability ■ Network structure/Gantt chart ■ Objects ■ Operations ■ Paging/communication ■ Partners ■ Permits ■ Planning board ■ Printing ■ Production resources and tools ■ Qualifications ■ Refurbishment ■ Relationships ■ Scheduling ■ Settlement rule ■ Suborders ■ Task list selection
Capacity requirements planning	■ Available capacity ■ Capacity leveling ■ Capacity requirements ■ Evaluation of work centers ■ Overload ■ Planning board (tabular or graphical) ■ Shift sequence
Completion confirmation	■ Collective confirmation ■ Error pool ■ Individual time confirmation ■ Measurement documents ■ Overall completion confirmation ■ Time sheet
History	■ Document flow ■ Historical orders ■ Material where-used list ■ Notifications ■ Operations ■ Orders
Maintenance planning	■ Costing (general and object-based) ■ Cycle sets ■ Deadline monitoring ■ Maintenance item ■ Maintenance plan ■ Maintenance strategy ■ Mass change ■ Overviews for object, maintenance plan, and operation ■ Scheduling ■ Scheduling overview ■ Task lists
Standard analyses	■ Breakdown analyses ■ Cost analyses ■ Damage analyses ■ Exception analyses

Standard analyses, cont'd	■ Location ■ Manufacturer ■ Object classes ■ Object statistics ■ Planner groups ■ User-defined analyses
Work clearance management – work approval	■ Catalogs ■ Change documents ■ Controlling applications ■ Controlling orders ■ Controlling release for execution ■ Documents ■ Enhanced approvals ■ Location ■ Long text ■ Object list ■ Partners ■ Printing
Work clearance management – application	■ Catalogs ■ Change documents ■ Controlling orders (standard model) ■ Controlling release for execution (standard model) ■ Controlling work approvals (enhanced model) ■ Documents ■ Enhanced approvals ■ Location ■ Long text ■ Object list ■ Partners ■ Printing <i>Specifically for the work clearance application:</i> ■ Controlling operational work clearance
Work clearance management – document	■ Catalogs ■ Change documents ■ Documents ■ Enhanced approvals ■ Locations ■ Long text ■ Multiple use of tags ■ Object list ■ Partners ■ Printing ■ Untagging adjustment <i>Specifically for the work clearance document template:</i> ■ Library of work clearance documents <i>Specifically for operational work clearance documents:</i> ■ Checks (for example, conflict verification) ■ Controlling operational processes ■ Controlling work clearance applications ■ Operational protection ■ Simulations

LIFE-CYCLE COLLABORATION WITH mySAP PLM

LIFE-CYCLE COLLABORATION WITH mySAP PLM TODAY

World markets watch the continuous emergence of increasingly diverse products, driven by fast-moving, market-of-one forces demanding more and better personalized products – but only if the price is right.

As quickly as products and markets are changing, so is the competition. Today, a competitive edge must be sharper than ever for a company to succeed.

Productive communications through efficient product design and project collaboration has surfaced as a weapon in the battle for competitive advantage. Collaboration puts everyone in a company and in the extended enterprise on the same page, which is where companies need to be to squarely face the challenges of today and tomorrow.

The companies that most need a powerful collaboration solution are those that focus on product innovation, product development, and product maintenance. The companies that require more teams and more team collaboration than any other are those that make the products the world buys. And the demands on these companies are relentless – more products, faster, of higher quality, yet at a lower cost.

In answer to these challenges, dependable, flexible, high-performance, and easy-to-use communication and collaboration tools are helping global manufacturing and production companies to efficiently plan, make, and maintain new and successful products of all types. With these tools, companies can meet the challenges of conceiving, developing, and delivering new, quality products to market as fast and efficiently as possible.

With superb collaboration capabilities at their disposal, development teams are all using the same tools and accurate, up-to-date information to collaborate not only with each other, but also with suppliers, vendors, other partners, and customers worldwide. These teams are seamlessly sharing product and project information of all kinds – project plans, product structures,

analyses and specifications, parts information, notifications, and service bulletins. They are sharing it in all kinds of formats, too – from e-mail messages to drawings and diagrams to spreadsheets and thousands of documents.

This tightly integrated level of collaboration, while increasing efficiency, speeding time-to-market and volume and lowering costs, is also changing the way companies do business. For example, by using collaboration solutions, companies eliminate many face-to-face meetings that include expensive travel. They find that they save money and time by using the newest collaboration tools, such as state-of-the-art Web conferencing tools, which can be just as effective – and sometimes more effective – than face-to-face meetings.

Companies are finding all the newest tools of product design and project collaboration in the life-cycle collaboration (LCC) capabilities of mySAP Product Lifecycle Management. They are not only finding the newest tools, but also the right tools. With them, business team members are effective team players, suppliers and distributors are fully supported, and customers stay in touch throughout the product life cycle. Now, a company's employees are equipped to:

- **Collect and organize** product and project information in a logical, project-based framework
- **Share information** over the Web and collaborate with internal and external partners on a one-to-one, one-to-many, and many-to-many basis of communication
- **Execute** business process transactions throughout an entire project with an enterprise-wide portal – a single point of access for everybody to the information they need aggregated to precisely fit their roles

Companies can also expect mySAP PLM and its collaborative solutions to result in good communication. The results of that are easy to see:

- Efficient, effective product development cycles
- Clear and unambiguous activities and goals

- Consistent and reliable data
- Immediate and precise response to changes in the market or consumer needs

By enabling team members to share information with rapid precision, think fast, and respond just as quickly, mySAP PLM and its powerful collaborative solutions help companies stay competitive with a razor-sharp winning edge.

OVERVIEW OF LIFE-CYCLE COLLABORATION WITH mySAP PLM

To industries that communicate about complex products, assets, and project information within cross-enterprise processes and with external partners, mySAP PLM offers a complete, integrated solution for life-cycle collaboration. It provides both extensive design and project collaboration capabilities, including a full range of function-specific features (see the section titled “Key Elements of Life-Cycle Collaboration with mySAP PLM” on page 78), as well as the platform to support them and the people who use them.

Virtual Collaboration with cFolders and cProjects

A Web-based cooperation application in mySAP PLM called Collaboration Folders, or cFolders, creates a virtual collaboration environment in which everyone else can share, review, and exchange information. cFolders emphasizes support for collaborative product development in R&D and engineering.

cFolders are, together with cProjects, part of cProject Suite in mySAP PLM. cProject Suite delivers the collaborative design and project management capabilities it takes to compete successfully today.

cProjects delivers rich collaboration functionality to the project management capabilities of mySAP PLM. cProjects supports users at every step, including support of the stage-gate methodology (see the section entitled “New and Enhanced Life-Cycle Collaboration Features” on page 34).

Project Suite is dedicated to promoting communication and collaboration between groups of people who work together but apart – often dispersed around the globe. It enables teams within cross-enterprise processes to easily set up virtual work areas on the Internet where they can exchange information by using only a Web browser. Cornerstones for successful projects include strong commitment management, well-defined roles, and solid deliverables.



Figure 32: cProject Suite with mySAP™ PLM

The Elements of cFolders

cFolders hones collaborative skills and abilities with sophisticated features and functions:

- Hierarchical structuring of collaboration through various objects: folders, documents, data sheets, materials, bills of material (BOMs), technical assets, and custom-defined (or generic) objects
- A sophisticated authorization concept for access control
- An integrated redlining and markup tool for 2D and 3D object viewing
- WebEx real-time Web conferencing integration
- A full search functionality
- Notification and status management for process control
- Integration with mySAP PLM and mySAP Supplier Relationship Management

- Support for distributed content server architecture
- Large file handling via the FTP protocol

Good for Business

cFolders promotes communication and collaboration by substituting classic communication via e-mail, phone, fax, and face-to-face meetings with collaborative virtual work teams. It reduces misunderstandings and communication deficiencies and improves data consistency. It replaces unstructured communication with structured data using redlining, markups, data sheets, and secure areas. And cFolders provides synchronous communication capabilities, such as graphical information.

cFolders adds business value to a company by:

- Reducing design cycle times, as well as potential redesign, by enhancing collaboration and communication
- Accelerating collaboration with Web-based access, subscription, and notification mechanisms
- Allowing effective process control by using status management, which can trigger actions (notifications, authorization changes), depending on the status
- Improving communications flow and reducing miscommunications, which reduces engineering errors
- Supporting real-time collaboration over the Web
- Avoiding supplementary costs by providing integrated visualization and redlining tools
- Reducing the administrative load with a powerful and flexible authorization concept
- Enabling back-end integration with SAP R/3
- Supporting decision making by allowing users to easily compare supplier bids
- Increasing capacity resulting from faster time-to-market

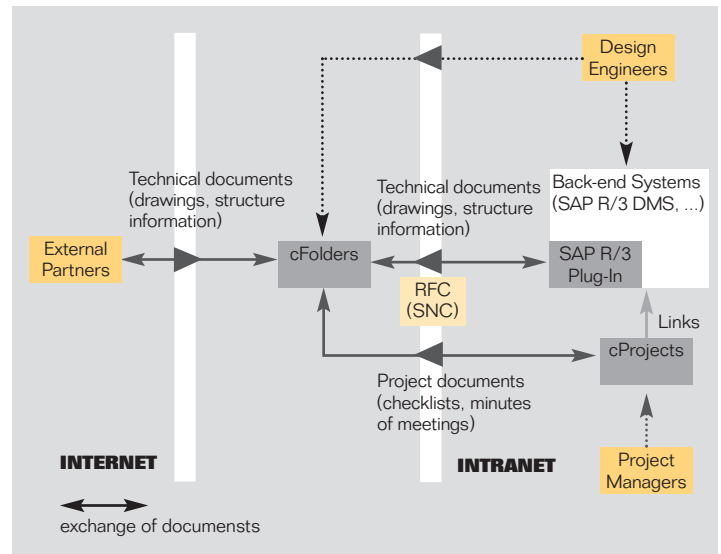


Figure 33: cProjects and cFolders Architecture

cProjects for Web-Based Project Management

cProjects is the mySAP PLM Web-based tool for collaborative project management. cProjects enables companies to realize innovative ideas in development projects, simplify internal processes, implement recognized quality standards, and reduce costs incurred by errors.

The cProjects cross-industry application supports the entire process of a development project – from conception to planning and quality checks and through to the completion of the project. It allows company employees to communicate with external partners or lead customers by using cFolders as an imbedded tool for collaboration. And cProjects supports internationally recognized standards (like QS-9000) and methods like Advanced Product Quality Planning (APQP), which was developed originally in the automobile industry.

In addition to enabling users to manage and track schedules, activities, and costs throughout the project life cycle, cProjects supports the stage-gate methodology, which defines milestones between project phases. A formal approval process between the subsequent phases ensures that clear decisions are made, and it

documents whether a project can enter the next phase. Using this methodology, users can ensure that all required tasks in one project phase have been completed before the next phase can begin.

The Elements of cProjects

cProjects supports project management efforts with the following features and functions:

- Flexible project structuring by phase with stage gates, tasks, and checklists
- Full scheduling functionality, including Gantt charts
- Templates for projects, checklists, and documents
- Collaborative work on tasks with external partners using cFolders
- Connection to business objects via links
- Role-oriented team creation and resource planning based on availability and skills
- Predefined evaluations
- Workflow integration
- Integration with the Controlling and Financials components of SAP R/3

cProjects at Work

cProjects adds business value in many dimensions:

- By using templates, cProjects standardizes the development process.
- By tightly integrating with cFolders, cProjects improves communication with external partners.
- By offering access directly from cProjects to all objects needed for the project, such as documents, materials, or opportunities – regardless of where they are stored – the cProjects application increases development efficiency.
- cProjects optimizes resource planning based on availability and skills, enables effective project management with a broad variety of predefined evaluations, and improves decision making.
- Using the phased approach to project management – stage-gate methodologies – cProjects ensures decisions are transparent.

Full Design and Project Collaboration Support for NPDI

All of the extensive life-cycle collaboration capabilities of mySAP PLM, including cProject Suite, are extended to the comprehensive, closed-loop new product development and introduction (NPDI) process. SAP's solution for this process efficiently and successfully designs, creates, and brings new products to market by uniting people with information in one streamlined business process. Through its life-cycle collaboration and analytics capabilities, mySAP PLM enables the logical, fast decision making required by the intensive NPDI process.

Open to the Extended Supply Chain

With cProject Suite, suppliers and partners do not need direct access to a company's PLM or ERP system, nor do they need any locally installed software to collaborate with all external and internal partners. All they need is a Web browser and a user ID in the system running cFolders, making it easy for large organizations to collaborate with small partners.

Even external users with low bandwidth Internet connections can upload and download large files to their cFolders server via the FTP protocol. Although anyone can access project information, anytime and anywhere, mySAP PLM is just as secure as it is open. All data passing between SAP Web Application Server and a browser uses Secure Socket Layer (SSL) encryption. For even tighter network security, companies can use reverse-proxy technology.

Finally, because cFolders and cProjects are shipped as add-ons to SAP Web Application Server, compatibility with any existing SAP infrastructure is ensured.

Better Collaboration Means Better Results

Setting up virtual work areas not only reduces travel-related costs, it improves data consistency – and that can considerably reduce product run times. mySAP PLM makes many other important benefits of better collaboration possible, as well. On an enterprise level, these include:

- Shorter lead times in all project phases, resulting in faster time-to-market and time-to-volume
- Higher quality products with greater market appeal
- Lower product development costs
- Less unnecessary repetition of product development processes

KEY ELEMENTS OF LIFE-CYCLE COLLABORATION WITH mySAP PLM

mySAP PLM enables life-cycle collaboration, and collaboration enables benefits – from improving customer service and increasing revenues to lowering working capital and reducing operating costs. For complete details about these benefits and the life-cycle collaboration features of mySAP PLM, visit www.sap.com/solutions/businessmaps/solutionmaps.

The following sections describe the key life-cycle collaboration features of mySAP PLM.

Design Collaboration with cFolders

mySAP PLM offers full design collaboration support. For example, hierarchical structuring allows users to collect and structure product and project information in a logical, collaboration-based framework. The folder structure within a collaboration project can be hierarchically structured by subfolders, which can contain different kinds of objects: documents, discussions, data sheets, bookmarks, materials, BOMs, custom-defined generic objects, and text.

Users can access documents in their virtual work areas, or import them from local computers and the **document management system (DMS)** in mySAP PLM. Folders can contain many types of documents, from CAD drawings and Microsoft PowerPoint presentations to spreadsheets and Microsoft Word documents.

In any folder for any user, a sophisticated, object-oriented **authorization concept** controls access rights and capabilities to cFolders objects, such as BOMs and documents.



Figure 34: Authorization Concept with cFolders

The comprehensive authorization control is optimized to work in dynamic project teams. It guarantees that each collaboration partner can only edit those objects for which the partner is allowed to work. Once a project is set up, the authorization mode is initially set for each member of the collaboration project and is passed on to all subordinate objects. This mode, at the highest level of a collaboration project, can later be overwritten for selected objects within the hierarchy. A full document revision history assures the orderly aggregation of accurate information. While documents are being processed, they can be locked to other users.

The document **version control** mechanism within cFolders enables all collaboration partners to track the change history of a document. In addition to version control, a **notification and subscription service** ensures all partners are actively informed of all relevant changes within objects. Anyone on the project team can use the notification service to learn about all changes to the project or any documents by subscribing to these documents. The notification service is especially useful to team members working locally, but who rarely or never meet.



Figure 35: Notifications with cFolders

Document handling includes direct editing of all file types supported by a front-end PC. Checked-out documents are stored in the list of favorites, and documents can be locked during check-in and checkout.



Figure 36: Document Handling with cFolders

Work-saving collaboration templates let users create a template based on a previous collaboration. This means, for example, that users can reuse a complicated collaboration project structure, which reduces setup time for new collaboration.

An alias concept integrated into cFolders allows the same object to possess different names. This is useful when dealing with different companies that call a specific object by another name.

Back-end integration allows users to exchange information with partners, suppliers, and customers using cFolders without giving them access to a company's back-end systems. Objects used in back-end processes, such as documents, materials, and BOMs stored in SAP R/3, can be published in cFolders. This enables external users without an SAP R/3 system and using only a browser to create and maintain material BOMs. External sources can easily add data or change the information, and then users can seamlessly move those changes from cFolders into a back-end system.

WebEx integration enables the synchronous, real-time sharing of information over the Internet. A cFolders interface allows users to access the WebEx meeting functions and then organize and track these meetings in their cFolders meeting inbox. WebEx can seamlessly display documents on screen in multiple locations simultaneously so applications, documents, and drawings are available to everyone at the same time.

To extend the benefits of mySAP PLM's life-cycle collaboration capabilities to many industries, cFolders offers two cooperation scenarios. The **collaboration scenario** uses a single joint-work area to support cooperation between all internal and external people engaged in product development. The **competitive scenario** provides a separate area for rival suppliers and assigns individual authorization settings to each during competitive bidding situations. Note that this capability also connects cFolders to end-to-end e-sourcing processes.



Figure 37: Collaboration Structure with Competitive Scenario in cFolders

To support engineering and design efforts, the solution provides an **integrated graphical user interface**, called ECL Viewer, for markup and redlining in 2D and 3D CAD drawings.

Specifications often require quantifiable product characteristics to be provided. Formalized **data sheets** simplify this process. They allow users to import a classification system to structure the characteristic values they want to provide in a bidding process. Users can include imported classes and characteristics from the SAP R/3 classification system or other classification schemes. In addition, mySAP PLM makes it easy to compare data sheets because it presents an overview of all data in Microsoft Excel.

During the competitive bidding process, users can create separate rival **supplier work areas** using individual authorization settings to manage the information exchange. In addition, the cFolders competitive scenario is integrated with the mySAP Supplier Relationship Management bidding engine for **end-to-end e-sourcing** processes.

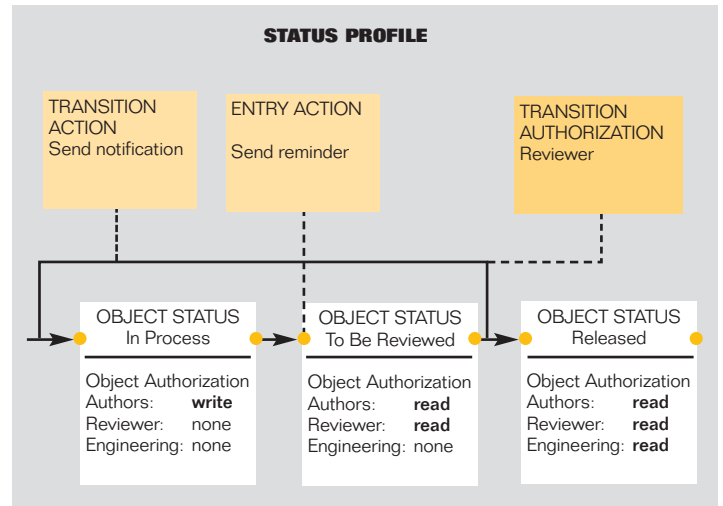


Figure 38: Status Management Process Control with cFolders

Status management features enable project leaders to create – on the fly – a simple process control, which otherwise would have to be achieved with bulky workflow tools and the help of expensive consultants. A status profile consists of freely definable statuses and the transitions between them. A **status profile** can be assigned to documents, data sheets, materials, BOMs, folders, areas, and collaborations. Status management features work with approval workflows. When a status changes, it automatically creates a notification, such as “to be approved” or “approved.” mySAP PLM offers complete **change authorization** functions, as well, permitting read-only access when changes are being processed. The right to make status transitions is user dependent and can be easily assigned by the collaboration project leader.

Because mySAP PLM supports **distributed content servers**, users can choose the nearest content server to decrease network traffic and improve access speed. And for users with low bandwidth Internet connections, mySAP PLM handles large files using FTP.

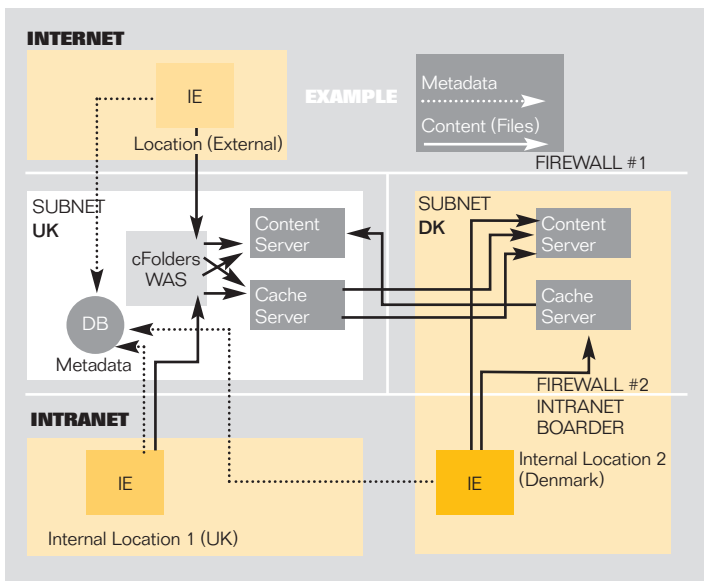


Figure 39: Example – cFolders and Distributed Content Server Architecture

Collaborative Project Management with cProjects

Users can handle the entire range of development project activities – from concept and planning through execution and the closing of a project – with mySAP PLM’s collaborative project management capabilities.

To simplify and standardize the creation of an operational project, users can create **project templates** that include all objects and checklists.



Figure 40: Project Structuring with cProjects

Users can set up **project teams** identifying all project participants, including internal employees and external business partners. They can structure projects using clearly defined **phases**, assigning checklist items and tasks to each phase.

Users can plan **quality gates** to achieve a tightly controlled transition between phases. All goals in the phase, which need to be met before starting the next phase, must be approved. Milestones during the phases can be modeled using checklist items or tasks. The checklists serve as control lists. The items in a checklist are relevant to the approval of a phase. Therefore, checklist items must be completed before the phase can be approved.

With **task planning**, users can set up tasks to plan and schedule required processing steps, and they can assign them to the responsible people. The status of a task is documented by a processing status and by reporting actual values.

cProjects **scheduling** capabilities calculate the earliest and latest start and end dates of project elements (although checklist items have only end dates).

Because documents are vital to any development project, **document templates** help users create new documents. Documents consist of metadata – for example, a short text, the person responsible, and the status – and the document content itself.

With an **interactive bar chart** (a Gantt chart) users can view and change checklist items and tasks, depending on the status of the project.

They can use **object links** to link an existing business object, such as a specification, BOM, work breakdown structure (WBS), or an inspection plan, to a project element.

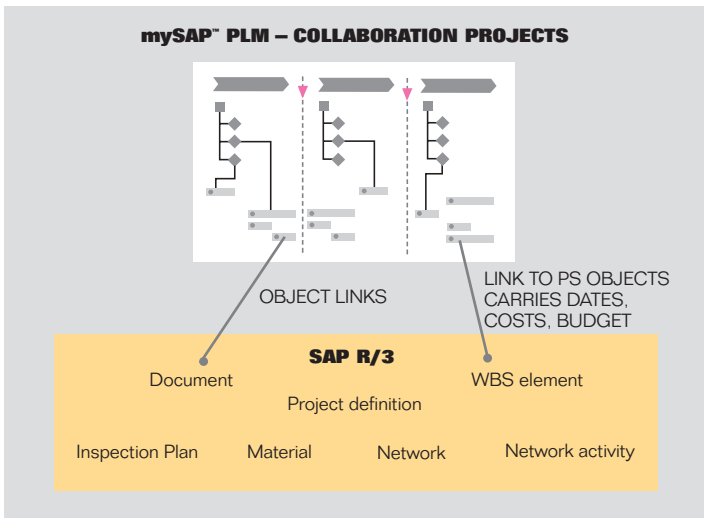


Figure 41: Object Links in cProjects

These objects do two important things: they document the product development process, and they grant access to information about product development. Users can link business objects of a different origin – for example, from a different system – to a project, as well as display data about these objects in the application or include them in project evaluations. This is close to reality. Many companies operate one central system for engineering, but have other systems in each plant that also contain data relevant to the product development process.

To monitor completion of tasks, the employees issue a **confirmation**. This can also be done using the Cross-Application Time Sheet (CATS) application.

An approval is a part of a phase in a development project. It refers to a clearly defined processing state. During an approval, decision makers document that the goals in the phase have been achieved. An **approval document** in PDF format is generated, and the completed phase is frozen – protected from subsequent modification. The approval document contains all information relevant to the approval. All decision makers are responsible for the approval, either not to sign the approval or sign it with a signature – individually granting or rejecting the approval.

cProjects also supports the APQP methodology, a stage-gate driven approach, which is required by the ISO TS 16949 standard in the automotive industry. APQP includes the use of a control plan to maintain process steps, tools, and product characteristics. Moreover, users can use this control plan to generate routings and inspection plans in the SAP R/3 back-end system.

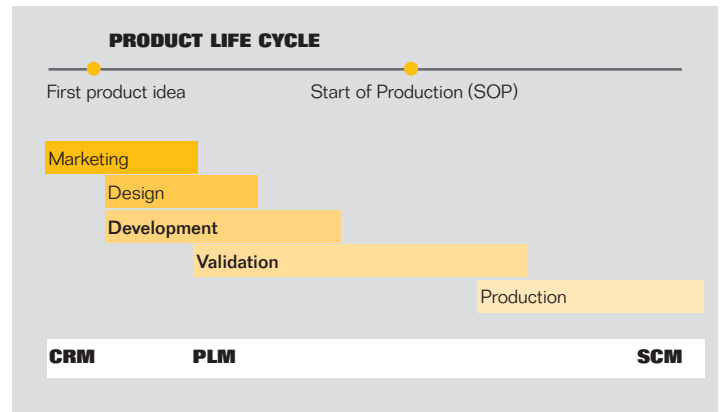


Figure 42: cProjects and the APQP Methodology

An integral part of cProjects is **role-based project resource management**. With resource management, users define, in the planning stages of the project, which roles are working in the project, such as project manager, project accountant, or design engineer. This definition also includes the required skill set to fulfill each position. Sophisticated search functionalities allow users to find the right person for the job, based on skills, availability, and organization. Parallel to the role definitions, the roles are assigned specific responsibilities and participation in areas of the project, such as certain tasks. This makes it clear which person is responsible for which task and when the results are due.

Based on the information in project resource management, project managers stay up to date on the project costs. Using innovative **integration with mySAP Financials** (patent pending), the solution automatically creates cost collectors based on organizational dependencies. The solution calculates resource allocations maintained in project resource management and summarizes them into a project cost and, importantly, also to a

project revenue plan. The information is directly available in cProjects and can be used in project execution for plan-actual comparisons.

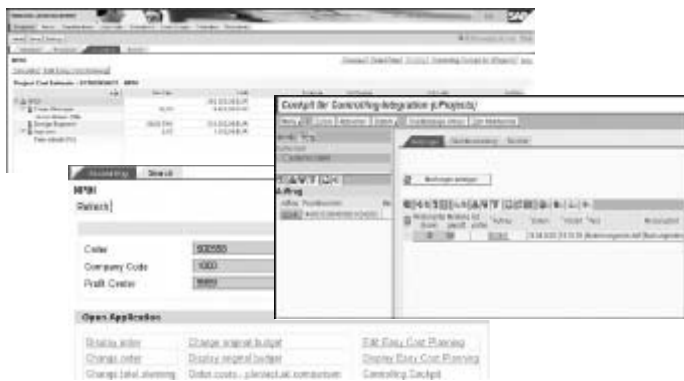


Figure 43: Integration of cProjects with mySAP® Financials

Quality Collaboration

mySAP PLM makes it easy for customers and vendors to exchange and process quality-relevant data using the Internet. This applies to quality certificates, inspection results, and notifications respectively corrective and preventive actions.

QUALITY CERTIFICATE PROCESSING AT GOODS RECEIPT

The quality certificate process begins when a material is ordered requiring a certificate upon which the customer and vendor have agreed to exchange quality management data electronically. When the order has been processed, the vendor creates an electronic certificate for delivery to the customer. mySAP PLM can automatically include the data in the electronic certificate in a goods receipt inspection lot at the customer. This process is often referred to as quality data interchange.

QUALITY CERTIFICATE ACCESS ON THE INTERNET

When material is ordered that requires a certificate, the vendor creates an electronic quality certificate for the delivery. This certificate can be tailored to a customer's need in terms of inspection characteristics, descriptions, and layout of the certificate form. The vendor can then put this certificate in the Internet instead of printing it out and sending it together with the goods

delivered. Finally, the customer or recipient can retrieve the electronic quality certificate over the Internet according to deliveries or batches. The customer or recipient can retrieve the electronic quality certificate over the Internet according to deliveries or batches.

RESULTS RECORDING OVER THE INTERNET

With an **Internet transaction server**, a customer can provide a vendor with an **HTML entry screen** on the Internet. The vendor can then record and save the inspection results on the Web. The results are transferred directly to mySAP PLM. This service is extremely useful for source inspections or inspections obtained from third-party facilities, such as commercial laboratories.

QUALITY NOTIFICATIONS ON THE INTERNET

Using quality **notifications**, users can contact business partners to request permission to deviate from the specification. If a vendor part does not meet specifications, for example, a company can document the part as defective in a quality notification and then send the Web link for approval directly to the customer. Users can also create a complaint or internal service request using an Internet service. In this case, the request coming from the Internet is transferred into a quality notification in the SAP R/3 back-end system. The customer can display the processing status of the notification at any time on the Internet.

Analytics

mySAP PLM's support of **product design optimization** logically begins in the early development stages. Using the product design optimization tool, mySAP PLM is integrated with the business analytics capabilities within the SAP Strategic Enterprise Management component, enabling **design-to-cost engineering** and early **cost optimization**. The ability to calculate product characteristics and design alternatives early on provides important information for the product design team.

To monitor and control project data, the **project information system** provides flexible and comprehensive functions, enabling users to evaluate one project, partial projects, or multiple projects. The solution provides overview reports and reports containing varying degrees of detail. The system is designed to meet the needs of both project management and project personnel. Predefined **InfoCubes** in SAP Business Intelligence simplify information access and organization.

The **quality information system** is used to condense and evaluate inspection data and problems using various standards and reports. It also uses predefined InfoCubes from SAP Business Intelligence.

The **reporting and asset information system** uses maintenance management capabilities to offer a broad range of reporting features for technical objects and operative instruments, such as orders and notifications. The system also provides strategic evaluations for a company, including mean time to repair (MTTR) and mean time between repairs (MTBR) analyses.

Asset business content is also available from SAP Business Intelligence. Notification- and order-based InfoCubes and queries provide important information on asset availability, performance, and costs.

Web reporting capabilities are included for all areas of PLM analytics using the SAP Business Intelligence data warehouse. This makes available accurate and up-to-date information in the right context. Its architecture allows it to fit seamlessly into the varied IT landscapes of SAP and other software systems so data can be retrieved from a variety of sources.

Analytics are provided for **environment, health, and safety** waste management life-cycle analyses. These analyses list the total amount of waste that must appear in a waste life-cycle analysis, according to the waste code and point of origin. The analyses also contain the description of the waste, the justification for waste disposal, and the final disposal facility. The

1,000-employee quota is used to evaluate the total number of accidents and to determine how frequently accidents occur per 1,000 employees.

Powerful Enterprise Portal and Powerful Content

When colleagues and team members reach consensus on project information, the necessary business transactions can be executed. To enable this, SAP provides a business execution framework – SAP Enterprise Portal. Users only need a Web browser to access this highly functional, yet easy-to-use portal.



Figure 44: Enterprise Portal for Global Collaboration with mySAP™ PLM

The portal provides one-stop access. SAP Enterprise Portal provides a single point of access to the applications, services, tools, and information people need and use every day, all day. The solution is highly flexible, so users can customize the information they receive to specifically suit their needs and their role in the company. The portal can be customized for all users – in or out of the company – and their roles in or with the company and the project.

mySAP PLM offers capabilities that enhance a company's ability to communicate and collaborate in unique and versatile ways, like iViews – small applications that display information in vari-

ous formats, such as text, figures, graphs, reports, tables, and charts. A set of iViews is included in each of many portal content packages, called business packages, which enrich the portal experience.

iViews do far more than provide personalized, role-based content to each user. They draw real-time data from applications to provide users with interactive access to a wide range of business functions.

Although iViews content comes predefined, it can also be customized to fit individual user and entire user group needs. To find content provided by SAP that fits those needs and import it into the portal, the iViews Studio – www.iViewStudio.com is a one-stop resource. Here, users can search the content catalog for items of interest.

mySAP PLM goes a step further, complementing portal technology with a full array of portal content packages, known as business packages, designed to meet the diverse needs of different user groups. Business packages are predefined collections of role-based portal content – ready-to-use solutions that minimize time-to-deployment, streamline common business processes, and provide immediate productivity gains. Each business package contains a set of adjustable iViews.

SAP has created many content-rich, multilingual business packages. A few examples include the following:

- The **Business Package for Assets** focuses on information about equipment, functional locations, documents, malfunction reports, and maintenance orders.
- The **Business Package for Products** allows easy access to information relating to materials, BOMs, and documents.
- The **Business Package for Projects** contains program and project management capabilities and functions drawn from mySAP PLM.
- The **Business Package for Collaboration by cFolders** allows effective cooperation between customers, suppliers, and partners in product development.
- The **Business Package for mySAP PLM Connectivity** provides iViews specifically for quality management and environment, health, and safety management.

Access to documents stored in the document management system of mySAP PLM is also possible through the knowledge management platform of the enterprise portal. The full document can be accessed, including the content and the attributes of a document, as well as links to other objects. The federated search feature allows documents to be found across repository borders in different repositories, such as file servers, Web servers, and different document management systems. A repository manager provides the integration with mySAP PLM's life-cycle data management capabilities.

To maximize the impact of a company's Web presence, it is important to deliver a range of immediately **recognizable benefits** to users. With its one-stop, portal access to tools and information and with its personalized, role-based portal content enhanced with business packages, mySAP PLM puts users at the center, supporting them throughout their day by:

- Providing content tailored to the exact needs of each user and user group
- Offering iViews, which can be combined to optimize the benefits to various user groups
- Focusing on easy-to-use, easy-to-read iViews that provide all the important information required by each individual user
- Providing intuitive, easy-to-use interaction between available iViews
- Presenting the content in a unique and easily recognizable corporate design

Through the portal, users almost instantly begin to recognize and reap the time-saving and revenue-making benefits of collaboration made possible by mySAP PLM.

NEW AND ENHANCED LIFE-CYCLE COLLABORATION FEATURES

As collaborative solutions continue to dominate the competitive landscape, SAP continues to add robust life-cycle collaboration capabilities to mySAP PLM to enable companies to communicate and collaborate with increasing effectiveness. The following sections describe the new and enhanced life-cycle collaboration features for 2004.

Enhancements to cFolders

The cFolders application provides the following new and extended functions and features as of cFolders 3.00.

Enhanced Authorization

Enhanced authorization features include the following:

- **Authorization logic**
Local object authorizations now override inherited authorizations only of the same type (user, user group, or role). This prevents, for example, a local authorization from a user group with the activity *Read* from overriding an inherited user authorization with the activity *Administration*.
- **Authorizations tab page in the user settings**
Using the authorizations tab, system administrators can define whether users are authorized to create and delete collaborations, status profiles, and user groups.
- **Guest authorizations in the competitive scenario**
Partners and suppliers can view a public area without having their own separate work areas.
- **Authorizations for fields in generic objects and document categories**
System administrators can now decide who is authorized to work on certain fields (see below).

New and Enhanced General Functions

New and enhanced general functions include the following:

- **New history and overview functions**
New history and overview functions make collaboration more transparent. Users can display the history of everything that has happened to a folder or object in a folder, when it happened, and who carried out each activity.
- **Archiving support**
System administrators can improve system performance by using SAP Data Archiving to archive collaborations that are no longer required.
- **Enhanced document handling**
Users can choose to transfer several files simultaneously between a folder in cFolders and a local directory.

New Generic Object Types and Document Categories

To better reflect a company's business needs and those of its partners, users can now create their own company- or industry-specific object types, for example, a document structure or technical asset. They can also create their own document categories, such as certificates. A document category is a document with additional data. For example, a certificate could have the additional data pupil, teacher, and date of issue.

The use of status and authorizations for generic object types, and authorizations for document categories, enables users to reflect company business processes. Companies can also model business objects according to their specific requirements and those of their business partners and suppliers with whom they share these objects.

Once users have created a generic object type or a document category, they can create and edit objects of this new type or documents of this new category at any time in cFolders.

Only system administrators can create generic object types and document categories. They are created in XML format, adhering to a predefined XSD description, and are stored with all other cFolders system settings in the WebDAV directory. For more information, see the *cFolders Configuration Guide*.

Enhanced Status Management

Enhanced status management features include:

- **Restricting the use of status profiles**
To allow users to define that certain fields in a generic object cannot be changed when the generic object has a particular status, users can now restrict the use of a status profile to an object type. They can set an indicator to protect these fields in each status, if required.
- **Object authorizations per status**
On the detail screen of each status in a status profile, users can now define object authorizations to determine who is authorized to work on an object. Object authorizations override all other inherited or local authorizations.
- **Protected versions**
On the detail screen of each status in a status profile, users can set an indicator to protect versions of an object. If someone changes a protected object, the system creates a new version.
- **Status log**
Transparency in status management is ensured. Users can now display a complete overview of all the statuses that an object in cFolders has had, when they were set, and by whom. Users can display the status log for all cFolders objects that are relevant to status management, including collaborations, areas, folders, documents, data sheets, materials, BOMs, and generic objects.

Future Plans for cFolders

Future enhancements to cFolders will include the following features:

- Integration with Microsoft Windows Explorer (in which cFolders acts as a virtual drive in Windows Explorer)
- Integration with the knowledge management platform of SAP Enterprise Portal
- Integration with central user management

- Integration with the service order capabilities of mySAP Customer Relationship Management
- Integration with the supplier portal of mySAP Supplier Relationship Management
- Security enhancements, including encryption

Enhancements to cProjects

The following sections describe the new and enhanced features for cProjects in 2004.

Subtasks enable users to split a task into numerous sublevels and assign smaller work packages to employees.

Usability enhancements allow documents, collaborations, and object links to be visible in the table view so users can directly navigate to these objects. In addition, a table view displays project roles and resource assignments. And users can search for documents using full text retrieval.

Users can change the interface by adding or deleting fields from the screen without modifying the data.

Accounting integration enables users to carry out project controlling. Users can plan project costs and revenues based on assigned resources. Confirmations are posted as actual costs to an internal order and are therefore integrated with the SAP R/3 accounting capabilities. In parallel, users can also make confirmations in the CATS application, which are then posted to the internal order in the Controlling component of SAP R/3.

Control plan features lets users generate inspection plans and routings in an SAP R/3 back-end system.

Microsoft Project integration allows users to download a project into Microsoft Project. With this they can share information with partners who do not have access to cProjects without revealing confidential information.

When a company uses cProjects together with SAP's workforce management solution, the resource planning engine determines the most suitable and available resource for each role and task. It automatically compares the availability of resources and issues a hit list from which to choose resources. Users can also start a resource search independent of a specific project.

Staffing features let users advertise unstaffed project roles in an iView in the enterprise portal.

Workforce analytics let users manage, control, and evaluate project staffing and resources.

Future Plans for cProjects

Future enhancements to cProjects will include:

- WebService@cProjects – opening cProjects for the Enterprise Services Architecture
- Enhanced integration with the document management system
- Exception-based alerting
- Project status reporting
- Management of multiple projects and subprojects
- Project versions
- Uploads from Microsoft Project

LIFE-CYCLE COLLABORATION WITH MYSAP PLM: FEATURE OVERVIEW

Design collaboration	■ Hierarchical structuring of collaboration projects ■ Document check-in and checkout mechanism ■ Object versioning ■ Sophisticated authorization concept for access control with automatic access control according to the status of an object ■ Integrated visualization using a redlining and markup tool for 2D and 3D object viewing ■ WebEx integration ■ Full text search functionality ■ Notification and status management for process control ■ Support of content encryption ■ Integration to mySAP PLM via the document management system and BOMs ■ Integration with Microsoft Windows Explorer with cFolders as the virtual drive ■ Integration to mySAP SRM with cFolders as the container for technical specifications in the mySAP SCM bidding engine and sourcing cockpit ■ Integration with SAP Enterprise Portal with cFolders collaborations visible through the knowledge management platform of the portal ■ Support for distributed content server architecture for reduced network traffic to the central server ■ Large file handling via FTP and low bandwidth support ■ Large set of APIs available
Collaborative project management	■ Flexible project structuring ■ Scheduling ■ Templates for projects, checklists, and documents ■ Collaborative work with external partners using cFolders ■ Connection of any business object via links ■ Web-based access to projects ■ Team creation and resource planning based on availability and skills ■ Predefined evaluations ■ Workflow integration ■ Support of stage-gate methodologies and APQP ■ Support for control plans according to QS 9000 development phases and the definition of phase output as checklists ■ Tasks with duration, responsibilities, and relationships ■ Task hierarchies to break down work packages ■ Graphical maintenance of dates and relationships using Gantt charts ■ Integration with SAP Easy Cost Planning ■ Integration with Financials ■ Integration with Controlling ■ Integration with Microsoft Project
Quality collaboration	■ Document exchange with external partners ■ Recording inspection results via the Internet ■ Exchanging certificates of analysis via the Internet ■ Quality data interchange for incoming certificates ■ Collaborative quality notification processing
Analytics	■ Reporting through SAP Business Intelligence ■ Predefined business content ■ Analytics covering: - Quality management - Enterprise asset management - Program and project management - Product cost design estimates
Enterprise portal content	The following mySAP PLM content packages are available: ■ Business Package for Assets ■ Business Package for Products ■ Business Package for Projects ■ Business Package for Collaboration by cFolders ■ Business Package for mySAP PLM Connectivity ■ Business Package PLM Analytics

QUALITY MANAGEMENT WITH mySAP PLM

QUALITY MANAGEMENT WITH mySAP PLM TODAY

Companies around the world continue to improve their abilities to listen to and understand their customers. Tools to analyze and evaluate customer demographics and demands are continuously refined. As a result, customer demand for more product personalization and more product quality keeps increasing, and new and diversified products are being produced at a record pace.

To meet these demands successfully, manufacturers in every industry are cutting time-to-market, while maintaining a firm grip on costs. At the same time, they are focusing more than ever on product and process quality. Meeting or exceeding customer expectations for quality is more critical than it has ever been to a company's competitive success.

For these companies, quality is not an afterthought. It is at the forefront of their product life-cycle processes. Quality is engineered into the design of their products from the very start to the end. They collaborate with their partners on quality all along the way. And quality – in both products and processes – is constantly improved to enable companies to keep and obtain new customers and to outperform their competition.

These companies also realize that achieving quality requires a fully supported e-business platform and the ability to communicate with a global quality network. The focus has shifted. The new paradigm focuses internally on the skills of empowered employees and externally on efficient business relationships. The efficient flow of internal and external information is essential to outdistancing competitors.

For all of these reasons, companies in every industry are taking a comprehensive approach to quality management – an approach that integrates information and processes across all departments and corporate boundaries, and involves everyone in the quest for quality – from employees on the plant floor to supply chain partners on the other side of the world.

OVERVIEW OF QUALITY MANAGEMENT WITH mySAP PLM

mySAP Product Lifecycle Management is a powerful solution that offers a comprehensive, broad-based approach to total quality management. mySAP PLM provides the tools to control the quality of products and assets throughout their entire life cycles and to react fast to unexpected events.

mySAP PLM enables product quality through the seamless execution of all product-related processes. From product innovation and development through production ramp-up and change management through complete supply chain and manufacturing to service and maintenance processing, the solution ensures effective quality management.

mySAP PLM forms a complete quality loop that supports each step of each phase of the product life cycle:

- Quality engineering in the planning phase: evaluation, design and development, validation, and product launch
- Quality assurance and control in the implementation phase: procurement, production, final inspection, and storage
- Continuous improvement in the usage phase: sales and distribution, service, and disposal

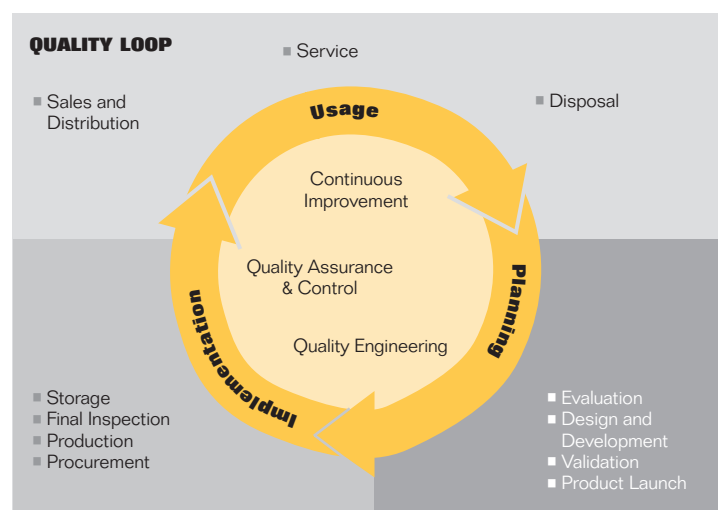


Figure 45: Quality Management in the Product Life Cycle

Improving the Quality of Quality Management

mySAP PLM provides the full complement of software functions needed by quality management experts to do their daily jobs. But it is not comparable to conventional computer-aided quality (CAQ) systems or laboratory information management systems (LIMS). It is far more. mySAP PLM is not an isolated system; it is integrated with other quality-related functional areas of the mySAP ERP solution to fully support the quality loop. Sharing consistent information and the direct, end-to-end control of business processes make total quality management a reality.

Internal Integration: Enabling Total Quality Management

Five cornerstones mark the seamless integration of mySAP PLM in a company.

First, the quality management capabilities of mySAP PLM are closely interlinked with other key functional areas and processes, including the engineering level during the design process and on the execution level along the whole supply chain.

Second, mySAP PLM is directly linked to various tools and resources that aid in the efficient management of business processes. Two of these tools are Webflow Engine, a tool for quickly and efficiently processing activities that arise from inspection lots and corrective actions, and ArchiveLink, a tool for storing documents, such as certificates, that are linked to application functions. mySAP Business Suite also offers many additional auxiliary functions, like document management or the central maintenance of authorizations, which are very important to fulfill legal requirements like 21 CFR Part 11 (which covers electronic records and digital signatures).

Third, with its open, standards-based architecture, mySAP PLM can cooperate with all other mySAP Business Suite solutions, such as mySAP Supplier Relationship Management, mySAP Supply Chain Management, and mySAP Customer Relationship Management, so companies can access and control important processes.

Fourth, using the SAP Exchange Infrastructure component of the SAP NetWeaver application and integration platform, mySAP PLM can also be integrated with non-SAP systems. That integration makes the sharing of quality information and true end-to-end business processes a reality.

And fifth, the role-based SAP Enterprise Portal becomes a cockpit and gateway to all the business processes and to all the information, functions, and services users need – at any time and from any location. The role-based portal can be flexibly personalized to individual users, allowing them to access functions and contents of their own work areas and to adapt their roles to suit their individual work preferences. Powerful functions, called iViews, supply users with the key figures and work lists relevant to their roles. Content-rich business packages are available, such as the analytic capabilities of SAP Business Intelligence, which provides a comprehensive overview of the key figures and evaluations to monitor quality performance.

mySAP PLM supplies several ready-made portal roles for the various tasks in quality management: quality manager, quality planner, inspection planner, quality inspector, notification processor, and test equipment manager.

External Integration: Enabling End-to-End Business Processes

mySAP PLM not only extends the reach of quality information, but the solution also gives companies the features they need to efficiently use that information. Internet services and mobile services for collaborative or remote scenarios are just two of the state-of-the-art features that the solution provides.

First, a company and its internal or external business partners can create service requests via the Internet, collaborate online to discuss and record information, search for suitable solutions, create and dispatch notifications, and track the processing status of requests.

Second, users can record inspection results on the Internet during a source inspection at a vendor site.

Third, vendors can supply quality certificates on the Internet for supplied goods; incoming certificates of analysis can also be managed. Companies can provide certificates and analysis data to their customers using electronic media.

And fourth, for the inspection of items that are not directly accessible, the solution supports the use of handheld devices. Mobile results recording is both simple and reliable. And users can control which inspection operations should be performed in the field with mobile devices.



Figure 46: Mobile Results Recording Using a Handheld Device

THE KEY ELEMENTS OF QUALITY MANAGEMENT WITH mySAP PLM

The functions of mySAP PLM can be grouped into three key elements: quality engineering, quality assurance and control, and quality improvement (See Figure 47).

QUALITY ENGINEERING	QUALITY ASSURANCE & CONTROL	QUALITY IMPROVEMENT
■ Quality manual (ISO 9000) ■ Advanced Product Quality Planning (APQP) ■ Production part approval process (PPAP) ■ Failure mode and effect analysis (FMEA) ■ Stability/shelf-life study ■ Inspection planning and specifications ■ Sampling procedures (ISO 2859, ISO 3951) ■ Recipe management ■ Document management ■ Master data and change management (ISO 10007) ■ Supplier management	■ Quality inspection ■ Inspection equipment interfacing ■ Statistical process control (SPC) ■ Laboratory Information Management (LIMS) ■ GxP compliance ■ Sample management ■ Batch management and traceability ■ Recipe management ■ Certificate of analysis ■ Gauge calibration ■ Quality cost	■ Audit management ■ Problem management ■ Corrective and preventive action (CAPA) ■ Solution database ■ Quality information management

Figure 47: Key Elements and Functions of Quality Management with mySAP™ PLM

Within a product life cycle, one area of functionality within mySAP PLM can be assigned to the engineering level. These functions – for example, advanced product quality planning (APQP) – are mainly applied during the development and design process. Another important part of the functions, such as quality inspection and problem management, can be assigned to the execution level. These functions are applied all along the whole supply chain, starting with procurement and going through warehousing to manufacturing and delivery and ending with customer service.

Quality Engineering

Quality engineering comprises the planning, analysis, and control of the product life cycle at all stages to maximize the quality of the process itself and the products it produces. The following sections describe the elements of quality engineering.

QUALITY MANUAL

To comply with the ISO 9000 standard, a company has to document its quality system by producing a quality manual. In the automotive industry, ISO 9000 is supplemented by such standards as QS-9000 and ISO TS 16949. In the food and drug industry, it is supplemented by good manufacturing practice rules and other regulations.

mySAP Business Suite offers several tools to maintain a quality manual: the knowledge warehouse or the document management system in mySAP ERP. The knowledge management warehouse contains the layout of a Web-based quality manual, which simplifies the creation and administration of a company's quality manual.

ADVANCED PRODUCT QUALITY PLANNING

APQP is a segment of the ISO TS16949 standard for the automotive industry that promotes typical methods and tools for product development and defect prevention. The cProjects application offers ideal support for APQP. This Web-based application improves the development process by focusing on quality planning and defect prevention at an early stage. It provides methods like quality gates and tools like control plan, as well as corrective and preventive action.

PRODUCTION PART APPROVAL PROCESS

The production part approval process is another segment of ISO TS16949. It defines the generic requirements for approval of production parts. The first purpose of this process is to determine whether the supplier properly understands a customer's engineering and design records and specifications. Another purpose is to ensure that the production process can produce products permanently meeting these requirements.

Several SAP tools support this process:

- cProjects with quality gates between the development phases
- Quality notifications to manage preventive or corrective action
- Quality information records to document the status of a supplier relationship and to release a supplied part in a strict sequence with several stages, such as *Prototype* or *Preliminary series*

FAILURE MODE AND EFFECTS ANALYSIS

Failure mode and effects analysis (FMEA) is a procedure in which potential failure modes of an item are analyzed and the individual risks are calculated. The method is suggested in the QS-9000 standard, but is also popular outside the automotive industry. Quality notifications can be used as the basis for a FMEA process.

cProjects can calculate a risk priority number for each checklist item as a product of its occurrence, detection, and severity, representing the final result of an FMEA.

STABILITY AND SHELF-LIFE STUDIES

Stability studies are used to test the quality and reliability of products over predefined periods of time under controlled environmental conditions. mySAP PLM offers a complete set of tools for planning and performing stability studies.

INSPECTION PLANNING

An inspection planner identifies the inspection characteristics and defines the inspection methods and inspection operations.

mySAP PLM offers advanced functions for inspection planning, such as the engineering workbench and the product structure browser. The engineering workbench is a sophisticated tool for professional planners of routings and inspection plans. The product structure browser displays all hierarchically related master data, superordinated, as well as subordinated. Standard operation procedures (SOP) as documents are supported in an inspection plan on operation level and in the inspection method. Users can generate an inspection plan from a control plan that was created in cProjects as part of APQP.

SPECIFICATIONS

A specification states the requirements to which a given product must conform. Requirements are described by quality inspection characteristics. Inspection plans can be used to represent specifications on the plant level, whereas material specifications can serve as global quality standards across plants.

SAMPLING PROCEDURE

A sampling procedure indicates the sampling sizes and associated acceptance or rejection criteria to be used for valuating an inspection characteristic.

mySAP PLM supports standardized sampling systems, such as ISO 2859 (attributive inspection) and ISO 3951 (variable inspection), as well as self-defined sampling procedures. Sampling plans according to ISO 2859-1 and ISO 3951 are available and ready for use. The solution also supports dynamic modification of the sample size and skip lot. Modification rules according to ISO 2859-1 (normal – tightened – reduced inspection) and 2859-3 (varying skip rate) are available as well.

RECIPE MANAGEMENT

The process industry needs recipe management capabilities during the development process. mySAP PLM supports structured product specifications and recipe definition from the first product idea down to the production line level. mySAP PLM now has a dedicated solution for recipe management.

DOCUMENT MANAGEMENT

Creating, maintaining, and retrieving documents in a safe and secure way is essential for quality documents. mySAP Business Suite offers several tools for document management. For quality documents, the document management system of mySAP PLM is an optimum choice because it is tightly integrated with the master data of supply chain management and quality management. Companies can also use the knowledge warehouse for document management, for example, to process and publish their quality manual.

MASTER DATA AND CHANGE MANAGEMENT

Creating, maintaining, and retrieving master data are required for quality management and for all the other functional areas, as well. With mySAP PLM, quality data can be embedded in general master data, such as material master or vendor master data. Other master data is needed for quality inspections, such as inspection method, inspection characteristic, and catalogs. The solution fully supports standard operation procedures and change management according to configuration management systems that correspond to ISO 10007. It also provides powerful audit trail capabilities.

SUPPLIER MANAGEMENT

Supplier management capabilities ensure that supplied products will be fit for use with minimal inspection and corrective action. mySAP PLM supports primary activities to:

- Define product quality requirements
- Evaluate alternative suppliers
- Select suppliers
- Conduct joint quality planning
- Cooperate with suppliers during the execution of a contract
- Obtain proof of conformance to requirements
- Certify qualified suppliers
- Conduct quality improvement programs, as required
- Create and use supplier quality ratings

mySAP PLM handles all of these aspects. For example, using a data set that is related to vendor and material, a supplier relationship can be initiated, released step-by-step, or blocked. Using this data set, quality agreements can also be managed.

Quality Assurance and Control

Quality assurance comprises all the planned and systematic activities implemented within the quality system. It is used to establish confidence that a product or service will fulfill requirements for quality.

Quality control provides the operational techniques and activities used to fulfill requirements for quality. Often, however, the terms “quality assurance” and “quality control” are used interchangeably, referring to the actions performed to ensure the quality of a product, service, or process.

QUALITY INSPECTIONS

Inspecting means measuring the characteristics of a product and comparing the results with specified requirements to determine whether conformity is achieved for each characteristic.

mySAP PLM supports the generation of inspection lots of various origins, like goods receipt, goods movement, and goods issue. It supports in-process inspection, recurring inspection of batches, and many other inspection types.

Results can be recorded with inspection characteristics values, which can be related to inspection points, samples, batches, or serial numbers. In addition, results can be recorded with nonconformity data, which can be related to inspection lots, inspection operations, or inspection characteristics. Results can also be recorded via the Internet or mobile devices so users can record the results of field inspections and inspections of difficult-to-reach objects.

When an inspection lot is released from quality inspection and a usage decision is made, follow-up actions, such as batch classification, dynamic modification of the inspection scope, or stock postings, can be performed in the background.

INTERFACES WITH TEST EQUIPMENT

Special equipment may be required for inspection operations, which can deliver the results electronically.

If a company needs to connect measuring subsystems, mySAP PLM offers a standardized quality management inspection data interface, together with a transfer protocol for downloading inspection specifications and uploading inspection results. SAP has already certified several third-party vendors for correctly handling the interface protocol. However, only a few software partnerships exist in the area of test equipment interfacing via external subsystems and not for interfacing a full-blown LIMS or CAQ system.

STATISTICAL PROCESS CONTROL (SPC)

Statistical techniques are widely used to manage quality control processes. The term “statistical quality control” is often used interchangeably with “statistical process control.”

mySAP PLM supports Six Sigma approaches for analyzing measured values and nonconformance data. For example, users can run various quality control charts, check the data for exceptions by applying the Western Electric rules, and calculate capability indicators cp and cpk . Moreover, they can perform Pareto analysis, trend analysis, and analysis of self-defined exceptions. In cases when they need to verify a distribution or significance tests or other highly sophisticated mathematical tools, they can transfer the data to external tools for statistical analysis via a standardized statistical interface.

LABORATORY INFORMATION MANAGEMENT SYSTEM

A laboratory information management system facilitates the testing of samples that routinely pass through an analytical lab. It manages the complete routine, from sample log-in and testing until final reporting. It organizes the inspection results into specific report formats to meet regulatory requirements.

mySAP PLM can functionally compete with best-of-breed LIMS. Supporting the trend of shifting from isolated data islands to global information platforms, this enterprise LIMS reaches far beyond the borders of traditional LIMS. Because of its tight integration in the supply chain components of mySAP ERP, it offers functions for inventory control and batch tracking, as well as problem handling and management of corrective actions. Enterprise-wide transparency and consistency can be achieved if a company uses mySAP PLM for laboratory information management. And no extra technical platform is required.

GxP COMPLIANCE

GxP is an acronym for Good Clinical/Laboratory/Manufacturing Practice (GCP, GLP, GMP), which are guidelines for enterprises regulated and supervised by the U.S. Food and Drug Administration (FDA) or other authorities.

mySAP PLM provides all the functions necessary to fulfill such requirements. For example, it supports document control and electronic records (corresponding to 21 CFR Part 11), including electronic batch records, digital signatures, audit trails, and batch traceability. SAP's development and service organizations are ISO 9000 certified.

SAMPLE MANAGEMENT

Usually in a laboratory, inspected items are samples, not inspection lots. Two cases are to be distinguished. First, in industrial laboratories, samples originate from inspection lots and represent the quality of inspection lots. The sample drawing process can be planned. Second, in commercial laboratories, samples may be delivered in an unplanned way. They do not need to have any relation to inspection lots and even the inspection specifications may vary with every order.

mySAP PLM provides all the required sample management functions, including sample drawing (primary samples, pooled samples, reserve samples) and control, recording sample analysis data, and reporting the results by means of a certificate of analysis. Users can create and maintain individual sample master records for various sample types, such as goods receipt samples, production samples, environmental samples, and competitor samples.

BATCH MANAGEMENT AND TRACEABILITY

Batch traceability is required in cases where products may vary in quality if they originate from different production runs. mySAP PLM can provide identification, classification, and inventory management for different batches of a material. It offers both bottom-up and top-down analyses, which show the final products into which a batch of raw material has flowed or which batches of raw materials have flowed into a final product. It might even be necessary to make the life cycle of individual items fully transparent. In this case, traceability is achieved using serial numbers.

CERTIFICATE OF ANALYSIS

A certificate is related to an inspection lot, a batch or a delivery and is used to document the quality of the material or product.

mySAP PLM creates all types of certificates as suggested by European standard EN 10204. The solution automatically selects the form, contents, recipients, language, and the delivery form of a certificate, depending on material, customer, or other criteria. Outgoing certificates are usually created for a delivery item and printed at delivery time. Structured and textual certificate data can also be transferred to the recipient via the quality data interchange function. mySAP PLM also supports the management of incoming certificates, which arrive together with supplied material. Certificate data received via the quality data interchange function can be automatically recorded as inspection results, which are linked to an inspection lot.

GAUGE CALIBRATION

Calibration is the comparison of a measurement instrument of unverified accuracy to a standard instrument of known accuracy to detect any deviation from the accuracy specification.

mySAP PLM supports the complete calibration process, especially calibration inspections, in accordance with the ISO 10012 and other standards. It generates equipment calibration orders on a regular basis, provides the calibration instruction, enables results recording and results history, and it can adjust the status of the equipment, depending on the inspection result.

QUALITY COSTS

Usually, companies identify and separate out several types of quality-related costs. For example, the appraisal cost is associated with inspecting a product; the cost of poor quality is associated with poor quality of products or services; and prevention costs arise from efforts to minimize the cost of poor quality.

mySAP PLM records efforts and the calculation and collection of quality costs, which users can analyze using standard controlling tools like order analysis or cost center analysis.

Quality Improvement

The ongoing improvement of products, processes, or services to achieve customer satisfaction is essential for production and service enterprises. mySAP PLM offers important quality improvement capabilities.

AUDIT MANAGEMENT

An audit is the examination and assessment of a quality system, a process, or a product to ensure compliance with given criteria.

mySAP Business Suite contains a comprehensive audit management tool that covers all aspects of the ISO 19011 standard: planning and performing audits, defining and monitoring corrective actions, and managing auditor qualifications. This highly generic tool can be used for various kinds of assessments in areas like quality management, environmental management, safety and security management, financial management, or risk management.

PROBLEM MANAGEMENT

mySAP PLM supports notification and analysis of nonconformities, defects, and failures related to products, processes, or services with a dedicated quality notification tool. Notifications are a flexible way to record, process, and monitor all unplanned events and the related corrective actions. The Internet service request is an extension of the quality notification tool that makes it possible to record quality notifications via the Web using variable templates, depending on the scenario.

CORRECTIVE AND PREVENTIVE ACTION

Corrective and preventive action features enable the implementation of solutions to eliminate identified problems, as well as the activities taken to improve a process to prevent future occurrences.

The quality notification tool supports the necessary corrective actions. Corrective measures can be related to the problem in general, as well as to individual nonconformities. A complete eight discipline (8D) process can be performed based on a quality notification. 8D processes are commonly used in the automotive industry for defect handling and prevention. The cProjects application of mySAP PLM supports extensive 8D processes, which take on the habit of a corrective project.

SOLUTION DATABASE

The solution database of mySAP Business Suite contains information that describes an action intended to eliminate a problem indicated by one or more symptoms. Its design allows solutions to be defined using various information sources (such as free-text description, actions that must be performed, and attachments) and links it with symptoms. The solution database can be accessed from within the problem management tools. If a solution can be immediately found in the solution database, users may not even need to create a problem notification.

QUALITY INFORMATION MANAGEMENT

mySAP Business Suite provides a rich set of methods and tools for analyzing quality data. The data is delivered into the data warehouse in SAP Business Intelligence, which is part of mySAP Business Suite, and into SAP Logistics Information System (SAP LIS), which is part of mySAP ERP. The solutions aggregate the operational data and calculate key figures to be used for quality analysis. Both systems provide standard and user-defined quality analyses of key figures (quality scores), performance indicators (cpk), trends, exceptions, correlations, Pareto diagrams, and more. These results can be used for vendor evaluation, internal and external benchmarking, balanced scorecards, and other methods of quality analysis. The data warehouse is user friendly, with features like Web reporting and freely definable parameters for aggregation and analysis.

NEW AND ENHANCED FEATURES FOR QUALITY MANAGEMENT WITH mySAP PLM

Quality Engineering

Two quality engineering enhancements have been made. First, the APQP capabilities of cProjects has been enhanced. Object links are provided to quality inspection and quality notification. Second, recipe management can handle inspection characteristics, trials, and stability studies using building blocks for optimized processing.

Quality Assurance and Control

SAP R/3 Enterprise has been enhanced for several practical requirements:

- Inspection results can be copied from one inspection lot to another.
- Users can choose to create isolated inspection lots instead of partial lots at batch split in a Sales & Distribution delivery.
- During stock posting in the usage decision, users can choose to display different units of measure.
- Inspection processing with handling unit-managed materials will be enhanced.
- New business add-ins (BADIs) improve the flexibility for customer-specific software enhancements.

Quality Improvement

The following quality improvement enhancements have been made:

- The quality manual can be maintained with SAP Easy Document Management.
- Stability studies can be analyzed with SAP Business Intelligence.
- The audit management solution is available in SAP R/3 Enterprise.
- Complaint processing in the automotive industry via iPoint Internet Agent will be supported by a partner solution. The Internet agent searches for complaint messages on the OEM portals of automotive manufacturers and can generate a quality notification in the supplier's SAP R/3 back-end system.

Main Trends for Future Development

SAP's first goal is to improve quality management during product development processes by APQP and other methods. This is a challenge because it demands high flexibility and tight integration in a heterogeneous system environment.

The second goal is to improve the collaborative processes, for example, the 8D procedure in complaint handling, concession requests, quality inspections as services, and certificate data interchange. Here SAP will use its best-of-breed tools, such as cProject Suite, and enrich them with Web services that work through a portal. This is also a challenge because heterogeneous systems must be interlinked by an exchange infrastructure.

QUALITY MANAGEMENT WITH mySAP PLM: FEATURE OVERVIEW

Feature Group	Feature
QM in development	■ APQP ■ Control plan ■ Inspection plan ■ Specification management ■ Recipe management ■ Stability studies ■ Flexible inspection specifications
QM in procurement	■ Vendor release ■ Status of supplier relationships ■ Production part approval process ■ Goods receipts with inspection lot processing ■ Dynamic modification of inspections and skip lot ■ Certificate processing ■ Quality data interchange ■ Complaints against vendor with quality notifications ■ Vendor evaluation ■ Supplier audits ■ Stock control and stock posting
QM in production	■ In-process control with inspection lot processing ■ Post-process control with inspection lot processing ■ Internal problem handling with quality notifications ■ Batch classification ■ Sample management ■ Stock control and stock posting ■ Connection of electronic test equipment ■ Statistical process control ■ Process audits
QM in sales and distribution	■ Goods issue with inspection lot processing ■ Certificates of analysis for deliveries ■ Returns processing with inspection lot processing ■ Customer complaints
QM as service	■ Calibration planning and execution ■ Customer-specific quality analysis (contract analysis) ■ Performing audits as service
Continuous improvement	■ Quality manual administration ■ Audit management ■ Corrective and preventive actions ■ 8D processing using quality notifications ■ Solution database ■ Quality analysis with the quality information system

ENVIRONMENT, HEALTH, AND SAFETY WITH mySAP PLM

ENVIRONMENT, HEALTH, AND SAFETY WITH mySAP PLM TODAY

Global businesses are contending with increasing complexity today, not only in products and processes, but also in the growing number of regulations – particularly in the area of environment, health, and safety (EH&S). It is a key part of the product life cycle that frequently faces complex and critical changes.

To begin with, as fast as products and processes are changing – which is all the time – EH&S processes must follow them. Process changes often result in changes in work areas, which can immediately impact issues of industrial hygiene and safety, as well as occupational safety for the employees – two of the primary concerns of EH&S.

At the same time that products and processes are continuously changing, so are the laws and rules that regulate them. It can be time consuming to keep up with changing regulations, which must be carefully tracked, because they influence many things: shipping requirements, waste-disposal restrictions, hazardous substance control, and product labeling and documentation to name just a few. They impact practically all areas of the company, too – from purchasing to new product development and manufacturing to service and maintenance to sales and distribution.

It is not easy, and EH&S issues continue to increase in importance. Failure to comply with regulations can result in high fines and penalties. The increasing number of new laws and regulations, together with more complex products, is why the significance of comprehensive and effective environment, health, and safety capabilities has never been greater than it is today.

But neither have the benefits of EH&S capabilities been greater. Safer products and safer customers and communities, safer workplaces and reduced risk to employees, easier and faster full compliance with regulations, lower costs and optimized business processes, and an enhanced corporate image are just some of the resulting benefits.

In today’s rapidly changing environment, mySAP Product Life-cycle Management helps companies to take advantage of those benefits. It provides the tools and integrated information companies need to manage today’s sophisticated EH&S processes – in all areas of a company and throughout the complete product life cycle.

OVERVIEW OF ENVIRONMENT, HEALTH, AND SAFETY WITH mySAP PLM

The EH&S capabilities of mySAP PLM are deployed by many companies in different industries, such as the automotive, chemical, pharmaceutical, and consumer goods industries. mySAP PLM is a cross-industry solution that supports environment, health, and safety project managers and employees to manage the many faces of safety – from handling hazardous materials and disposing of waste to assuring the health and safety of employees, the community, and the environment.

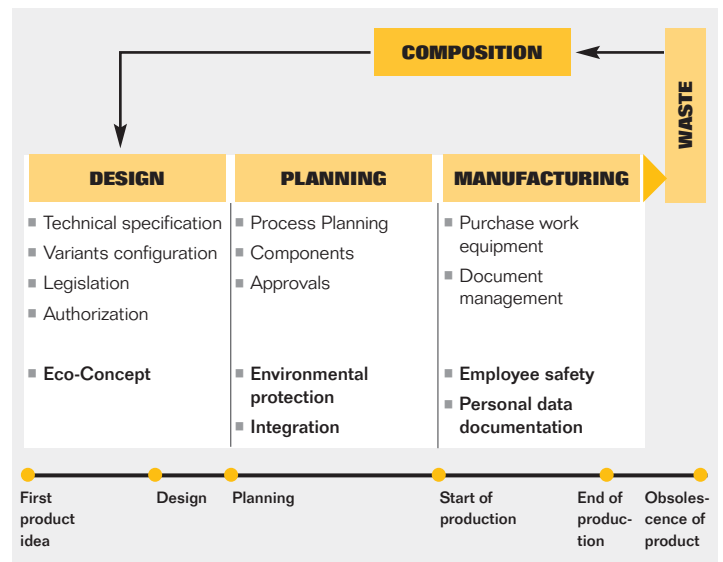


Figure 48: Environment Protection and the Product Life Cycle with mySAP™ PLM

Designed for EH&S Professionals

mySAP PLM supports the full spectrum of activities and tasks belonging to environment, health, and safety professionals:

- Industrial hygiene and safety
- Occupational health
- Product safety and hazardous substance management
- Dangerous goods management
- Waste management

EH&S: Complete, Proactive, and Open

Using the EH&S capabilities of mySAP PLM, companies can track vital information, including changing regulations. The solution supports EH&S requirements from beginning to end of the product life cycle, easily manages data and all types of documents, and, through tight integration with all product- and process-related information, automates and optimizes business processes.

mySAP PLM enables a company to proactively manage the health and safety of the entire enterprise. The solution makes the full range of product, plant, process, and safety information available at all times and in any location.

mySAP PLM is open to information, and the information is open to the whole enterprise. It allows a company to manage EH&S processes globally, and it can easily adapt to a company's unique processes and requirements.

Because mySAP PLM is built on an architecture based on open standards, it allows open communication and collaboration with internal employees and external business partners. mySAP PLM is also open to the Internet. It provides a state-of-the-art, Web-based enterprise portal with content specifically designed for EH&S professionals. To provide users with the information they require, the portal provides centralized, role-based, context-driven Web access to all the information, applications, tools, and services they need to do their jobs.

Finally, the open connectivity of mySAP PLM allows companies to integrate mySAP PLM with non-SAP and SAP solutions. That means companies can use EH&S capabilities in such areas as human resources, finance, and supply chain management, boosting the efficiency and effectiveness of EH&S efforts across the enterprise.

Full Integration to Full Information

mySAP PLM allows users to access and use all relevant product, plant, process, and safety information – at any time. The integration of EH&S with the other solutions of mySAP Business Suite broadens a company's ability to gather accurate, up-to-date information from such sources as purchasing, human resources, and materials management. This saves time in data searches, improving the ability to make accurate decisions, and enables the proactive health and safety management of a company.

EH&S integration with mySAP PLM helps companies in many overall ways, as well:

- Because accurate, comprehensive information flows smoothly across an enterprise, mySAP PLM improves cooperation between departments.
- mySAP PLM automatically updates dependent data – for example, a new standard operating procedure when hazardous substance data changes – saving time and eliminating errors.
- mySAP PLM can automatically track safety measures.
- mySAP PLM allows for complete documentation of all processes, such as incident and accident management.
- The use of worldwide, standardized procedures and documentation ensures consistency and completeness.
- Actual, up-to-date company data improves the cost-effectiveness of analyses.

Complete and Personalized Support

mySAP PLM supports EH&S professionals completely. It uses innovative approaches to deliver company-wide, integrated information, such as the role-based enterprise portal and a highly flexible specification management database. To help users complete day-to-day tasks and handle long-term projects,

mySAP PLM provides all the information and tools they need to use that information, including complete sets of EH&S-specific functions. In addition, mySAP PLM delivers a full body of EH&S knowledge and content to enhance the profession and help professionals excel at the things they do every day:

- **Enhance employee safety**

To identify and minimize risks to employee health, as well as comply with government safety regulations, mySAP PLM provides complete industrial hygiene and safety and occupational health functionality. (For more details, see the section entitled “Key Elements of Environment, Health, and Safety with mySAP PLM.”)

- **Support preventive healthcare**

To track employee healthcare, as well as monitor the results of medical examinations and diagnosis, the occupational health functions include a medical service that records all relevant health surveillance protocol data, such as medications, consultations, and follow-up appointments. It supports all other functions necessary to occupational health professionals – from proposal lists and scheduling to creating occupational health documents, identifying exposure groups, and a full array of other occupational health-specific functions.

- **Store and manage EH&S data**

The solution’s basic data and tools include a highly flexible specification management database that enables companies to track substances, exposures, packaging, dangerous goods classifications, and waste codes. Data can easily be changed and logged to reflect legal changes. Environmental protection and work safety data can be made available across the company. A key element is EH&S Expert, a rule-based expert system that calculates secondary data. EH&S basic data and tools also provide comprehensive search and output functions for quick access to information.

- **Ensure product safety and hazardous substance management**

In addition to the specification management database – a major component of EH&S – mySAP PLM helps ensure product safety with specific product-safety functions. These include automatic product composition calculation, rule-based material safety

data sheets (MSDS) authoring using EH&S Expert, automatic MSDS generation, global label management capabilities, and MSDS shipping. Other elements of hazardous substance management include a substance information search system, blocking and releasing hazardous materials for purchasing, and hazardous substance inventory reports.

- **Manage documents**

With mySAP PLM, users can produce, maintain, and manage EH&S documents of all kinds, including MSDS, transport emergency cards (tremcards), standard operating procedures (SOPs), and labels. Flexible product safety functions let users create reports manually, or specific events can automatically trigger reports, such as a MSDS when a customer buys a product. When specification data changes, the solution creates and ships reports automatically. This ability, combined with a central information repository, makes mySAP PLM especially well-suited to hazardous-substance management.

- **Track dangerous goods**

mySAP PLM stores and manages regulatory data, groups dangerous goods classifications, performs dangerous goods checks during transport, and easily creates and maintains dangerous goods documentation. This data can be automatically supplied to other solutions for use in delivery notes and packing lists. The solution checks to see which transport – road, rail, sea, or air – is appropriate, and ensures shipments leave the company premises only if they comply with regulations.

- **Manage waste disposal**

mySAP PLM handles the efficient disposal of both hazardous and nonhazardous waste, ensuring compliance with all relevant waste disposal laws and regulations. Waste basic data management features assist users to manage waste generators. Other functions help users create hazardous waste permits, choose suitable disposal firms, allocate disposal costs among internal departments, and ensure authorized waste quantities are not exceeded. And users can quickly create all required disposal documents and reports, such as waste manifests, waste approval, waste life-cycle analysis, waste management concepts, or business management reporting.

KEY ELEMENTS OF EH&S WITH mySAP PLM

The key functional elements provide the features and functions that enable users to manage the entire range of EH&S processes and give them the ability to react quickly and comply with new and changing regulations. The key elements of EH&S within mySAP PLM are:

- Basic data and tools
- Industrial hygiene and safety
- Occupational health
- Product safety
- Hazardous substance management
- Dangerous goods management
- Waste management

Basic Data and Tools

mySAP PLM provides the text, reporting, and label management capabilities, as well as the databases, the expert system, and the EH&S Web interface to fully support EH&S information requirements.

The central capability of the solution's basic data and tools area is specification management, which employs a highly flexible database tool for managing the property data of numerous EH&S-related data objects, including substances, agents, dangerous goods classifications, waste catalogs, and packaging material specifications. The flexibility of managing the database extends to standard objects, which can be enhanced by users with no limitations. The specification management database is also used in recipe management processes.



Figure 49: Specification Management with mySAP® PLM

To determine risk and calculate the composition of a product, basic data and tools include **EH&S Expert** and **EH&S Easy Expert** to provide rule-based expert system calculations of secondary data, such as risk classification and risk and safety phrase determination.

The **specification information system** provides quick search capabilities using simple or complex criteria for each specification object.

EH&S textual information is managed by a library-based **phrase management system**, which provides a standard interface to phrase library content providers. **Report management** enables any EH&S-related document, such as MSDS, standard operating procedure, tremcards, hazardous substance labels, or waste manifests, to be created and stored in the document database. **Global label management** manages different types of labels that can be related to EH&S processes or other business processes.

The additional available **EH&S Web interface** provides Internet access to the EH&S document database, which contains MSDS, tremcards, and labels.

Industrial Hygiene and Safety

Several key features of EH&S within mySAP PLM support the full range of industrial hygiene and safety processes. These begin with the **work area management** feature, which is at the heart of the industrial hygiene and safety capabilities in mySAP PLM.

The work area provides a central location and reference point in which to conduct numerous processes, including these:

- Conducting, evaluating, and managing risk assessments
- Recording agents and maintaining, evaluating, and managing an exposure log
- Organizing, performing, and evaluating site inspections
- Generating and managing standard operating procedures
- Organizing, evaluating, and managing health and safety briefings
- Managing incidents and accidents, including the creation of legal reports and statistics and managing the incident and accident log

The work area allows industrial hygiene and safety professionals to exactly map an organization in terms of industrial hygiene and safety information and processes. Work area integration with the other mySAP PLM capabilities allows users to create references to existing structures, such as functional location structures in mySAP PLM's enterprise asset management capabilities, storage location and work center structures in mySAP Supply Chain Management, and organizational management objects in mySAP Human Resources.

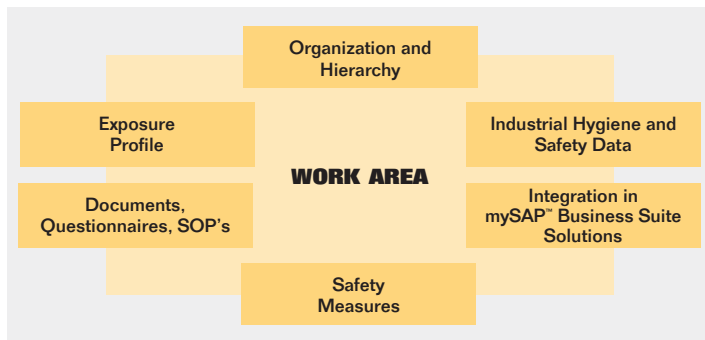


Figure 50: mySAP™ PLM Work Area and Related Information

Work area management also enables you to create complete **exposure profiles** and manage the safety information of specific physical work areas, use questionnaires and checklists to standardize and manage data collection, and quickly roll out safety measures. A pattern feature leverages safety data across work areas.

Special **pattern work areas** are work areas that do not have a reference to a location or organizational level. The pattern function uses the principle of multiplication to create standardized exposure profiles and risk assessments.

Risk assessment is a major process within EH&S. It allows users to determine, analyze, and prevent all risks and exposures in employees' working environment.

The solution can track risks by many means, such as while taking control measurements or setting up exposure profiles. If a risk is detected in a physical work area, the solution triggers a risk assessment and helps users carry it out with the activity scheduling, documents, and measurements needed.

When assessing risks, including the specific rating, severity, and probability or risk of exposure, users can use a number of different methods to assign values. These include standard methods, such as measured value-reference value comparison, the evaluation questionnaires, or comparison with standardized criteria when risks cannot be quantified by measurements. A measurement management feature enables the processing and management of more detailed analyses.

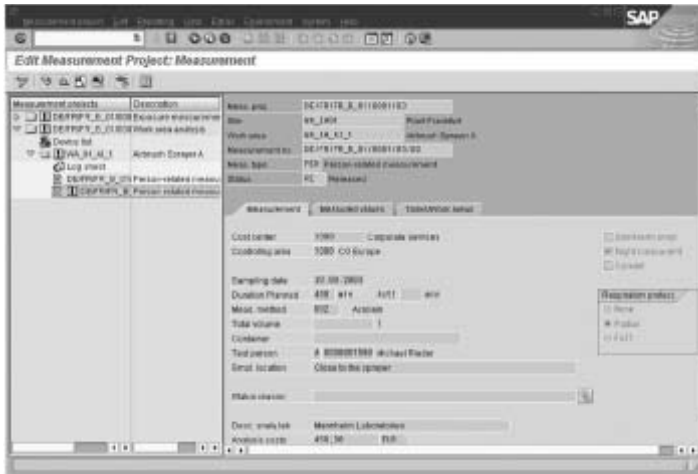


Figure 51: Measurement Projects with mySAP™ PLM

As part of the solution's risk management functionality, an **exposure log** enables users to systematically track all agents in the enterprise. The exposure log can track all kinds of exposures, such as hazardous substances, noise, lighting, and climate. Exposure log data is used to perform evaluations and create reports.

For handling and working with hazardous substances and agents, users can create comprehensive SOPs with mySAP PLM. Agent information in the SOP is obtained from the specification database, whereas workplace, installation, process, or task data is derived from the work area database.

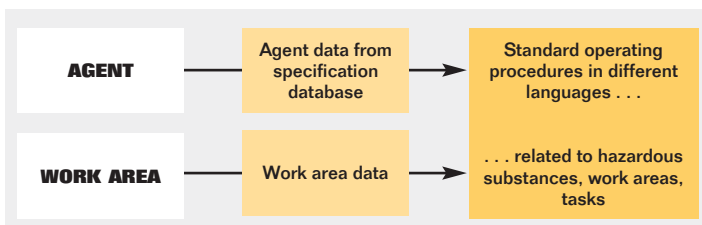


Figure 52: Information Sources for Standard Operating Procedures

Because SOPs are integrated into the document management capabilities of mySAP PLM, the solution provides full status management and document versioning. SOPs can also be created in any language by applying reusable phrases in multiple languages.

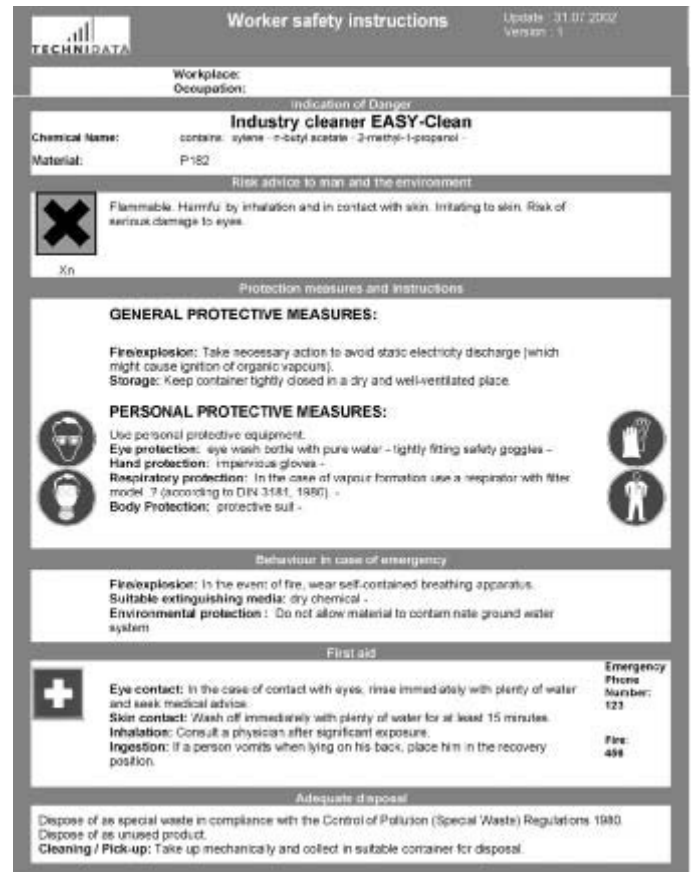


Figure 53: Example of Standard Operating Procedure

With **document assignment** features, users can assign pictures, maps, and documents stored in the document management system directly to work areas or to incident and accident log entries.

To help companies do everything they can to prevent accidents, mySAP PLM provides an **incident and accident log** to record and manage incidents and accidents. Users can generate and manage accident reports – for example, Occupational Safety and Health Administration (OSHA) reports – then initiate safety measures and track their status. With the classification system in mySAP PLM, users can customize the log with other types of information, such as costs of medical treatments, repairs, and prevention measures.

INJURIES AND ILLNESSES INCIDENT REPORT (EXAMPLE)

This Injury and Illness Incident Report is one of the first forms you must fill out when a recordable work-related injury or illness has occurred. Together with the Log of Work-Related Injuries and Illnesses and the accompanying summary, these forms help the employer and OSHA develop a picture of the extent and severity of work-related incidents.

Within seven calendar days after you receive information that a recordable work-related injury or illness has occurred, you must fill out this form or an equivalent. Some state workers' compensation, insurance, or other reports may be acceptable substitutes. To be considered an equivalent form, any substitute must contain all the information asked for on this form.

According to Public Law 91-596 and 29 CFR 1904, OSHA's record-keeping rule, you must keep this form on file for five years following the year to which it pertains.

If you need additional copies of this form, you may photocopy it as many times as necessary.

COMPLETED BY: WOLFGANG MILLER
TITLE: HIS PROFESSIONAL
PHONE: EXT. - 7787
DATE: 11/25/2003

OSHA's Form 301

Attention: This form contains information relating to employee health and must be used in a manner that protects the confidentiality of employees to the extent possible while the information is being used for occupational safety and health purposes.

 U.S. Department of Labor
Occupational Safety and Health Administration

INFORMATION ABOUT THE EMPLOYEE

1) Full name Michael Rieder
2) Street 108 Preston Road
City: Somerville
State: MA
ZIP: 60486
3) Date of birth 05/10/1959
4) Date hired 01/01/1994
5) Sex Male

INFORMATION ABOUT THE PHYSICIAN OR OTHER HEALTH CARE PROFESSIONAL

6) Name of physician or other healthcare professional
Dr. Maria Browning
7) If treatment was given away from the work site, where was it given?
First-aid area
8) Was employee treated in an emergency room?

☐ Yes
☒ No

9) Was employee hospitalized overnight as an in-patient?

☐ Yes
☒ No

INFORMATION ABOUT THE CASE

10) Case number from the log DEMO_02
(Transfer the case number from the log after you record the case.)
11) Date of injury or illness 1/10/2003
12) Time employee began work 14:30:00
13) Time of event 12:00:00

14) What was the employee doing just before the incident occurred?

Describe the activity, as well as the tools, equipment, or material the employee was using. Be specific.
Examples: "Climbing a ladder while carrying roofing materials." "Spraying chlorine from hand sprayer." "Daily computer key-entry."

15) What happened? Tell us how the injury occurred.
Examples: "When ladder slipped on wet floor, worker fell 20 feet." "Worker was sprayed with chlorine when gasket broke during replacement." "Worker developed soreness in wrist over time."

16) What was the injury or illness? Tell us the part of the body that was affected and how it was affected; be more specific than "hurt," "pain," or "sore."
Examples: "strained back," "chemical burn, hand," "carpal tunnel syndrome," "hearing loss."

17) What object or substance directly harmed the employee?
Examples:
"concrete floor," "chlorine," "radial arm saw." If this question does not apply to the incident, leave it blank.

18) If the employee died, when did death occur?

Date of death:
Completed by: Wolfgang MILLER
Title: HIS Professional
Phone: EXT. - 7787
Date: 11/25/2003

Figure 54: Example of OSHA Accident Report Form

The capability offers streamlined and flexible **reporting and statistical analysis** features. For example, hand-written accident reports are a thing of the past. With mySAP PLM, users can generate them with the push of a button. They can produce statistical reports to determine where there is an increased risk of accident. Users can access the 1,000-employee quota or the one-million-hour quota statistical tools, and they can export log data into other statistical programs or the data warehouse of SAP Business Intelligence.

mySAP PLM also provides a comprehensive electronic **injury and illness log**, meeting legal requirements to securely maintain confidential information for several years. The injury and illness log records far more information than traditional first-aid logs. Also, data can easily be transferred to incident and accident management, allowing first-aid centers and accident administration to work together more effectively.

To provide a central information source for industrial hygiene and health professionals, mySAP PLM includes an **industrial hygiene and safety information system**. It provides fast access

to the information typically needed on a regular basis, including standard reports, such as the hazardous substance register, accident statistics, and risk assessment overview reports. Easy to use, the system even comes with a preconfigured common reports template: just click and go.



Figure 55: Example of Enterprise Portal for Occasional Users

The mySAP PLM enterprise portal enhances the role of EH&S professionals with the content-rich Business Package for Environment, Health, and Safety. It includes convenient iViews, small applications that assemble information from various sources that make it fast and easy for users to:

- Search for substances
- Search for materials with substance relationship
- Maintain a personalized substance portfolio
- Search and display substances with certain properties
- Keep EH&S documents like MSDS and safety instructions
- Search in work areas

For any given task, mySAP PLM allows users to define protection goals and **safety measures**. With this capability, users can prevent risks and exposures by triggering and tracking various measures, which can be technical, organizational, or people-related measures.

Site inspections play a key role in any systematic, preventive company safety policy. Their main purpose is to determine to what extent company-wide health protection and safety standards have been fulfilled. Site inspections also emphasize the importance of health and safety to managers and employees. mySAP PLM provides all the EH&S tools necessary to fully plan, perform, and document site inspections.

Regulations often require companies to train employees in the basics of workplace safety and first aid. This is frequently achieved through **on-site health and safety briefings**. The health and safety briefing capabilities of mySAP PLM draw on the comprehensive training and event management functions in mySAP Human Resources. The solution identifies exactly which employees require a health and safety briefing. It also tracks employees and flags those who need to participate in updated sessions.

The **questionnaire** is a familiar tool to industrial hygiene and health professionals. mySAP PLM provides a generic tool for creating specific questionnaires. These include site inspections and risk assessment checklists, which can be created on the basis of a multilingual question catalog. Questionnaires can be filled out directly in the system, or they can be printed, completed by hand, and then scanned back into the system.

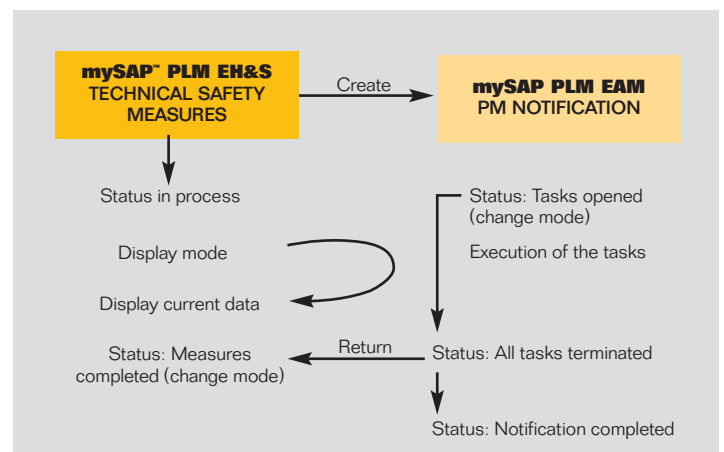


Figure 56: Integration of EH&S with the Enterprise Asset Management Capabilities of mySAP PLM

Occupational Health

The key elements of occupational health offered by mySAP PLM are designed to equip occupational health professionals with the tools and functions they need to provide for the health and well-being of employees. Few people have a more important goal.

A primary EH&S tool within mySAP PLM to achieve this objective is the **health surveillance protocol** — a set of medical examinations and tests a person takes so risks can be recognized and dealt with. These exams are given at regular intervals and are required by law in most countries. The health surveillance protocol functions in mySAP PLM allow users to specify and assign many different protocol parameters, such as the category of the protocol (a new hiring), the type of protocol (exposure-, task-, or age-related), protocol trigger levels, protocol frequency and regulations, and other necessary functions.

A protocol **proposal list** is generated in a report that determines the relevant health surveillance protocols and triggers for a particular person based on the actual exposure situation.

Occupational health specialists are also responsible for in-company **first-aid treatments**, as well as training people to render first aid. To record and document the treatments, mySAP PLM utilizes an **injury and illness log**, which can be transferred to incident and accident management for administrative purposes.

Occupational health professionals must be able to document and evaluate all examination results and diagnoses in a reliable manner. Using the comprehensive **medical service** capabilities of mySAP PLM, they can record all relevant health surveillance protocol data, including examinations, medical test results, diagnoses, restrictions, consultations, medications, and appointments.

All medically relevant information, such as batch numbers, is easy to enter, and upcoming appointments can be entered automatically. **Vaccination management** functions allow for the documentation of third-party vaccinations to provide a full patient history record. The **scheduling of medical services** for a person includes the option of sending invitations and questionnaires to people with appointment dates. It is also possible to complete a questionnaire while editing a medical service (see Figure 57).

To enable users to flexibly map all the required occupational health processes in a company, the **master data** can define objects individually, including the exposure group, agent, examination, health surveillance protocol, question catalog, and questionnaire.

An **exposure group** is a group of people who are exposed to the same agent and the same agent exposure ratings, for example, a reference value of less than 10%. mySAP PLM makes it easy to assign exposures and protocols to persons.

Also supporting occupational health professionals in the documentation and evaluation of examinations and results are a **selection of reports**, including legally required reports, such as workers' compensation association statistics. In addition, with the data contained in the medical service, several specific **occupational health documents** can be created, such as medical certificates, restriction lists, and the protocol log. And **brief consultation** enables users to easily document routine, day-to-day medical tasks.

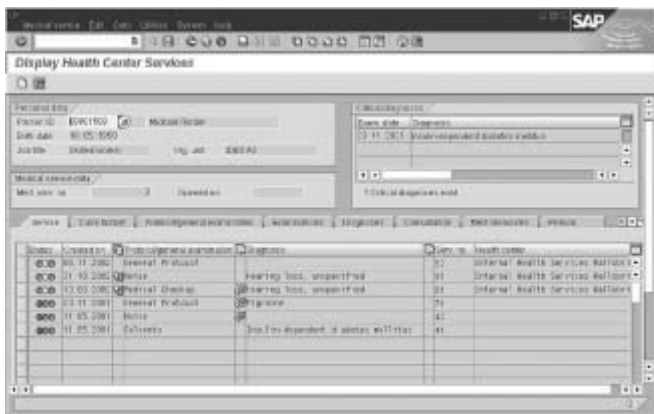


Figure 57: Screenshot of the Edit Medical Service

Product Safety and Hazardous Substance Management

Companies that buy, use, produce, or sell hazardous substances must protect their employees, their communities, and the environment from the hazards these substances could create. Around the world, many laws and regulations govern the use, production, and distribution of hazardous substances, which is why efficient hazardous substance management is of ever-growing importance to companies worldwide. However, the benefits outweigh the cost. Rigorous safety and environmental standards can improve a company's image, increase market opportunities, and reduce costs and risks.

Companies can achieve these standards and manage complex requirements for health and safety in the environment and at work with mySAP PLM. The solution is designed to optimize hazardous substance management and other business processes in a company, facilitate automatic compliance with regulations, and minimize risks.

Whether a company only buys hazardous substances, only makes or uses them, or does a combination of both, mySAP PLM provides two functional areas to manage them. First, for enterprises that purchase hazardous substances, such as processing companies, functions are grouped to provide **hazardous substance management**. Second, for enterprises that produce and sell hazardous substances, functions are grouped to ensure **product safety**.

Hazardous substance management with mySAP PLM helps users manage the following tasks:

- Checking for alternative substances before purchasing a hazardous substance
- Blocking hazardous materials in purchasing
- Packaging and identifying hazardous substances in accordance with regulations
- Storing hazardous substances safely
- Managing up-to-date MSDS and those that come from suppliers of hazardous substances
- Determining all hazardous substances that occur and creating a hazardous substance inventory
- Determining and assessing risks to protect employees
- Monitoring hazardous substance concentrations in the workplace
- Arranging, undertaking, and monitoring the success of protection measures
- Creating SOPs for work areas and substances
- Conducting employee health and safety briefings about hazardous substance handling

If a company uses, produces, or distributes hazardous substances, mySAP PLM enables it to carry out the following tasks:

- Classifying and identifying the hazardous substance
- Creating labels, MSDS, and other documents
- Shipping MSDS before delivery or when the hazardous substance is delivered
- Subsequent shipping of MSDS that have been changed

The Features of Hazardous Substance Management

mySAP PLM makes available a complete range of features and functions to manage every aspect of the hazardous substances that a company purchases. A highly flexible **substance management** database manages the property data of substances and agents. To enable a company to always know where agents are in the enterprise, which hazardous substances are used, and where they are kept, the solution provides **work areas** that store all hazardous substance and agent data and any other related information.

mySAP PLM manages the process of importing **MSDS**, which are delivered as paper documents or electronically into the EH&S document database. MSDS are linked to the purchased hazardous substances.

The solution's rule-based **expert system** functions calculate secondary data, such as risk classification, risk and safety phrase determination from substance properties, as well as the composition of a product.

All **hazardous substance-related documents**, including MSDS, standard operating procedures, tremcards, and hazardous substance labels, are created and stored in the document database.

Hazardous substance inventory functions report the actual stock level and other details of the hazardous substances in a company. **Hazardous material release** functions can block or release material for purchases based on its hazard properties.

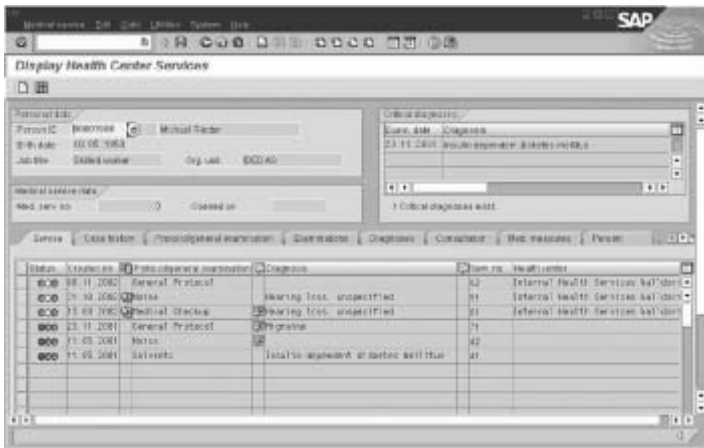


Figure 58: Hazardous Substance Register

Simple and complex criteria can be used to conduct data searches using the **substance information system**. The results can be displayed online in a list view, or they can be downloaded to Microsoft Excel.

The Features of Product Safety Management

mySAP PLM includes the features and functions to ensure product safety throughout a company.

Any company that uses, produces, or sells hazardous substances has to enter and manage enormous amounts of data. Therefore, this data must be easy to enter and managed centrally. The solution's **specification management database** makes this possible.

The specification database is at the core of the EH&S capabilities of mySAP PLM. Here, users enter and describe substances and agents of any sort as specifications, which are identified by name, number, and chemical formula. Users can also store user-defined properties, such as compositions, measured values, and texts. A single interface – **the specification workbench** – makes it easy to enter, display, edit, and use all specification data. Regulatory data from content providers can also be entered into the database.

The **product composition** of substance ingredients is calculated automatically from the BOM in production planning. **Rule-based MSDS authoring** through the expert system also supports the calculation of the product composition. In addition, MSDS can be created automatically and stored in the document database for the product.

MSDS shipping functions allow MSDS to be shipped to customers via integration with the Sales & Distribution component. Upon completing all changes to the MSDS, the current version is shipped to any recent recipient. Manual shipping is also supported.

A comprehensive **global label management** capability manages many different types of labels, whether they are related to EH&S or to other products. The labels can be printed directly from the business processes, for example, from outbound delivery.

Product safety is further supported by the **EH&S Web interface**, which provides access to the document database containing all MSDS, labels, and tremcards.

Dangerous Goods Management

Dangerous goods are tightly regulated in many countries. To comply efficiently and accurately with these regulations, a flexible, central information source is required.

mySAP PLM provides an EH&S **specification management database** in which to enter and update all regulatory data concerning the dangerous goods in a company.

Using **dangerous goods classification** functions, users can group the classification of a product together for each regulation and then link this classification to any material that is identically classified.

The **dangerous goods checking** functions, which are integrated with the sales, distribution, and transportation functions, allow users to check against regulations, depending on the mode of transport and the route. For example, users can check packing requirements, transportation approval, mixed loading, and limited quantities.

All **dangerous goods documents** can be stored and managed in the EH&S document database. The solution can automatically supply document data to other solutions for use in creating and printing delivery notes and packing lists. Users can also easily create and print tremcards for road and inland waterway transportation.

Figure 59: Example of a Tremcard

Waste Management

The proper, efficient disposal of hazardous and nonhazardous waste is of major concern to companies and governments. Local, national, and international waste disposal laws and regulations require companies to put in place stringent hazardous and nonhazardous waste management and disposal standards and methods. mySAP PLM offers comprehensive EH&S waste management features and functions to help companies meet or exceed these standards.

The **waste basic data management** tool helps companies manage waste generators within the enterprise. In addition, it contains the functions to manage partner relationships with waste transport companies, disposal companies, and authorities. Waste basic data management tools also manage all disposal channels.

Waste disposal management tools allow companies to process on-site and off-site disposal. They enable the transportation of hazardous waste as dangerous goods. And they manage disposal documents in accordance with the hazardous and nonhazardous waste disposal regulations integrated into mySAP Supply Chain Management.

mySAP PLM's waste management functions and features also include the definition and tracking of all **hazardous waste permits** approved by authorities and issued to a company.

To enable companies to always know where waste is located, the type of waste, and how much there is of it, mySAP PLM provides **waste inventory management** functions so companies can track all waste generators, waste movements in and out of the company, and complete inventory by quantities.

Finally, the cost of waste must be considered. mySAP PLM **waste-costing** capabilities help companies efficiently manage waste disposal costs and settlements. mySAP PLM provides the tools that determine the most cost-effective waste storage and disposal options.

NEW AND ENHANCED EH&S FEATURES

SAP has added many important functions and processes to all key EH&S functions of mySAP PLM. Particular emphasis is placed on integration and usability. The enhancements are available in SAP R/3 Enterprise PLM Extension 2.00 and are new for 2004.

The new enhancements in industrial hygiene and safety and occupational health add business value because they:

- Simplify the editing of exposure ratings for controlling health surveillance protocols
- Expand data retrieval within risk assessment
- Map the complete measurement management process in industrial hygiene and safety
- Assign costs to industrial hygiene and safety measures
- Enhance the editing options for incidents and accidents
- Allow a personal exposure log to be created
- Support the brief consultation and vaccination processes in occupational health

Enhancements in Industrial Hygiene and Safety

The enhancements for this year include the following:

- **Integration of exposure profile ratings**
Users can now transfer exposure ratings for the work area to respective infotype 1403 organizational management objects, such as exposure group and position. Data is entered in one, central location, simplifying the process and eliminating redundancies.
- **Risk assessment: assessing tasks**
Users can now also assess tasks in organization management.
- **New measurement management component**
This feature allows users to monitor different agent types. They can plan measurements (or samples) and evaluate measured values, helping to minimize risks while complying with regulations. Users can store and manage all measurement-related data easily and flexibly, perform advanced project planning, evaluate and compare measurement values, determine overexposures, and produce documents, such as measurement plans and reports, log sheets, and personal monitoring reports.

- **Assigned costs**

Costs can now be assigned to industrial hygiene and safety measures so companies can accurately estimate the total costs.

- **Work area management**

Users can now determine an overall exposure rating for work areas, as well as a received rating from the exposures one hierarchical level below the work area in question.

- **Incident and accident management**

Users can now create period-based accident reports, such as those prescribed by the OSHA in the United States (OSHA 300 and OSHA 300A documents). They can also assign root causes to an accident or incident, allowing cause analyses to be more detailed. And a reportable event indicator has been added to identify incidents in which no persons were injured.

- **Personal exposure log**

Risk assessments can now specify the person to whom a measurement applies. Users can then create an overview report of all measurement data for all work areas in which the person was employed.

Enhancements in Occupational Health

Occupational health enhancements for this year include a vaccination service and a brief consultation service. The **vaccination service** provides a quick, easy way to enter all medical information, such as batch numbers, with an automatic generation function for follow-up appointments. Third-party vaccinations can also be documented. The **brief consultation service** allows users to document and evaluate routine, daily medical tasks, such as providing medication or advice by phone, e-mail, or in person.

European Union Directive 95/46/EC

Member states are now required by law to monitor and document all access to sensitive personal data. mySAP PLM fulfills the requirements of the directive with programs available for incident and accident management, the injury and illness log, and for occupational health.

Enhancements in Basic and Logistics Functions

The new enhancements in basic and logistics functions improve functions and processes in basic data and tools, product safety, dangerous goods management, and waste management. These enhancements add business value because they:

- Simplify the editing of specification data
- Accelerate the creation of consistent labels
- Make it easier to create document templates
- Create legally required reports quickly and easily
- Handle dangerous goods classes 1 and 7
- Clarify sets of rules for the EH&S expert rule engine
- Simplify waste management processes to save costs

The enhancements for this year include the following:

■ New specification workbench

The workbench allows users to edit specification data from a single interface. Users can also use tables to enter multiple data records.

■ New global label management functions

The creation of suitable labels is now simplified and automated. The enhancements simplify logistics data and support users who are not familiar with legal requirements for labels.

■ Microsoft Word integration

Microsoft Word is now integrated into SAP GUI and contains numerous new functions, including the ability to edit document templates from a single interface.

■ Easier legal reporting

Extractors from SAP Business Intelligence now make it easier to create special, legally required EH&S reports, such as inventory lists for fire department, inventory lists according to accident regulations, annual dangerous goods transportation reports, and hazardous substance inventories. It also supports the Superfund Amendments and Reauthorization Act (SARA) for reports SARA 311 and SARA 312 in the United States.

■ Dangerous goods classes 1 and 7

mySAP PLM now enables companies to manage dangerous goods classes 1 and 7 (explosive and radioactive dangerous goods).

■ Explanation component in EH&S Expert

The enhancement helps users understand complex sets of rules.

■ Integration with SAP for Utilities

The waste management functions in mySAP PLM can now be used by the waste management component in SAP for Utilities, optimizing processes and cutting costs.

■ Open interface for content providers

Regulatory data from content providers can be entered via the EHS Open Content Connector using a XML-based interface.

ENVIRONMENT, HEALTH & SAFETY WITH mySAP PLM: FEATURE OVERVIEW

Feature Group	Feature
Basic data and tools	■ Specification management ■ Phrase management ■ EH&S Expert and EH&S Easy Expert ■ Report generation and report management ■ EH&S Web interface ■ Global label management ■ Specification information system
Product safety	■ Specification management ■ Automatic product composition calculation ■ Rule-based MSDS authoring ■ Automatic MSDS generation ■ Global label management ■ Material safety data sheet shipping ■ EH&S Web interface
Hazardous substance management	■ Substance management ■ Material safety data sheet import ■ Rule-based classification ■ Hazardous substance documents ■ Release of hazardous materials for purchasing ■ Hazardous substance inventory ■ Substance information system
Dangerous goods management	■ Dangerous goods regulatory data ■ Dangerous goods classification ■ Dangerous goods checks ■ Dangerous goods documents
Waste management	■ Waste basic data management ■ Hazardous waste permit ■ Waste disposal ■ Waste inventory management ■ Waste costing

Feature Group	Feature
Industrial hygiene and safety	■ Work area data management ■ Work area pattern ■ Work area integration ■ Risk assessment ■ Measurement management ■ Document assignment ■ Incident and accident management ■ Standard operating procedure ■ IHS information system ■ Safety measures ■ Health and safety briefing ■ Questionnaires ■ Site inspection
Occupational health	■ Medical service ■ Site inspection ■ Proposal list ■ Scheduling ■ Injury and illness log ■ Brief consultation ■ Vaccination ■ Questionnaires ■ Occupational health documents ■ Exposure group ■ Medical service accounting ■ Reporting ■ First-aid training

mySAP PLM FOR INDUSTRIES

mySAP Product Lifecycle Management is hard at work in thousands of companies in all kinds of industries. The following sections describe the versatility of mySAP PLM in helping companies to achieve their strategic objectives in various industries.

SAP FOR AUTOMOTIVE

The Automotive Industry: An Overview

The automotive industry is a traditional one. Many of the larger manufacturers have been in the business for more than 100 years. Automotive manufacturers compete globally. The trend toward make-to-order manufacturing – filling individual customer requirements rather than make-to-stock – has put new demands on manufacturers and dealers. Customers want what they want when they want it.

Over the years, globalization has led to dramatic consolidation in the industry and has put enormous pressure on automotive companies. It has prompted OEMs to intensify their focus on their core competencies and to outsource more and more production and development functions. That means that automotive suppliers must be able to link their processes and systems with those of the OEMs, as well as those belonging to their own suppliers and development and business partners. This drives the need for increased transparency and real-time information exchange across the entire value chain. Moreover, it pushes collaboration between the industry partners to new levels of sophistication and importance.

Trends in the Automotive Industry

One big challenge in the automotive industry arises from the fact that there are already huge overcapacities in the established vehicle-producing countries. New markets around the globe can be accessed most effectively when an OEM manufactures locally, due to such factors as local content rules, market entry barriers, and taxes. This requires huge investments in the plants of both the OEM and the OEM's suppliers, which must follow the OEM into the field to support local production. This also leads to even larger global overcapacities, along with cost and price pressures.

Thus, globalization has led to dynamic consolidation processes on the supplier side, as well as the OEM and retail side. In their attempts to increase shareholder value and return on assets (ROA), OEMs tend to focus on their core competencies and outsource more asset-intensive production and people-intensive development functions. As such, they are turning into original equipment designers (OEDs), much like OEMs in the high-tech industry.

These OEMs and OEDs concentrate on quality, final assembly processes, and brand management, while passing other responsibilities to business partners across the value chain. As a result, the balance of power changes within the supply and distribution chain.

Manufacturing the products customers demand – and manufacturing them to a high standard of quality – is just an entry point for competitive advantage. Constant product innovation, process excellence, and advanced technologies are key factors for ongoing competitiveness.

The Internet and other phenomena are also having a powerful impact on the automotive industry. The ability to log onto the Internet and order a book from Amazon.com or to configure a PC precisely as a user wants it at Dell.com is dramatically increasing the expectations of consumers. Automotive companies need to understand the impact of these other venues on their own industry and respond accordingly to evolving customer demands.

Increasing competition in the industry means that OEMs and their suppliers must be able to develop more sophisticated products faster than they ever have before. They are striving to create distinctive products through extraordinary design, or they are going back to former best sellers. The new Cadillac Sedan is an example of the former. Ford's Living Legends models are examples of the latter. They are also striving to reduce development cycles, while reacting even faster to individual customer demands.

The stronger dependencies among the business partners bring with them the need to collaborate much more effectively. Effective collaboration will have an enormous impact on the processes and responsibilities of all the partners involved in the value chain.

Pain Points for the Automotive Industry

Pain points for the automotive industry come from many directions, such as the following:

- **Industry pressure**
OEMs face considerable industry pressures from consolidation, mergers, and acquisitions. Size and scale remain critical factors, and leading industry players are driving further consolidation.
- **Switch from make-to-stock to build-to-order**
Companies must be able to provide what customers want instead of selling what is in stock, and that puts increased emphasis on both locate-to-order and build-to-order processes. New products (both cars and services) will be based on customer preferences and configurations.
- **Increasing internal competition**
Ongoing OEM concentration within a multibrand environment leads to increasing internal competition between models. As a result of the consolidation process, few remaining OEMs have to act and compete on a global scale.
- **Overcapacity**
Make-to-stock processes result in overcapacities of more than 24 million units annually – the equivalent of 96 modern assembly plants, according to PriceWaterhouseCoopers in its *The Second Automotive Century* report. Capacity utilization has come down from 80% in 1990 to 69% in 1999.
- **Shrinking time**
Decreasing the amount of time it takes to respond to changes in demand volume is the most important factor for the majority of manufacturers.

■ **Accelerating new product introductions**

The product development life cycle must become shorter, and it needs to support various model launches in parallel. To reduce time-to-market for new products, companies must make effective use of platforms and strengthen the management of common parts.

■ **Electronic components**

Product innovation today is mainly driven by electronic components. The related processes in engineering, with huge amounts of changes in late development phases, as well as in service for upgrade and update purposes, differ greatly from mechanical components. OEMs and suppliers today have difficulties controlling these new processes and need flexible tools to manage complex product structures in combination with functional electronic networks.

■ **Frequency of product design changes**

There is significant pressure to increase the frequency of product design changes. The acceleration and rationalization of new product development cycles is important to OEMs because it is driven by new generations of consumers and the need to meet the differing needs of markets around the globe.

■ **Complexity and customization**

Customization can lead to several thousand variants for a given vehicle model – car body, color, engine, transmission, and interior. A model mix is made up of a number of configured materials and their relationships. This requires the management of high volume of data and complex product structures.

■ Outsourcing

The primary responsibility of OEMs is to establish and foster their brands. As a result, there is an increased trend among OEMs toward outsourcing engineering and manufacturing processes as they focus their core competencies on the overall design of the vehicle. The interdependencies that result from these outsourcing activities require a high level of coordination and cooperation between engineering, manufacturing, and logistics tiers in both planning and execution. Even minor delays at one point in the manufacturing process can have a ripple effect across downstream processes. In an asset-intensive environment, frequently with constrained capacity, such disruptions are expensive.

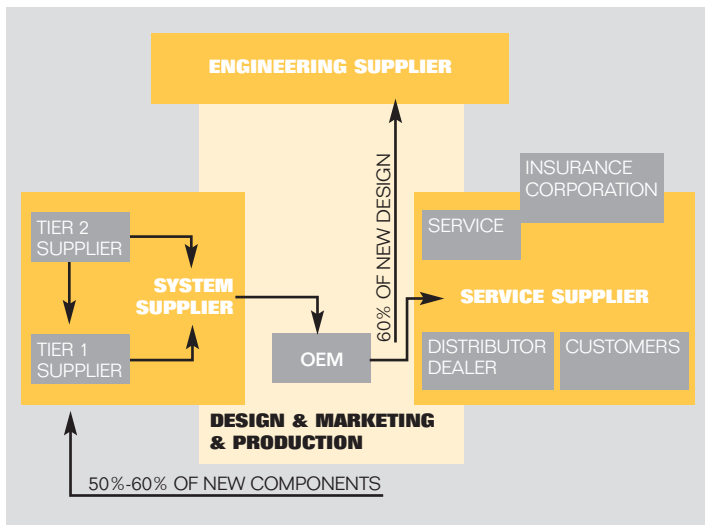


Figure 60: Outsourcing to Business Partners in the Automotive Industry

The Automotive Industry Ecosystem

To understand the kind of product life-cycle management capabilities that an effective automotive ecosystem requires, it helps to know how a number of key processes might work in such an ecosystem. When you understand the tasks and services that the various organizations perform, you can understand how they deliver value to the marketing, planning, engineering, manufacturing, distribution, selling, maintenance, and recycling efforts that are associated with the life cycle of a vehicle.

The automotive value chain contains a significant number of business partners. Globally, there are 13 mega-manufacturers whose revenues exceed U.S. \$100 billion. There are also several thousand suppliers in different tiers and a large number of importers. In addition, there are thousands of dealers in the world's major markets. All of these players are linked in a highly complex network.

Moreover, this entire network is supported by a large number of outsourced services, including engineering and design, transportation, materials handling and storage, asset maintenance, vehicle service, parts, aftermarket accessory sales, and ancillary services, such as leasing, finance, insurance, and rental services.

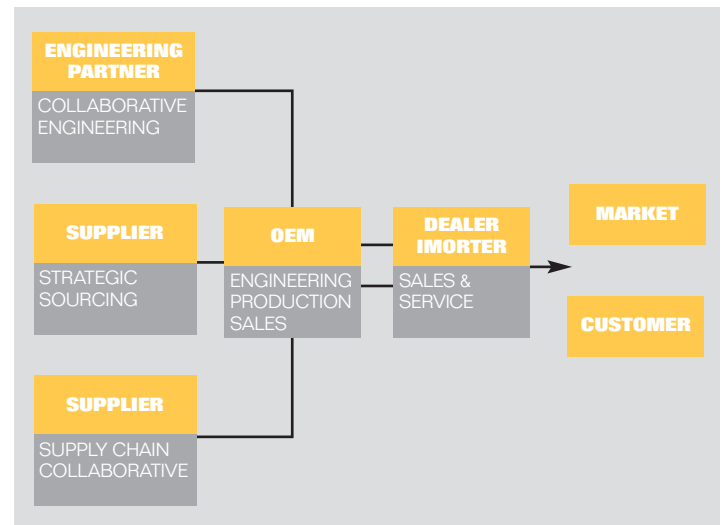


Figure 61: The Automotive Ecosystem

mySAP PLM Today

Increased competition in the automotive industry forces companies to react as fast as possible to the individual needs of their customers to maintain and increase customer satisfaction and retention. Both technology and market innovation are key factors to success.

The SAP for Automotive portfolio of solutions, together with mySAP Product Lifecycle Management, supports the automotive industry by enabling collaboration across business processes and enterprise boundaries.

Innovation is based on efficient project management, knowledge management, and collaboration. mySAP PLM provides a complete product life-cycle management solution – from first idea throughout the production cycle and distribution phase to the maintenance and obsolescence of a product.

SAP for Automotive solutions are tailored to the specific needs of this industry, emphasizing integration and allowing global virtual teams to facilitate collaborative engineering on new product developments.

Collaborative engineering can be executed simultaneously among internal teams, partners, and strategic suppliers. mySAP PLM enables all parties involved to effectively plan, monitor, and document engineering projects. It also provides access to the latest product, asset, and project data, thus increasing the interconnectivity of employees, partners and suppliers throughout the entire product life cycle. Engineering projects can be managed efficiently, both within companies in engineering teams, as well as across company borders, coordinating different engineering partners.

Meeting the Needs of Key Industry Segments

Today, mySAP PLM provides a full range of capabilities designed to help OEMs and suppliers to reduce time-to-market through collaboration.

For **OEMs**, the solution offers:

- Target and requirement management to define specifications that need to be fulfilled in later development phases. By linking targets to specific elements of the functional product structure, every engineer has the actual information on hand to design components or modules. The solution also provides easy ways to monitor targets versus planned and actual values.

- Integrated product and process engineering capabilities that support activities from the early planning phase to product launch and that provide consistent master data throughout the entire product life cycle. The solution tightly integrates engineering and manufacturing processes, including engineering change management for highly configurable products.
- Integration with concurrent costing capabilities that allow cost estimation for different product and production concepts, even if not a single part number has been defined.
- Complete project management functions to plan and track the different steps during the product development phase. This includes planning and monitoring of costs, timelines, and resources, with full integration to all mySAP Business Suite solutions.

For **automotive suppliers**, the solution offers:

- Collaboration with external business partners to share documents and product structures, as well as implement stage-gate processes. This offers the ability to manage commitments and checklists, improving product quality from the beginning.
- An interface to computer-aided design (CAD) and conversion tools makes engineering data available to every employee who needs this information, especially in the early phases. At the same time, developers have access to the most up-to-date product data.
- In the area of environment, health, and safety, mySAP PLM provides complete capabilities to ensure product safety and help companies manage dangerous goods, waste, and hazardous substances. This also includes the requirements for OEM regarding the International Material Data System (IMDS).

- A supplier portal, which simplifies the exchange of documents between a manufacturer and its suppliers and gives suppliers access to information on manufacturers' systems. A standard Web browser connects suppliers to OEMs' systems, enabling suppliers to track orders, deliveries, and settlements, as well as produce shipping notices. This Internet-based approach can be a low-cost alternative or enhancement to the electronic document interchange (EDI) protocol, which is widely used in the automotive industry.

Business Benefits for Automotive Companies

The benefits of mySAP PLM directly address the challenges the automotive industry faces today. They include the following:

- mySAP PLM provides a complete product life-cycle management solution – from first concept throughout the life cycle of the assets, including the maintenance and shutdown of the assets.
- mySAP PLM enables full and effective collaboration across business processes and enterprise boundaries.
- mySAP PLM enables a company, its employees, and its partners to effectively plan, monitor, and document engineering projects, and it provides access to accurate, up-to-date product, asset, and project data.
- mySAP PLM increases the interconnectivity of employees, partners, and suppliers throughout the entire product life cycle. Engineering projects can be managed efficiently, both within companies in engineering teams, as well as across company borders, smoothly coordinating different engineering partners.
- mySAP PLM is powered by back-end integration. It provides access to the newest information from the engineering process, such as the latest specification documents, planned resources for prototyping, or current cost and weight estimates, to pave the way for the ensuing project phases.
- mySAP PLM provides Web-based access, subscription, and notification mechanisms.
- mySAP PLM supports both online and real-time collaboration to shorten rework cycles.

mySAP PLM: New Features for 2004

mySAP PLM is complemented by a number of new features and new components designed to help automotive companies manage integrated processes along the product life cycle.

The **SAP xApp Resource and Program Management (SAP xRPM)** packaged composite application provides a complete picture on planned and ongoing projects, how these projects are doing, and how to assign the right resources to the projects. It reduces project failure rates, speeds up delivery cycles, and reduces personnel costs, which enables a faster time-to-market.

The **SAP xApp Product Definition (SAP xPD)** packaged composite application ensures a complete, efficient, and faster process for new product development and introduction. The solution helps to identify innovative products that match a company's strategy and provides a means of gathering and consolidating product ideas by making use of ideas from all touch points.

The **SAP Master Data Management (SAP MDM)** component stores, augments, and consolidates master data, while ensuring consistent distribution to all applications and systems within the IT landscape. SAP MDM drastically reduces data maintenance costs and accelerates the execution of business processes by ensuring cross-system data consistency.

SAP FOR ENGINEERING, CONSTRUCTION, AND OPERATIONS

The Engineering, Construction, and Operations Industry: An Overview

SAP for Engineering, Construction & Operations (SAP for EC&O) delivers a comprehensive suite of integrated business solutions designed to meet the specific challenges of businesses operating in the following industries.

Plant contracting enterprises perform turnkey projects in which engineering, procurement, and construction (EPC) are key elements, as well as servicing, operating, and maintaining plants. They design and build process-engineering structures, such as power and process plants, manufacturing facilities, transportation structures and systems, and water treatment plants. Such projects are engineering and project management intensive. These enterprises generally execute or manage all phases of an EPC project and are supported by a variety of subcontractors and suppliers. The projects run as design-build-operate (DBO), build-own-operate-transfer (BOOT), build-own-transfer (BOT), and many other types of business models between private and public business partners.

The construction industry includes construction companies, general builders, special trade contractors, and subconstruction businesses. As in plant contracting, the business centers on projects and is performed by one or more general contractors supported by many subcontractors and suppliers. Service and operating aspects are becoming increasingly important for construction companies.

For **home-building** companies, the process often starts with the development of a piece of land and may include the development of the homes to be sold in the new community. Besides land development, other important steps include selling the homes – including the customer-specific configuration of a home – and subsequent processes like pricing. The homes themselves are usually built by subcontractors and are managed as a project by the home builder. Depending on the region and the profile of the company, many levels of prefabrication are possible.

Shipbuilding enterprises perform turnkey projects in the shipbuilding industry. They design and build sophisticated merchant and naval ships, such as container vessels, liquid natural gas carriers, bulk carriers, tankers, or passenger vessels. Shipbuilding projects are engineering and project management intensive, and these enterprises generally execute or manage all phases of a shipbuilding project.

Trends and Pain Points in the EC&O Industry

Numerous trends and pain points are influencing the EC&O industry today. Among them are the following:

- Competitive pressure from global expansion and competition for the lowest cost, fastest time-to-delivery, and highest quality drives companies to seek greater efficiencies and productivity and pursue global sourcing and contracting. It also requires them to closely monitor contract profitability.
- Owners and operators are demanding more and more turnkey projects covering the complete plant or building life cycle, from engineering through construction, operation, and maintenance. This drives contractors to faster and more accurate bidding, reduced fulfillment cycle times, and the efficient management of the volume and mixture of data, documents, and drawings of the plant.
- Enterprises are demanding common business practices across all internal and external offices and divisions. Hence, companies are trying to apply Web-based IT strategies to gain savings and competitive advantage throughout the communication process between business partners, inside and outside of the company.
- Depending on the complexity of a construction project, there is always a considerable risk regarding costs and scheduling.
- Almost half (47%) of all construction spending by owners and operators is actually for renovation. This drives the need to access information about facilities on time and to determine the on-time availability of materials, equipment, and spare parts. Other considerations include complying with regulatory and corporate requirements for managing and approving the work and using efficient and productive renovation processes that result in lower costs, shorter shut-down times, and higher quality.
- Over 70% of all construction contract revenue is subcontracted out. This drives contractors to global sourcing for materials, equipment, services, and subcontracting. Explicit material management and effective planning capabilities are helping to leverage quantities, as well as time-of-delivery, pricing, and discounts.

The EC&O Industry Ecosystem

To understand the kind of product life-cycle management capabilities that an effective engineering, construction, and operations ecosystem requires, it helps to know how a number of key processes might work in such an ecosystem. When you understand the tasks and services that the various organizations perform, you can understand how they deliver value to the planning, engineering, production, construction, maintenance, and service efforts that are associated with the life cycle of, say, a process plant, a cruise ship, or a complex building.

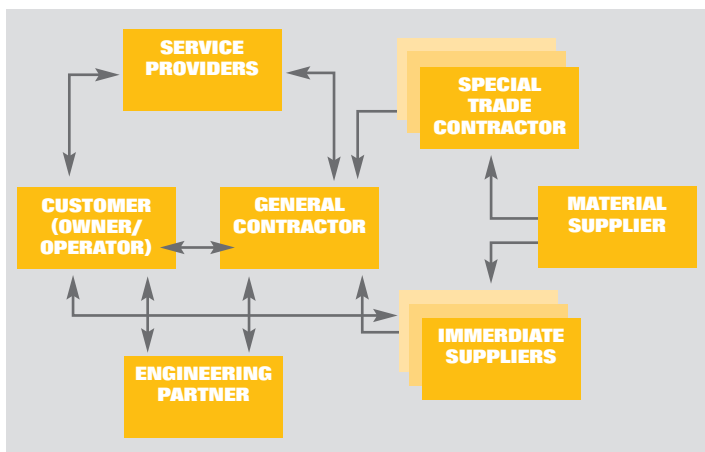


Figure 62: The Engineering, Construction, and Operations Ecosystem

mySAP PLM TODAY

The EC&O industry is currently facing low demand, which has led to constant pressures on revenues and margins – traditionally low in this industry. Only the product leaders and highly specialized companies are in a good position to keep up margins and revenues and to generate lasting shareholder value.

The SAP for EC&O portfolio of solutions, together with mySAP Product Lifecycle Management, supports the EC&O industry by offering a comprehensive project management solution and enabling collaboration across business processes and enterprise boundaries.

Innovation is based on efficient project management, knowledge management, and collaboration. mySAP PLM provides a complete product life-cycle management solution – from first idea through the assembly and construction phase and encompassing maintenance and service during the operation phase.

Collaborative engineering can be executed simultaneously among internal teams, partners, and strategic suppliers. mySAP PLM enables all parties involved to effectively plan, monitor, and document their work, providing access to the latest product, asset, and project data. It increases the interconnectivity of employees, partners, and suppliers throughout the entire product life cycle.

Meeting the Needs of the EC&O Industry

Today, mySAP PLM provides a full range of capabilities designed to help EC&O companies prepare for the challenges they face, including the following:

- Complete project management functions to plan, track, and execute the different steps during project execution. This includes planning and monitoring costs, timelines, and resources, with full integration to all mySAP Business Suite solutions.
- The document management system (DMS) in mySAP PLM integrates all documents in work processes with rich functionality, such as approval and change management, classification, Web-based access, and the ability to link documents to logistical processes. An easy-to-use DMS interface is integrated into the familiar Microsoft Windows Explorer environment.
- Collaboration with external business partners to share documents and structures, as well as implement stage-gate processes, which offers the ability to manage commitments and check-lists, improving quality from the beginning.
- An interface to CAD and conversion tools, which makes engineering data available to every employee who needs this information, especially in the early phases. At the same time, developers have access to the most up-to-date product data.

- Complete environment, health, and safety capabilities to ensure product safety and help manage dangerous goods, waste, and hazardous substances.

Business Benefits for EC&O Customers

mySAP PLM provides a complete product life-cycle management solution – from first concept throughout the life cycle of the assets, including the maintenance and shutdown of the asset.

mySAP PLM enables a company, its teams, and its business partners to effectively plan, monitor, and document engineering projects, providing access to accurate, up-to-date product, asset, and project data.

mySAP PLM increases the interconnectivity of employees, partners, and suppliers throughout the entire product life cycle. Engineering projects can be managed efficiently, both within companies in engineering teams, as well as across company borders, smoothly coordinating different engineering partners.

mySAP PLM is powered by back-end integration. It provides access to the newest information from the engineering process, such as the latest specification documents, planned resources for prototyping, or current cost and weight estimates, to pave the way for the ensuing project phases.

mySAP PLM: NEW FEATURES FOR 2004

mySAP PLM provides a number of new features and enhancements designed to help EC&O companies manage integrated processes throughout the product life cycle. For example, the solution is complemented by the SAP xApp Resource and Program Management (SAP xRPM) packaged composite application, which provides a complete picture of planned and ongoing projects, how these projects are doing, and how to assign the right resources to them. This reduces project failure rates, speeds up the delivery cycles, and lowers personnel costs – rushing time-to-market.

SAP FOR INDUSTRIAL MACHINERY & COMPONENTS

The Industrial Machinery and Components Industry: An Overview

The industrial machinery and components (IM&C) industry is extremely diverse, ranging from industrial manufacturers that make simple components, such as bearings, switches, or relays, to companies whose products are highly complex production machinery. Some manufacturers sell their product directly to end customers; others sell through a distribution network.

Another feature of the industry is its diverse manufacturing models. Companies may use make-to-stock, make-to-order, or engineer-to-order models. Typically, they engage in a combination of these manufacturing models. In addition, after-sales service becomes increasingly important as a key differentiation in the product offerings of companies and is mandatory for increasing customer loyalty and for the additional revenue stream it generates.

In the IM&C industry, key business drivers and respective solutions may vary depending on a company's manufacturing model, business process, or specific value chain. If you accept the proposition that companies must work toward excellence in customer relationship management, product life-cycle management, and supply chain management – customer intimacy, product innovation, and operational excellence – then the type of manufacturing model used will significantly impact how companies strategically rank key business drivers and the impact they have on their value chain.

In make-to-stock companies, for example, solutions supporting business processes that enable operational excellence and customer intimacy are perceived as more important than those enabling product innovation. That is not to say that product innovation is not important to the overall value proposition. But the way it is positioned will differ according to the company targeted by the sales effort.

The same can be said for the other manufacturing models. Make-to-order companies look for solutions enabling business drivers centered on innovation, customer intimacy, and operational excellence. Engineer-to-order companies emphasize customer intimacy, innovation, and operational excellence – in that order.

Trends and Pain Points in the IM&C Industry

The following trends and pain points can be identified in the industrial machinery and components industry:

- **Speed up product innovation cycles**

IM&C companies need to speed up the innovation cycle and reduce production lead times. They need to develop and market the right product at the right time and the right price. Companies that service this market segment thrive on technological expertise and market reputation. Production cycles are often measured in months; sales cycles can take years. They provide products that are crucial to their customers, determining customer success or failure.

- **Margin pressures**

Declining demand for investment goods leads to a lower utilization of capacities, tightening competition. This puts customers in a strong position to squeeze prices and buy from the cheapest supplier. Companies invest only if necessary, and there are strong arguments for the business value of the products.

- **Reduced investment**

The sluggish economic climate is forcing companies to cut costs and investments. Existing capacities are often sufficient to satisfy demand. To persuade their customers to invest under these circumstances, IM&C companies have to create additional value for them and convincingly demonstrate the ROI.

- **Increasingly demanding customers**

Customers expect a faster, more flexible, and more accurate response to their inquiries and needs. They are increasingly demanding greater choice and higher product quality. Factors influencing customers' purchasing behavior include price, innovation, reliability, quality, productivity, life-cycle operating costs, in-service performance, and market service. Long-term relationships can generate lucrative revenue streams for manufacturers in this market segment.

- **Revenue model expansion**

In today's economic environment, IM&C companies are searching for ways to diversify or grow their business to level off peaks and valleys in their revenue streams. One way to do this is by acquiring or expanding product portfolios to round out an the revenue model.

The IM&C Industry Ecosystem

To understand the kind of capabilities that an effective IM&C ecosystem requires, it helps to know how a number of key processes might work in such an ecosystem. When you understand the tasks and services that the various organizations perform, you can understand how they deliver value to the marketing, planning, engineering, manufacturing, distribution, selling, maintenance, and recycling efforts that are associated with the life cycle of a machine.

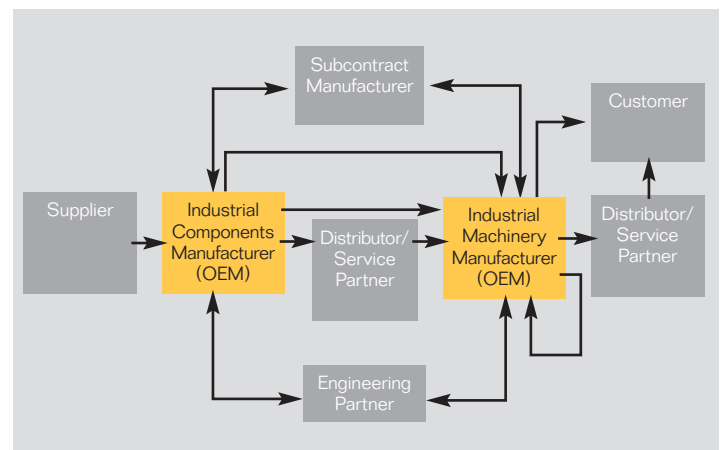


Figure 63: The Industrial Machinery and Components Ecosystem

mySAP PLM Today

Key drivers for all companies in the IM&C industry wishing to gain and retain product leadership are short product innovation cycles and fast time-to-market. To survive and thrive, companies must introduce appropriate products at the appropriate time and price. To drive product innovation, they need to revitalize research and development and invest in new product development.

The SAP for Industrial Machinery & Components (SAP for IM&C) portfolio of solutions, together with mySAP Product Lifecycle Management, supports the IM&C industry by enabling collaboration across business processes and enterprise boundaries.

Innovation is based on efficient project management, knowledge management, and collaboration. mySAP PLM provides a complete product life-cycle management solution – from first idea throughout the production cycle and distribution phase to the maintenance and obsolescence of a product.

SAP for IM&C is tailored to the specific needs of this industry, emphasizing integration and allowing global virtual teams to facilitate collaborative engineering on new product developments.

Collaborative engineering can be executed simultaneously among internal teams, partners, and strategic suppliers. mySAP PLM enables all parties involved to effectively plan, monitor, and document engineering projects. It provides access to the latest product, asset, and project data, thus increasing the interconnectivity of employees, partners, and suppliers throughout the entire product life cycle. Engineering projects and product development can be managed efficiently, both within companies in engineering teams, as well as across company borders, coordinating different engineering partners.

Meeting the Needs of the IM&C Industry

mySAP PLM provides a full range of capabilities designed to help IM&C companies to reduce time-to-market through collaboration, including:

- **Target and requirement management** to define specifications that need to be fulfilled in later development phases. By linking targets to specific elements of the functional product structure, every engineer has the necessary information on hand to design components or modules.
- **Integrated product and process engineering** capabilities that support activities from the early planning phase to product launch and that provide consistent master data throughout the entire product life cycle.
- **Complete project management** functions to plan and track the different steps during all project phases. This includes planning and monitoring costs, timelines, and resources with full integration to all mySAP Business Suite solutions.
- **Collaboration with external business partners** to share documents and product structures, as well as implement stage-gate processes. This offers the ability to manage commitments, improving product quality from the beginning.
- **Interfaces to CAD and conversion tools** that make engineering data available to every employee who needs this information, especially in the early phases. At the same time, developers have access to the most up-to-date product data.
- Comprehensive **quality management** capabilities that meet quality management needs throughout the product life cycle and along the supply chain. The solution supports not only the traditional tasks of inspection planning and processing, but also focuses on prevention and continuous process improvement through collaboration and sustained quality control.

Business Benefits for IM&C Companies

mySAP PLM enables IM&C companies to reap the following benefits:

- mySAP PLM provides a complete product life-cycle management solution – from first concept throughout the life cycle of the built assets, including the maintenance and shut-down of the asset.

- mySAP PLM enables a company, its teams, and its business partners to effectively plan, monitor, and document engineering projects, providing access to accurate, up-to-date product, asset, and project data.
- mySAP PLM increases the interconnectivity of employees, partners, and suppliers throughout the entire product life cycle. Engineering projects can be managed efficiently, both within companies, as well as across company borders, smoothly coordinating different engineering partners.
- mySAP PLM is powered by back-end integration. It provides access to the newest information from the engineering process, such as the latest specification documents, planned resources for prototyping, or current cost and weight estimates, to pave the way for the ensuing project phases.
- mySAP PLM provides Web-based access, subscription, and notification mechanisms.
- mySAP PLM supports both online and real-time collaboration to shorten rework cycles.

mySAP PLM: New Features for 2004

mySAP PLM provides a number of new features and enhancements designed to help IM&C companies to manage integrated processes along the product life cycle. mySAP PLM is complemented by two new solutions: SAP xApp Resource and Program Management (SAP xRPM) and SAP xApp Product Definition (SAP xPD).

SAP FOR HIGH TECH

The High-Tech Industry: An Overview

In the high-tech industry, short product life cycles and rapidly decreasing prices are the norm. This means that the value of components and finished products diminishes rapidly over time and products become obsolete very quickly. As a result, speed and flexibility are critical, and companies that do not have the right products to meet fast-changing customer demands lose their competitive edge fast.

In this environment, many companies are willing – and able – to pursue new business models and to consider new forms of partnership with other companies in their ecosystems. For example, many are pursuing the notion of “focus on what we do best and outsource the rest.” Other innovative approaches include providing self-service capabilities to customers, suppliers, and employees; collaborating on product development and demand planning, forecasting, and replenishment; and knowledge management and e-learning.

The High-Tech Value Chain

The typical high-tech value chain encompasses a significant number of business partners. These can include semiconductor and component manufacturers, distributors and resellers, electronic manufacturing service (EMS) providers, OEMs, and software providers.

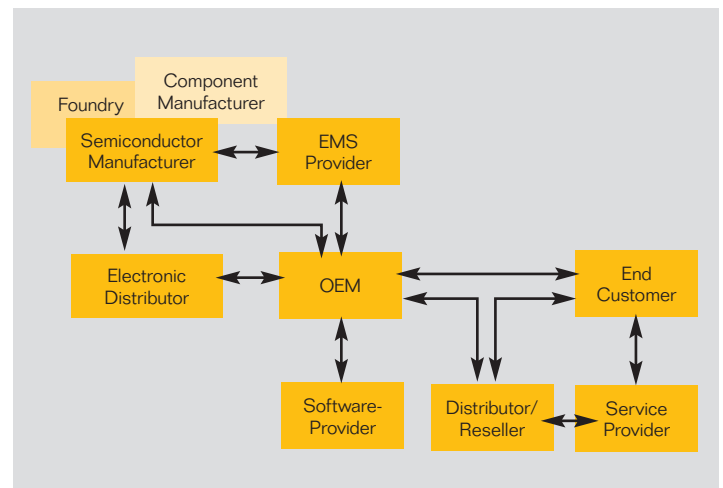


Figure 64: The High-Tech Industry Ecosystem

The key business processes of the value chain encompass product design, opportunity management, semiconductor manufacturing, channel management, component procurement, and product manufacturing, as well as sales-order processing and post-sales service and support. Such value-chain processes bring partners together to interact in a complex network and collaborate to design, plan, produce, deliver, and service a new high-tech product.

Trends and Pain Points in the High-Tech Industry

For the **semiconductor industry**, fluctuations in the business cycle, the asset-intensive nature of manufacturing, ever-shorter product life cycles, increasingly complex global supply chain management, price pressures driven by global competition, and the complexity of products represent the major challenges.

OEMs face shorter product life cycles and growing global supply chain management challenges. In addition, OEMs need to manage changing demand with multitier collaborative forecasting of capacity and component supply. They also need to meet increasing customer demands for customized products and services.

These forces have fueled an emerging trend toward outsourcing and the growth of the **EMS** or contract manufacturing industry — the world's fastest growing high-tech industry segment. However, this growth has contributed to increased competition in this segment. The key metric for success is a company's speed, flexibility, and cost control when manufacturing and delivering products to OEMs.

Along with this phenomenal growth, **EMS providers** are increasingly taking over a larger part of the high-tech supply and value chain. Today, EMS providers are continually expanding beyond their traditional manufacturing core competencies and participating in such business processes as design services, outbound logistics, and after-market services for consumers.

The realities facing the high-tech industry today include:

- **Rapid new product launches and shortening product life cycles**

The average life cycle of a PC is less than nine months, and the cost of a missed PC sale (the contribution margin) is 25%-30%. At the same time, products are increasingly becoming commodities and component prices are decreasing. In such an environment, effective obsolescence management can typically result in savings of 1%-2% of sales annually — that is, U.S. \$100 million to U.S. \$200 million can be saved on sales of U.S. \$10 billion.

- **First-to-market products earn highest profits**

In a world of short product cycles, companies that are first to market gain a significant advantage in ultimate market share and profitability. A network of companies is required to bring a new product to market with speed, and each company involved must work seamlessly with the other companies.

- **Global competition, global supply chains**

Companies are increasingly doing business with partners in other regions of the world to better serve local markets, move faster, and take advantage of cheaper sourcing opportunities. Companies have to be able to manage partners in real time worldwide. Increasingly, low-cost fulfillment is seen as a strategy for fending off competitors — or driving them out of business.

- **Highly complex product offerings and services**

Today, customers want products and services that meet their specific requirements, and they are less likely to accept make-to-stock products and standard solutions. They also want solutions that go beyond the core product to include high-value adjunct services, such as post-sales support. In that environment, companies must manage a growing number of SKUs and partner with other companies to provide customization and wraparound services. Customers are also making choices based not only on products, but also on the experience that they have when buying. Creating a superior customer experience is a growing area of competition.

- **Huge research and development investments**

To stay on top of the changes in technology, high-tech companies must continually make major investments in new technology and better processes. This is especially true in the semiconductor industry, which has huge investments in capital equipment for producing, testing, and assembling chips. Many companies must also invest in advanced research and development labs to ensure that the next generation of technology is being developed.

■ Innovation drives leadership

The rapid pace of advancing technology in the industry means companies must continually introduce new products that take advantage of cutting-edge technology – especially the processing and storage power of the latest semiconductor chips. The top companies in the industry are those viewed as being leaders in innovation.

mySAP PLM Today

To respond to all those challenges, high-tech companies are beginning to move beyond the traditional linear supply chain, with its inherent delays and costs, to create flexible, collaborative partner networks that can respond and adapt rapidly to changes. To help them make that transformation, SAP offers mySAP Product Lifecycle Management – a comprehensive solution that provides the tools that enable high-tech manufacturers to get the right product to market at the right time at the right cost. And that means increased profitability and competitive advantage.

Meeting the Needs of Key Industry Segments

mySAP PLM provides a full range of features and functions designed to help OEMs, EMS providers, component manufacturers, semiconductor manufacturers, and other players in the high-tech ecosystem.

Life-Cycle Collaboration (LCC) and Analytics

Collaboration is at the core of most activities in the high-tech ecosystem. From OEMs to semiconductor manufacturers, high-tech companies must work with members of their supply chain to meet their objectives, which means they must coordinate the activities of their different members.

In fact, collaboration is a strategic priority for high-tech companies in such areas as new product development where collaboration lends itself naturally to an exchange environment. It permits employees from the different ecosystem partners to effectively share relevant product and project information globally – information that is consistently and conveniently made available at a central location, while keeping supporting and detailed informa-

tion within the enterprise. The life-cycle collaboration capabilities of mySAP PLM provide OEMs and EMS providers with the ability to collaborate on designs, including the exchange of technical drawing documents, bills of material (BOMs), and manufacturer parts information in order to reduce time-to-market and to ensure manufacturability.

Life-cycle collaboration functionality supports the design registration process for semiconductor manufacturers, enabling collaboration with the distributors and OEMs early on during the design phase. This ensures a good product fit and a well-managed opportunity. This ability is important to the semiconductor industry in which constant innovation and increasingly shorter product life cycles are always a challenge.

Life-Cycle Data Management (LDM)

The life-cycle data management capabilities provided by mySAP PLM, such as engineering change management and order change management, are key for enabling a global solution that reduces time-to-market, cost-to-market, and cost-of-changes for the high-tech industry. Especially for high-tech companies, strong integration with electrical/electronic computer-aided design (ECAD) systems is very important because most high-tech product design is related to electronic components and assemblies.

Environment, Health and Safety (EH&S)

Toxic chemicals are often used in the semiconductor manufacturing and the imaging and photographic equipment manufacturing industries. Concern about the environment and the health and safety of employees and customers needs to be backed up by support in many areas. mySAP PLM provides a complete environment, health, and safety solution for high-tech companies.

Enterprise Asset Management (EAM)

mySAP PLM provides a complete asset life-cycle management solution for semiconductor and other asset-intensive high-tech manufacturers so they can achieve optimal use of their assets with the lowest costs and the best possible performance.

On average, semiconductor companies spend about 19% of sales on new plants and equipment. With an extremely high capital investment, a state-of-the-art fab (12-inch wafer) can cost in excess of U.S. \$3 billion to build. Given this level of spending, equipment utilization and optimization (maximizing ROI for capital investments) is a key measure of success for these manufacturers.

Quality Management

In the highly competitive industry of high-tech manufacturing, quality in processes and products is a key differentiator over the competition. With well-managed, quality processes, EMS providers, semiconductor manufacturers, and component manufacturers can provide cost-efficient, high-quality parts to OEMs, who in turn provide an equally high-quality finished product to end consumers.

Program and Project Management (PPM)

In today's fast-moving high-tech world, efficient program and project management during product development provided by mySAP PLM enables high-tech companies to innovate constantly and bring their new products to market first.

mySAP PLM: New Features for 2004

mySAP PLM provides a number of new features and enhancements designed to help high-tech companies successfully manage their product development and manufacturing processes. These include:

■ ECAD integration

For high-tech as well as other discrete industries companies with electrical/electronic design departments, enhanced ECAD integration capabilities increase manufacturability, as well as the improve the efficiency of the communication of electronic design data to the supply chain partners like electronics manufacturing services providers.

■ SRM-PLM integration

Bringing buyers and suppliers together and enabling them to communicate and collaborate efficiently is one surefire way to save costs and accelerate time-to-market. To achieve this efficiency, collaboration tools must deal with both the technical data and bidding data. This can't happen properly without a tightly integrated, simultaneous bidding scenario and design collaboration processes. When this capability is added to improved communication between engineering and purchasing teams, then a company has everything in place to optimize product costs during the early design phase and considerably shorten time-to-market.

■ Collaboration on new product development and introduction

Increased flexibility in defining industry-specific data within the cProject Suite application of mySAP PLM enables sharing and managing of dedicated processes within product design and production ramp-up. This includes, for example, process technology transfer between semiconductor companies or product change management between OEM and EMS providers.

SAP FOR CONSUMER PRODUCTS

The Consumer Products Industry: An Overview

Consumer packaged goods (CPG) is a mature industry. Globalization, keen competition, technology, and market changes, as well as shortened product life cycles, are the major challenges facing CPG companies today.

Twenty-five to 30 years ago, retailers had to deal with perhaps 150 new items per year. Today, it is nothing for a retailer to have to make new product acceptance decisions on thousands of new products. It is important to note that retail selling space has not grown at the same rate. Therefore, for every new product accepted by a retailer, it is likely a product that is currently in distribution will have to be discontinued to make room for the new product – in the warehouse, on the shelf, and in the merchandising plans.

The challenge of the CPG industry is to determine which new products to accept and offer to consumers. The reality is, most retailers will not accept all the new items offered by all the manufacturers, and only products that appropriately match consumer demand have a good chance to reach the shelves.

New items take a lot of time and money to design, develop, test, market, and sell. For most manufacturers, it is a major investment that requires due diligence in the research phase to improve the likelihood for success. Time-to-market is a key driver. However, the marriage of research and development, marketing, sales, logistics, and production represents a great challenge for manufacturers. Beyond internal cross-functional teams, best practice CPG manufacturers also include consumers, partners, suppliers, and retailers at an early stage of the development process.

To overcome the barriers associated with evaluating ideas for new products, developing them, and launching them, CPG manufacturers are looking for an integrated, closed-loop solution that can support the entire process – from product concept to sell-through – to ensure more successful, profitable new products and efficient processes.

Trends and Pain Points of the Consumer Products Industry

All companies recognize that supply chain improvements reduce costs, but the real winners understand the direct relationship between supply chain strength and selling success. The right products are required to attract consumers and meet their needs.

The battle for consumers' attention takes place at the store shelf. The manufacturer's goal is to convert consumers' attention into action and get the product into the shopping basket. The check-out counter – the point-of-sale or POS – is the moment of truth for a product. Although some blockbuster products carry the success of an entire company on their shoulders, consumer

tastes and preferences keep changing. To meet their demands, companies come out with new and enhanced versions of the same product. Therefore, all manufacturers must invest in the development of new products.

However, new product development requires enormous resources – both human and financial. At least one in four new product introductions fail. Therefore, the need to ensure more successful and profitable new products and efficient processes from CPG manufacturers, as well as retailers, is huge.

The challenge for manufacturers is to find the right products to launch in the market and to promote to end consumers. True innovation is difficult in the consumer products industry – and it's not enough. CPG manufacturers share the challenge of not only generating fresh ideas, but making them commercially successful as well.

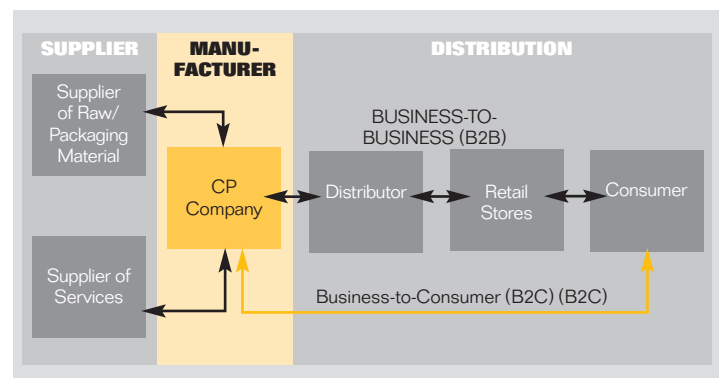


Figure 65: The Consumer Products Industry Ecosystem

The Consumer Products Industry Ecosystem

To understand the kind of product life-cycle management capabilities that an effective CPG ecosystem requires, it helps to know how a number of key processes might work in such an ecosystem. When you understand the tasks and services that the various organizations perform, you can understand how they deliver value to the marketing, planning, engineering, manufacturing, distribution, selling, maintenance, and recycling efforts that are associated with the life cycles of consumer products.

mySAP PLM Today

Helping the CPG industry in its new product development and sell-through efforts is SAP's commitment to the new product development and introduction (NPDI) process. Fully supported by mySAP PLM and other key capabilities and applications, NPDI is a comprehensive business process that enables the conceptualization, design, development, and launch of new products – all in a single, integrated, closed-loop process. SAP offers the capabilities CPG manufacturers require to develop quality products faster and more cost-effectively.

SAP supports all cycles of the NPDI process with capabilities, such as the following:

- Idea management
- Project management with stage-gate functionality
- Approval checkpoints
- Recipe management
- Collaborative scenarios
- External collaboration
- Document management
- Reporting
- Integrated quality management
- Promotion and demanding planning
- Integration to production
- Integration to distribution

Meeting the Needs of the Consumer Products Industry

CPG manufacturers need to think fast, respond quickly, and share information rapidly. SAP support for the new product development and introduction process enables CPG manufacturers to bring new products to market successfully and cost-effectively by orchestrating people and information into one streamlined process. This approach results in the following advantages:

- New products fit better with the corporate strategy
- New products better meet consumers' needs
- Visibility of all ideas for new products and early “go and kill” decisions that avoid wasted efforts

- More efficiency and effectiveness across the entire development
- Transparency in and easy access to all project-related documents
- Consistency of used and developed recipes
- Minimized risk of market failure
- Increased listing acceptance of retailers
- Aligned production and shipment processes that reduce time-to-market,
- Reduced reaction time by real-time monitoring of the product launch

Business Benefits for Consumer Products Customers

mySAP PLM benefits consumer products companies in many ways, including the following:

- mySAP PLM provides a complete product life-cycle management solution – from first concept throughout the life cycle of the assets, including the maintenance and shutdown of assets.
- mySAP PLM enables full and effective collaboration across business processes and enterprise boundaries.
- mySAP PLM enables a company, its teams, and its business partners to effectively plan, monitor, and document development projects, providing access to accurate, up-to-date product, asset, and project data.
- mySAP PLM increases the interconnectivity of employees, partners, and suppliers throughout the entire product life cycle. Development projects can be managed efficiently, both within companies, as well as across company borders, smoothly coordinating different engineering partners.
- mySAP PLM is powered by back-end integration. It provides access to the newest information from the development process, such as the latest specification documents, planned resources for prototyping, or current cost and weight estimates, to pave the way for the ensuing project phases.
- mySAP PLM provides Web-based access, subscription, and notification mechanisms.
- mySAP PLM supports both online and real-time collaboration to shorten rework cycles.

mySAP PLM: New Features for 2004

With SAP Recipe Management 2.1, SAP offers important functionality for development processes, especially for consumer products customers. With SAP Customer Relationship Management (SAP CRM) 4.0 and SAP Trade Promotion Management, another milestone for the new product development and introduction scenario was achieved in 2003. This powerful solution provides the rich functionality required to support consumer products companies in successfully developing and introducing new products into the marketplace.

Future Outlook

Currently, SAP is working on a strategic development project for a new capability – trial management – a new PLM scenario that supports the product development process in the research and development environment.

Beginning with the early stages of product design and development up to production, it is important to have the proper tools to prove product quality and to evaluate the production process by trial in a consistent manner. A new product passes through various iterations in the R&D process before it is widely produced. Therefore, recipe management capabilities and the management of samples in the quality management capabilities of mySAP PLM will be integrated into a workbench to manage trials. Trial management will allow developers to combine management of their creative work and the definition, testing, and comparison of related samples in a single environment.

THE TECHNOLOGY FOUNDATION

When analyzing their need for a product life-cycle management solution, companies must consider the benefits of integration. From an IT perspective, integrated systems let companies leverage their existing investments in technology and reduce the total cost of ownership of systems. From a business perspective, the integration of systems is a prerequisite for reaching beyond enterprise boundaries to link up with suppliers and customers. Integration is also vital to adaptability – to create a technology infrastructure that enables an organization to sense and respond to the market and that is flexible enough to support new business processes and keep up with changing business conditions.

The concept of integration has evolved in recent years. In today's dynamic business environment in which collaboration and responsiveness are paramount, companies need to do more than just link systems. They must address integration at several levels, and encompass **people, information, and processes**.

Unfortunately, that kind of comprehensive integration has been difficult to achieve. Most companies find themselves with a range of heterogeneous systems, and connecting those systems can be an expensive, time-consuming effort. What's more, the integration technologies that have been brought to market have tended to focus purely on the technological facets of integration and ignore the process and people parts of the equation.

SAP addresses those issues with SAP NetWeaver.

SAP NetWeaver

mySAP Product Lifecycle Management is powered by the SAP NetWeaver technology platform, a comprehensive integration and application platform that helps reduce companies' total cost of ownership (TCO). SAP NetWeaver facilitates the integration and alignment of people, information, and business processes across organizational and technological boundaries.

SAP NetWeaver easily integrates information and applications from virtually any source. It interoperates with and can be extended using the primary market technologies of Microsoft .NET, Sun's Java 2 Platform, Enterprise Edition (J2EE), and IBM WebSphere.

SAP NetWeaver is the technical foundation for the mySAP Business Suite family of business solutions and the SAP xApps portfolio of packaged composite applications. It ensures maximum reliability, security, and scalability, so mission-critical business processes run smoothly. This Web services-based platform offers a comprehensive, tightly integrated set of capabilities. And by providing preconfigured business content, SAP helps reduce the need for custom integration and shortens time to value. SAP NetWeaver takes companies into the future by laying the foundation for innovative, cross-functional business processes that help organizations grow.

SAP NetWeaver is also the technical foundation of Enterprise Services Architecture, the recently introduced SAP blueprint of a services-based, enterprise-scale, integrated business architecture. Enterprise Services Architecture combines SAP's enterprise applications expertise with the flexibility of Web services and other open technologies into a blueprint for complete and services-based business solutions. These solutions set new standards for usability, scalability, and adaptability, while significantly lowering TCO. Going forward, all SAP business solutions will be developed based on the Enterprise Services Architecture blueprint.

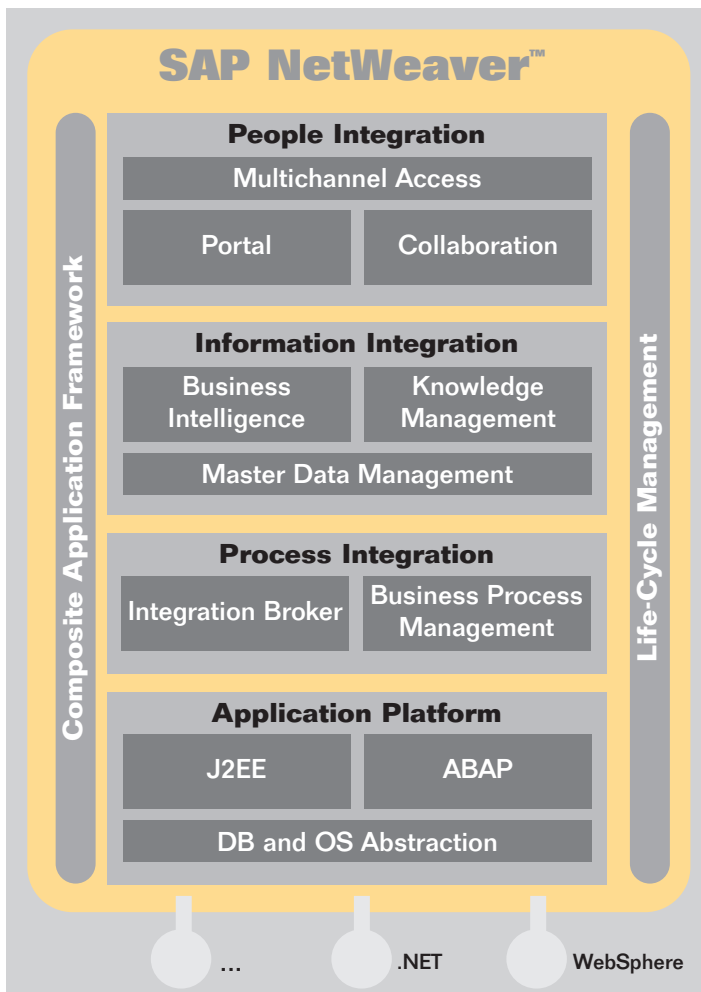


Figure 66: SAP NetWeaver™ Composite Application Framework

Support for the SAP xApps Portfolio

SAP NetWeaver enables SAP xApps composite applications – a new generation of applications that snap onto existing IT infrastructures. SAP xApps composite applications lower the cost of integration by delivering integration solutions as software products, reducing the cost of project work or custom development. Companies can compose and enhance SAP xApps composite applications to develop customized solutions without starting each development effort from scratch.

Openness and Interoperability

Based on open standards, SAP NetWeaver is fully interoperable with Java-based systems (IBM WebSphere in particular) and Microsoft .NET specifications, thus preserving SAP, partner, and customer investments in mission-critical business applications. This flexibility prevents vendor lock-in and lowers overall TCO. SAP is also working closely with the major standards bodies to drive the further evolution of open standard technologies with a particular emphasis on Web services.

mySAP PLM and SAP NetWeaver Capabilities

The following sections describe the SAP NetWeaver key capabilities that play important roles in enabling the features and functions of mySAP PLM.

Enterprise Portal

SAP NetWeaver's enterprise portal capabilities combine open, standards-based portal technology with a variety of ready-made business content – all delivered through a single point of access.

mySAP PLM is supported by a full-blown enterprise portal designed to intuitively accommodate the specific needs of each user. Because different users work differently – each placing different and specific demands on their workspace – the individual portals are preconfigured and customized for their role in the organization. The role-based enterprise portal gives everyone in an enterprise – including business partners – fast, simple, and user-friendly Web access to all of the internal and external applications, information, tools, services, and resources they need to perform daily tasks as efficiently, as organized, and as productively as possible.

mySAP PLM includes specific portal interfaces and menus that have been created to fit the specific roles encountered in product life-cycle management, such as roles for the project manager, maintenance engineer, design engineer, and so forth. In addition, to enhance the solution's portal technology, mySAP PLM provides content-rich business packages that are designed for specific user groups.

The knowledge a user needs is delivered through special iViews programs that assemble information from all sorts of different sources according to each specific role. The most important projects, for example, can be displayed, together with related networks, the documents used most often, or current messages. iViews are easy and quick to use, add, change, or remove.

For mySAP PLM users, the portal gives personalized access to the wide range of information, applications, and services people need to do their jobs based on their individual responsibilities within the supply chain network.

Collaboration

SAP NetWeaver supports real-time collaboration and dynamic communication within permanent communities and ad hoc groups. Real-time collaboration features – such as online meetings, chat, instant messaging, and application sharing – enable users to interact and exchange knowledge instantly. These collaborative tools enhance the value of information resources by allowing them to be accessed and shared in a contextual, cost-saving, team environment.

With mySAP PLM, collaborative tools can be extended across the enterprise and across the supply chain network, helping to accelerate processes and increase speed and responsiveness.

Business Intelligence

SAP NetWeaver's business intelligence capabilities include a complete range of data warehousing and analytical tools that enable companies to integrate, analyze, and disseminate relevant, timely information to a wide variety of business users.

mySAP PLM includes reports about all necessary objects of PLM via the SAP Business Intelligence solution. Predefined business content can be easily adapted and enlarged in various ways. The following areas are covered:

- Quality management (notifications, items, causes, tasks, actions, inspection lots, and inspection results)
- Enterprise asset management (notifications, items, causes, tasks, actions, maintenance orders, and service orders)
- Program and project management (costs, cash, progress, earned value, dates, and claims)
- Costing (product cost design estimate)

Virtually all phases in the management of the product life cycle benefit from the business intelligence capabilities of SAP NetWeaver.

Mobile Solutions

An open, standards-based platform for mobilizing enterprise applications, SAP Mobile Infrastructure is the technology behind SAP solutions for mobile business – the industry-leading enterprise mobility solution that delivers the power of mySAP Business Suite solutions to mobile devices. SAP solutions for mobile business provide complete business capabilities in connected and disconnected environments.

mySAP PLM supports mobile asset management to process complete maintenance-relevant business processes. Mobile asset management capabilities enable service teams to read, enter, and process data and information online or offline – anytime, anywhere. Using mobile devices, they can transfer the recorded information to a back-end system. This provides maintenance engineers with an increased scope of action, and helps to reduce read and data transfer errors.

Technology for Today and Tomorrow

SAP NetWeaver will evolve over time to meet the changing needs of SAP customers, the introduction of new technologies, and varying business conditions. From this point on, SAP NetWeaver will function as the underlying engine for all SAP products and

initiatives. It lays the foundation for innovative, cross-functional business processes that help companies grow. And as the foundation for the Enterprise Services Architecture blueprint, SAP NetWeaver enables the efficient integration of people, information, and business processes across technological and organizational boundaries.

Today, businesses need to go beyond simply linking systems to truly integrate people, processes, and information. SAP NetWeaver is a groundbreaking platform that enables that level of comprehensive integration – across the enterprise and with supply chain network partners – in a cost-effective, coherent fashion. SAP NetWeaver also provides a bridge between today and tomorrow with its practical approach to weaving existing technologies together into an infrastructure that will meet future demands for increased flexibility, speed, and efficiency.

CONCLUSION

Make it better. Make it faster. Make it affordable. Make it before the competition. And make it right so customers come back again and again.

The only way to do all of that and succeed is to do it together — people and processes working seamlessly to achieve well-planned, well-defined goals and objectives. To accomplish such uniformity from the beginning to the end of the product life cycle requires a complete answer — one unified solution that seamlessly integrates information and processes and allows everyone in the enterprise to effortlessly share and exchange project and process data. Only one such solution exists: mySAP Product Lifecycle Management from SAP.

In thousands of companies around the world, mySAP PLM is working the way people work — together — by providing the sophisticated applications, resources, services, and tools that people need to do their jobs and that companies need to compete and win. SAP is proud of the role this powerful solution is playing in helping to produce the thousands of diverse new products enriching global markets today. As these markets change and grow, SAP is committed to enabling companies to meet their needs by continually adapting mySAP PLM to the demands tomorrow brings.

SAP hopes the *mySAP Product Lifecycle Management Statement of Direction 2004* gives you a better understanding of mySAP PLM. For more information about mySAP PLM and the SAP market-driven PLM strategy for success, please visit the SAP Web site at www.sap.com/plm

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